Kolmogorov-Arnold Sieve Networks for High-Dimensional Estimation and Inference

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Abstract

This paper introduces Kolmogorov-Arnold Sieve Networks (KASN), a novel deep learning framework for nonparametric estimation and semiparametric inference in high-dimensional settings. By integrating the Kolmogorov-Arnold representation theorem into a sieve extremum framework and leveraging sparsity, our proposed method achieves an explicit asymptotic rate of convergence in the root-mean-square norm of $o_P(n^{-1/4})$. The sparse compositional structure imposed on the target function embeds multiple-index and latent factor models, capturing nonlinear interactions across covariates while preserving the dimension-free convergence rate. This framework delivers asymptotically normal plug-in estimators and facilitates nuisance function estimation within Double/Debiased Machine Learning (DML). These are areas where existing deep neural network architectures are less suited due to their black box nature, reliance on i.i.d. assumptions, and slower convergence rates. KASN overcomes these limitations through a learnable B-spline activation function structure that is theoretically compatible with stationary mixing processes.

Keywords: Curse of dimensionality, deep learning, double/debiased machine learning, Group Lasso, Kolmogorov-Arnold networks, latent factors, method of sieves, multiple-index models, time-series.

JEL classification: C14, C32, C45, G12.

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Full code with documentation is available at: https://github.com/samiabd8/SieveKAN, click here for the most recent version of this paper.

1 Introduction

In modern statistical and economic applications, flexible nonparametric function estimation methods (such as deep neural networks, kernel-based methods, and spline-based methods, among others) are increasingly important for modeling complex relationships arising from high-dimensional datasets. A fundamental challenge facing these methods in high-dimensional settings is the curse of dimensionality (CoD), which dictates that the sample size required for a given estimation accuracy grows exponentially with the dimensionality of the covariate space. Although approaches such as generalized additive models (GAMs) seek to mitigate the CoD through structural assumptions (e.g., additivity, limited degrees of interaction), these assumptions generally fail to capture the compositional and interactive nature inherent in many complex real-world relationships. Introduced by Liu et al. (2024), Kolmogorov-Arnold Networks (KANs) represent a promising alternative to standard deep neural network (DNN) architectures that inherently addresses some of these limitations. As the name suggests, the universal approximation capabilities of KANs arise from the Kolmogorov-Arnold representation theorem (KART; Kolmogorov, 1956; Arnold 1957), which posits that continuous multivariate functions can be represented as a finite composition of univariate functions and summation operations, regardless of the degree of interaction between covariates.

This paper develops a novel and theoretically grounded sieve extremum estimator based on deep KANs for nonparametric regression in high-dimensional settings featuring time-series data. By integrating KANs with the sieve estimation framework of Chen and Shen (1998) and leveraging a sparse version of the KART (where only a small subset of univariate components are non-zero), we achieve an explicit asymptotic convergence rate of $o_P(n^{-1/4})$ in the L^2-norm that does not depend on the dimension of the covariates (d). This directly mitigates the CoD in a manner similar to sparse GAMs when the target function admits a sparse compositional form. Although we assume the true sparse compositional target function is fixed for a given sample size n, our novel regularization method centered around the Group

Lasso (Yuan and Lin, 2006) is consistent with a data-driven form of model selection from the perspective of Barron et al. (1999). The Group Lasso is a regularization technique that encourages sparsity by penalizing entire groups of coefficients simultaneously (rather than individual coordinates as in the Lasso), enabling functional selection by pruning redundant activation functions in the resulting Kolmogorov-Arnold sieve network (KASN) extremum estimator.

The KASN framework can be understood as a flexible compositional extension of multiple-index models[1] (e.g., Ichimura and Lee, 1991; Hristache et al., 2001), where both the index transformations and their aggregation are nonparametric and learned from data; with sparsity governing which components are active (rather than fixing them a priori), as we formalize in the model assumptions. Our sieve estimation approach relies on a finite-dimensional space (Θ_n) that grows with n in order to control network complexity growth in tandem with functional sparsity, while simultaneously ensuring sufficient network approximation power.

To illustrate, consider the nonparametric regression model $Y_t = f_0(X_t) + U_t$ where given covariates $X_t = \{X_{t,j}\}_{j=1}^d$, the parameter of interest belonging to an infinite-dimensional space Θ is defined as: $\theta_0 \equiv f_0$ with $f_0(X_t) = \mathbb{E}[Y_t|X_t]$. We approximate Θ using a sequence of finite-dimensional sieve spaces $\{\Theta_n\}$ that are dense[2] in Θ as $n \to \infty$. Kernel-based approaches (e.g., Nadaraya, 1964; Watson, 1964) estimate f_0 directly but suffer from the CoD as d grows; single-index restrictions of the form $f_0(X_t) = f(X_t'\beta)$ circumvent this by reducing the problem to one dimension but confine the structure to be estimated to a single linear projection (Ichimura, 1993). Fully nonparametric sieve and/or spline methods (e.g., the various estimators outlined in Chen, 2007) impose smoothness but require the researcher to pre-specify the basis structure and interaction terms. A similar tradeoff is present in projection pursuit regression (PPR; Friedman and Stuetzle, 1981) methods and the aforementioned multi-index models, which extend to finitely many linear indices but

[1] See Horowitz (2009) for a comprehensive overview of this class of models and the single-index models they generalize.

[2] That is, any element in Θ can be approximated arbitrarily well by elements in the sieve for sufficiently large n: $\Theta_n \subset \Theta_{n+1}$ for all $n \geq 1$ and $\overline{\bigcup_{n\geq 1} \Theta_n} = \Theta$.

still require pre-specifying the basis and index structure. GAMs of the form $f_0(X_t) = \sum_{j=1}^{d} f_j(X_{t,j})$ mitigate the CoD through additivity, but rule out the complex compositional structures present in many applications. In contrast, KASN exploits the Kolmogorov-Arnold representation theorem to decompose multivariate dependencies into adaptive univariate compositions, estimating f_0 at the dimension-free $o_P(n^{-1/4})$ rate in the L^2-norm without pre-specifying the interaction structure; which is precisely the accuracy required for valid semiparametric inference.

More generally, the partially linear regression (PLR) model[3] introduced in Robinson (1988) (which also relies on kernel-based methods), where $Y_t = D_t\alpha_0 + g_0(W_t) + U_t$, introduces a finite-dimensional linear component α_0 such that $\theta_0 \equiv (\alpha_0, g_0)$. Estimating α_0 at a $\sqrt{n}$-rate requires first estimating the nuisance function g_0 at a sufficiently fast nonparametric rate[4], a task where the choice of method matters considerably when the controls are high-dimensional. Unlike the other referenced methods, our KASN extremum estimator $\hat{\theta}_n \in \Theta_n$ instead estimates the unknown function (whether f_0 in the general nonparametric regression setting or the nuisance function g_0) as a sparse compositional function learned directly from data. Our proposed estimator delivers the $o_P(n^{-1/4})$ rate required for $\sqrt{n}$-consistent inference on α_0, without pre-specifying the interaction structure (see Examples 1-2 in Section 3).

In contrast to Multilayer Perceptron (MLP) architectures, a widely-used class of deep neural networks (DNNs) where approximation power is tied to network width and length (potentially leading to intractable complexity growth), the expressiveness of KASNs hinges on the flexibility of the splines characterizing the learned activation functions. MLPs represent a class of flexible nonparametric estimators that has demonstrated strong empirical performance in a variety of machine learning tasks (e.g., regression, classification, natural language processing), largely due to their powerful approximation capabilities. Relative to single hidden layer ("shallow") network architectures, deep MLPs regularly achieve im-

[3]We formally define a partition of X_t into a treatment variable $D_t \in \mathbb{R}^{d_1}$ and controls $W_t \in \mathbb{R}^{d_2}$ in the next section.

[4]Specifically, a convergence rate of $o_P(n^{-1/4})$ in the L_2-norm, which ensures that the bias from estimating the nuisance function is asymptotically negligible (see, e.g., Newey, 1994).

proved empirical performance (see Goodfellow et al., 2016 for a comprehensive overview). Despite this, the application of MLPs for rigorous statistical inference and econometric analysis presents significant challenges. The most critical of which arises from the "black box" nature of these architectures, which hinders tractability[5] in a manner that models such as GAMs do not. Furthermore, much of the recent work attempting to bridge MLPs with the statistical theory required for inference (typically under the category of "Deep ReLU networks", or ReLU DNNs) relies on i.i.d. assumptions, rendering this class of DNNs less suited for the time-series data structures common in economic and financial applications. Lastly, commonly derived convergence rates in this literature require conditions that tie the smoothness of the target function to the dimension of the covariates (imposing degrees of differentiability that are generally unrealistic for even a modest number of regressors) in order to deliver the convergence rate required for semiparametric inference, as we discuss throughout.

Our proposed KASN extremum estimator $\hat{\theta}_n$ directly addresses these limitations and achieves a dimension-free (up to logarithmic factors[6]) convergence rate of $o_P(n^{-1/4})$ under sparsity, the aforementioned rate required for delivering $\sqrt{n}$-consistency in the semiparametric settings we consider, spanning the partially linear model above and the structural applications noted below. By verifying high-level conditions pertaining to stochastic equicontinuity and the Riesz representation theorem, we establish convergence and the asymptotic normality of smooth (linear or otherwise) plug-in functionals of $\hat{\theta}_n$. As a result, we are able to accommodate a wide range of practical applications involving time-series and semiparametric inference. Such applications include the nonparametric regression model example under a high-dimensional data-generating process (DGP) that we consider in simulations, as well as treatment effect estimation in partially linear structural models (e.g., the partially linear

[5]From the perspective of controlling the metric entropy of the network in high dimensions, which is a measure of function space complexity (e.g., in terms of covering numbers).

[6]As we show in our theoretical results covered in Section 3, the only dependence of the metric entropy on the covariate dimension is logarithmic in nature and asymptotically negligible; crucial for mitigating explosions in complexity as we move towards high-dimensional settings.

instrumental variables regression model, a semi-nonparametric example covered in Chen, 2007). The KASN estimator consistently demonstrates strong performance across Monte Carlo simulations of these settings and an empirical application involving macroeconomic financial time-series data. In each setting, our novel regularization method centered around the Group Lasso effectively prunes redundant univariate functions in the architecture, crucial for controlling the metric entropy. We also demonstrate how our estimator can be used for nuisance function estimation in the Double/Debiased Machine Learning (DML) framework of Chernozhukov et al. (2018). DML has emerged as an indispensable tool in high-dimensional statistics for constructing valid inference for finite-dimensional parameters. We demonstrate how DML offers a bias-robust alternative in high-dimensional regimes ($d \gg n$) and outline a cross-fitting procedure that permits stationary processes[7], including those covered by the KASN model assumptions.

Through the tractability of our KASN architecture featuring learned B-spline activation functions, our proposed framework bridges the gap between deep compositional networks and multiple-index models (as we demonstrate in Section 2.4). Although in theory it is possible to obtain similar dimension-free rates of convergence for existing sieve-based methods (e.g., penalized splines) by leveraging sparsity, the distinction lies in specification versus discovery. To our knowledge, no existing penalized sieve method jointly achieves: data-driven structural recovery using deep network architectures, theoretical guarantees under time-series dependence, and a convergence rate of $o_P(n^{-1/4})$. Traditional sieve methods place the burden of structural intuition on the researcher, requiring one to pre-specify the basis (including knot locations, polynomial degrees, and specific interaction terms), thus encoding structural assumptions that are often rigid in nature. In contrast, our proposed framework delegates this architectural search to the data through the learned activation functions and our aforementioned regularization method. As a result, the model recovers the active compositional components without requiring a priori knowledge of the interaction structure. This flexibil-

[7]Namely, the "neighbors-left-out" cross-fitting method of Semenova et al. (2023)

ity comes at the cost of significantly increased computation time relative to convex methods such as the Lasso (a tradeoff common in DNN estimation), although our implementation is designed to efficiently employ the processing power of modern computer architectures[8].

The goal of this framework is to reopen the research frontier on sieve artificial neural networks (aNNs), a foundational data-driven methodology (White, 1990; Chen and Shen, 1998; Chen and White, 1999) for flexible nonparametric estimation and semiparametric inference, as we discuss in the literature review.

1.1 Related Literature

Many of the recent contributions to the literature on DNNs revolved around networks using the Rectified Linear Units (ReLU) activation function, $\sigma(x) = \max\{0, x\}$, due to their convenient and relatively simple properties. The manner in which this activation function truncates negative inputs can be viewed as a natural induction of sparsity[9] in activation maps (as opposed to in weights), in addition to mitigating the vanishing gradient problem (discussed later on in the context of our KASN architecture). Yarotsky (2017, 2018) develop the approximation theory for this class of DNNs, while Farrell et al. (2021) and Schmidt-Hieber (2020) propose rigorous frameworks for their application in semiparametric inference and nonparametric regression, respectively. The former derives nonasymptotic high probability bounds for DNNs, while the latter considers a sparsely connected[10] architecture and obtains an oracle inequality.

Linking the improved performance of networks when additional layers are added in MLPs to the KART, Schmidt-Hieber (2021) presents an interpretation of this representation theorem using deep ReLU networks under simple modifications (namely, smoothness of the tar-

[8]Through the use of graphics processing units (GPUs). We expand on how our proposed architecture benefits from this with respect to speeding up computation times in Appendix A.4.

[9]Ensuring that only neurons with non-negative inputs ($x \geq 0$) remain active, preserving important features across forward passes. This differs from the *structural* sparsity relevant to this framework and methods such as weight decay.

[10]With respect to the network weights, where a compositional structure assumption is imposed on the target function.

get function; resulting in learnable Lipschitz continuous inner functions). This constituted a breakthrough in the application of the KART in aNNs, once thought to be irrelevant[11] in this context. Although several papers in the time between Kolmogorov (1956) and Schmidt-Hieber (2021) have attempted to show the usefulness of the KART in aNNs (e.g., Sprecher, 1996, 1997), these were limited to network structures with two hidden layers and fixed activation functions. Unlike Schmidt-Hieber (2021), where the intuition regarding the KART is inspired by ReLU DNNs, our KASN architecture and extremum estimator are motivated by the literature concerning single-index and multiple-index models (Ichimura and Lee, 1991; Ichimura, 1993; Hristache et al., 2001). This bridges the compositional structures of DNN architectures to the econometrics literature on dimension-reduction, through shifting the emphasis from the regularity of the inner functions in the KART to the compositions of the outer aggregations.

Applying the crucial insight from Schmidt-Hieber (2021), Liu et al. (2024) propose an alternative to MLPs in the form of KANs, which derive their universal approximation capabilities from the KART rather than the Universal Approximation Theorem (Hornik et al., 1989). While both of these DNN architectures are fully-connected, KANs replace the fixed activation functions in MLPs with learnable activation functions in the form of B-splines. The authors demonstrate how by defining the layers of KANs in terms of matrices of univariate functions, the KART can be generalized to arbitrary widths and depths. As a result, KANs are effectively architectures with MLPs on the outside and splines on the inside, the latter contributing to significantly improved accuracy as shown by Liu et al. (2024) in a wide range of applications. Deriving generalization bounds for KANs, Zhang and Zhou (2024) show how it is possible to guarantee that the bound scales with the L^1-norm of the coefficients when the activation function is composed of B-spline basis functions.

In both of these KAN frameworks, ℓ_1-regularization plays an essential role. While these papers lay the groundwork for KANs with learned activation functions in the form of B-

[11] e.g., Girosi and Poggio (1989), titled: "Representation Properties of Networks: Kolmogorov's Theorem Is Irrelevant" attributed to the otherwise fractal nature of the inner functions.

splines, our proposed framework introduces a new regularization procedure centered around the Group Lasso ($\ell_{1,2}$-regularization, targeting groups of coefficients *across* activations) and achieves dimension-free rates of convergence for the KASN M-estimator. Gouriéroux and Monfort (2025) formalize the identification issues and representation non-uniqueness inherent in Kolmogorov-Arnold and MLP architectures[12] and relate multi-layer networks to stochastic state-space models.

In a major contribution to the literature on sieve extremum estimation (Grenander, 1981), Chen and Shen (1998) obtain convergence rates for these nonparametric and semiparametric estimators involving time-series data; in addition to establishing the $\sqrt{n}$-consistency and asymptotic normality of plug-in sieve extremum estimates with respect to smooth functionals[13]. The method of sieves entails estimating a function belonging to a possibly infinite-dimensional parameter space Θ using a sequence of approximating parameter spaces (growing in complexity as $n \to \infty$, defined as the sieve space Θ_n) that are dense in Θ. This sieve M-estimator framework derives theoretical results using a quasi-maximum-likelihood estimation (QMLE) method (White, 1982; Gouriéroux et al., 1984) which does not suffer from the issues associated with infinite-dimensional maximum likelihood (namely, slow convergence rates and potential inconsistency under misspecification). Crucially, the extension of sieve extremum estimators to time-series data in Chen and Shen (1998) allows for the application of this method to a wide range of economic applications that would otherwise not be possible under the i.i.d. assumptions made in the existing literature (e.g., Shen, 1997). Among the applications involving time-series models considered to demonstrate this sieve extremum estimator, rates of convergence for nonlinear state-space models using aNNs are obtained. Chen and White (1999) obtain improved approximation rates (explicitly $o_P(n^{-1/4})$ in the L^2-norm) for shallow sieve aNN estimators building directly upon this framework. An outline of regularization approaches in nonlinear (finite-dimensional or otherwise) sieve estimation can

[12] Directly addressed by the KASN framework, as discussed after stating the theoretical results in Section 3.

[13] A "plug-in" (or substitution) estimator is a method of estimating a functional of interest by substituting unknown population distributions or nuisance functions with their estimated counterparts; see Shen (1997) for a general overview.

be found in Chen (2007). Shen et al. (2023) investigate the asymptotic properties of shallow sieve network estimators and discuss how their results can potentially be extended to DNNs with Lipschitz continuous activation functions. To the best of our knowledge, there does not exist a framework integrating the method of sieves with modern DNN architectures in order to establish an explicit $o_P(n^{-1/4})$ rate of convergence of such networks.

The remainder of this paper is organized as follows. Section 2 defines the target function space and outlines our proposed KASN extremum estimator, including the key parameters of the KASN architecture and its assumptions. Section 3 presents our main theoretical results on approximation error and convergence rates. We present results of our Monte Carlo simulations for our DML (semiparametric mean regression) model in addition to a simple nonparametric regression in Section 4. Section 5 features our two empirical applications involving macroeconomic financial variables, including a forecasting exercise. We conclude with a summary and a discussion of directions for future research in Section 6.

2 Kolmogorov-Arnold Sieve Networks

We begin by defining the relevant metric spaces and their properties, in addition to the notation used throughout. Although we adopt the notation of Chen and White (1999), our approach diverges in the characterization of the target function space. Rather than assuming functions in this space admit a Fourier representation[14], we impose the more standard assumption that it takes on the form of a Sobolev space. As we discuss in Section 2.4, we impose a comparable yet local structural assumption on the univariate functions (Lipschitz continuity), rather than the global spectral assumption of Chen and White (1999).

[14]Central to obtaining dimension-independent rates of convergence in their framework, see Section A.5 in the Appendix for a full exposition.

2.1 Notation

Given a strictly stationary sequence $\{Z_t\}_{t=1}^n = (Y_t, X_t')'$ with dimension $(1+d)$ and joint probability measure P_0, where $\{X_t\}_{t=1}^n$ is a sequence of covariates (potentially including lags), let μ be a probability measure on $\mathbb{R}^d$ (the marginal distribution of $X_t \in \mathbb{R}^d$).

We define $L_2(\mu)$ as the Hilbert space of all measurable functions $h : \mathbb{R}^d \to \mathbb{R}$ such that $\int_{\mathbb{R}^d} |h(x)|^2 d\mu(x) < \infty$, equipped with the following norm

$$\|h\|_{L_2(\mu)} = \left(\int_{\mathbb{R}^d} |h(x)|^2 d\mu(x) \right)^{1/2}$$

and to simplify the notation, let $\|\cdot\| \equiv \|\cdot\|_{L_2(\mu)}$ unless stated otherwise. In order to characterize the space of the target function ($f_0 \equiv \theta_0 \in \Theta$ for nonparametric regression[15], where Θ may more generally represent a class of regression functions, probability densities, and other structural mappings of interest), we define the multi-index[16] $\bar{\lambda} = (\bar{\lambda}_1, \ldots, \bar{\lambda}_d)^\intercal$, which represents a vector of non-negative integers with order $|\bar{\lambda}| = \sum_{i=1}^d \bar{\lambda}_i$. Let $D^{\bar{\lambda}}$ represent the weak partial derivative operator corresponding to $\bar{\lambda}$, where:

$$D^{\bar{\lambda}} h(x) = \frac{\partial^{|\bar{\lambda}|} h(x)}{\partial x_1^{\bar{\lambda}_1} \ldots \partial x_d^{\bar{\lambda}_d}}$$

The target function space is defined as a Sobolev space $\mathcal{F}_d^{m+1} \equiv W_2^{m+1}(\mu)$. In addition to consisting of all measurable functions $h : \mathbb{R}^d \to \mathbb{R}$ such that $h \in L_2(\mu)$, this space contains its weak derivatives $D^{\bar{\lambda}} h \in L_2(\mu)$ for all multi-indices $\bar{\lambda}$ with $|\bar{\lambda}| = \sum_{i=1}^d \bar{\lambda}_i \leq m+1$. The norm for a Sobolev space $W_2^{m+1}(\mu)$ is defined as:

$$\|h\|_{W_2^{m+1}(\mu)} = \left[\sum_{|\bar{\lambda}| \leq m+1} \left(\|D^{\bar{\lambda}} h\|_{L_2(\mu)} \right)^2 \right]^{1/2}$$

[15] Whereas $\theta_0 \equiv (\alpha_0, f_0)$ in the aforementioned semiparametric applications, e.g., the partially linear regression model (Robinson, 1988).

[16] A vector specifying the number of partial derivatives to be taken with respect to each corresponding input variable.

where $m \geq 0$ is an integer governing the smoothness of the target function $f_0 \in \mathcal{F}_d^{m+1}$. Specifically, the value $m+1$ is the upper bound on the order of the multi-index[17] $\bar{\lambda}$ dictating the maximum order of partial derivatives that are required to be square-integrable (i.e., belonging to $L_2(\mu)$).

As we demonstrate in our model assumptions, this definition allows us to quantify the smoothness of the true function we seek to estimate. As a result, we are able to link the properties of the proposed KASN architecture that features (learned) spline activation functions to this function space. Critically, this target space allows us to work towards convergence rates where the dependence on the smoothness parameter m is absorbed into constants; unlike typical "minimax" optimal convergence rates[18] (Stone, 1982) commonly found in the literature on deep ReLU networks. For example, a common minimax rate of convergence in nonparametric estimation takes on the form of $n^{-m/(2m+d)}$, which is prohibitively slow in many potential high-dimensional settings. By avoiding such a dependency, our approach is a key step towards mitigating the CoD in DNN architectures applied in semiparametric inference.

Although our analysis relies on the smoothness provided by the Sobolev space, we are only able to obtain convergence rates in the L^2-norm rather than in the stronger Sobolev norm (the latter implying convergence in the former). We will demonstrate why this is the case in the context of spline approximation theory that our key results hinge on, as rates of convergence in the Sobolev norm would not be independent of m under the KASN architecture. Nonetheless, this weaker form of convergence is sufficient for the applications we consider, including important extensions[19]. Furthermore, given how quickly convergence rates involving non-static conditions on m breakdown in high-dimensional settings, this

[17]For example, when $d = 2$ and $m = 0$, the norm sums derivatives of order 0 (the function $h \in L_2(\mu)$ itself, where $\bar{\lambda} = (0,0)$) and order 1 (where $\bar{\lambda} = (1,0)$ or $\bar{\lambda} = (0,1)$, representing $\frac{\partial h}{\partial x_1}$ and $\frac{\partial h}{\partial x_2}$). This yields three terms in total.

[18]Where the smoothness degree of the target function m is tied to d; see Schmidt-Hieber (2020) for a full discussion in relation to DNNs.

[19]Namely, obtaining asymptotic normal linear plug-in estimators, where the nonparametric component is estimated using our proposed KASN extremum estimator.

tradeoff is justified by how it enables the proper theoretical grounding for analysis in these settings.

2.2 KASN Estimator

We assume the target function $f_0 \in \mathcal{F}_d^{m+1} \equiv W_2^{m+1}(\mu)$ admits a sparse compositional structure, where only a small subset (of cardinality s_0) of the univariate functions in the KART are non-zero. This form of sparsity arises naturally in many settings involving high-dimensional data and we thoroughly verify it computationally across Monte Carlo simulations and our empirical applications. For example, in a financial risk model predicting the probability of a given borrower defaulting, a researcher might collect hundreds of variables based on available data. It is reasonable to expect that only a few covariates (e.g., including payment history and debt-to-income ratios) truly impact odds of default, with complex and not necessarily additive nonlinear relationships arising between this small subset of variables. Formally, the KART states:

Definition 1 (Kolmogorov-Arnold Representation Theorem)**.** For a multivariate continuous $f : [0,1]^d \to \mathbb{R}$ and given univariate functions $\phi_{i,j} : \mathbb{R} \to \mathbb{R}$, $\Phi_i : \mathbb{R} \to \mathbb{R}$, we have:

$$f(X_t) = f(X_{t,1}, X_{t,2}, \ldots, X_{t,d}) = \sum_{i=1}^{2d+1} \Phi_i \Bigg(\sum_{j=1}^{d} \phi_{i,j}(X_{t,j}) \Bigg) \tag{2.1}$$

where the inner functions $\phi_{i,j}$ represent univariate feature transformations of each input $X_{t,j}$; the inner sum $\sum_{j=1}^{d} \phi_{i,j}(X_{t,j})$ aggregates these into a scalar summary (which can be viewed a nonlinear generalization of the linear index $X_t'\beta$ in single-index models) and Φ_i acts as a nonlinear link function on that summary. Together, this two-level structure reduces a d-dimensional estimation problem to at most $2d+1$ one-dimensional problems, regardless of the interaction structure of $f(\cdot)$. As we discuss below, our sparse variant uses only $s_0 \ll 2d+1$ active components, which is central to mitigating the CoD. In order to ensure regularity of these univariate components, we impose smoothness and Lipschitz continuity assumptions

in addition to the smoothness of the target function $f_0 \in \mathcal{F}_d^{m+1}$.

In the context of the possibly infinite-dimensional parameter space $\mathcal{F}_d^{m+1}$, this sparse compositional assumption is analogous to those made in sparse GAM frameworks[20], as it entails approximating a multivariate function using a subset of univariate functions[21]. As we demonstrate throughout this paper, a KASN can be viewed as a generalization of these models to sparse compositional structures (thus allowing for complex interactions), embedded within a DNN architecture. In terms of multiplicative interactions, Liu et al. (2024) demonstrate how even a simple two-layer KAN structure is capable of approximating functions such as $f(x_1, x_2) = x_1 x_2$. In this example, it can be observed that the KAN computes $x_1 x_2$ by leveraging $2x_1x_2 = (x_1 + x_2)^2 - (x_1^2 + x_2^2)$, where the spline functions learn the individual parts of this decomposition. The main consequence of this sparse compositional assumption imposed on the target function $f_0 \in \mathcal{F}_d^{m+1}$ is that it allows for the dimension of the covariates, d, to grow with the sample size at even a polynomial rate: $d = O(n^{\zeta_d})$ for any $\zeta_d > 0$.

The main consequence of this structural assumption is that it facilitates achieving explicit asymptotic convergence rates in the L^2-norm that do not depend on d at all. The ability of our proposed framework to permit covariate/input dimensions that scale as the sample increases without causing the metric entropy to grow uncontrollably[22] makes it a very promising direction for the application of sparse deep networks in high-dimensional settings; namely due to how assumptions commonly made in high-dimensional models such as $s_0 = O(1)$ (which implies that the target function can be approximated by a fixed number of univariate components: e.g., $f(X_t) \approx \sum_{j=1}^{s_0} f_j(X_{t,j})$ in GAMs) can be overly restrictive[23] in many practical applications. In GAMs, s_0 represents the number of covariates that have a non-zero functional effect on the response variable. As a result, we mitigate the CoD by

[20]Such as Raskutti et al. (2011), Meier et al. (2009), and Friedman (1991).

[21]Also referred to one dimensional, or "1D" functions.

[22]By leveraging spline approximation theory, in addition to the method of sieves and structural sparsity, as we discuss in the next section.

[23]To the contrary, we allow s_0 to grow slowly in tandem with the *size* of the KASN architecture as $n \to \infty$.

exploiting sparsity in a manner similar to sparse additive models and bound the network entropy by using a sieve estimation approach. We note that a similar claim is also made in Liu et al. (2024), one that is debatable for general KAN classes in the absence of sparsity and appropriate mechanisms for controlling complexity[24]. Without these theoretical underpinnings, the network complexity *implicitly* scales with d and renders general KAN classes vulnerable to exponential growth of said complexity, especially in high-dimensional settings. As we demonstrate throughout, our resulting convergence rate is independent of d and its impact on the metric entropy is shown to be asymptotically negligible. Going forward, we use the symbol $\asymp$ to denote asymptotic equivalence[25]. Furthermore, let $\mathbb{E}[\cdot]$ and $P(\cdot)$ denote the expectation and probability under measure P_0 (the same applies to $Var[\cdot]$ and $Cov[\cdot]$) henceforth.

We integrate our proposed KASN architecture in the sieve extremum estimator framework of Chen and Shen (1998), which allows for time-series data and avoids the reliance on the otherwise restrictive i.i.d. assumption (e.g., as in Schmidt-Hieber, (2020) and Farrell et al., (2021), among many other DNN frameworks designed for nonparametric regression). This is of significance as a wide range of the potential high-dimensional applications a researcher might be interested in using DNNs for generally involve time-series data. In addition to obtaining asymptotic normal plug-in estimators, we consider semiparametric applications in the double/debiased machine learning framework introduced by Chernozhukov et al. (2018). Our proposed KASN extremum estimator[26] achieves the required rate of convergence (precisely $o(n^{-1/4})$ in the L^2-norm) for nuisance function estimation under Neyman-orthogonal scores in DML.

For the strictly stationary process $\{Z_t\}_{t=1}^n = (Y_t, X_t')'$ and the true probability measure P_0, where $\theta_0 : \mathbb{R}^d \to \mathbb{R}$ is defined as the maximizer of the population expected criterion

[24]The sparse compositional form of the target function central to our framework is one of many potential examples.

[25]Where $a_n \asymp b_n \iff a_n \gtrsim b_n$ and $b_n \gtrsim a_n$ as $n \to \infty$. This is equivalent to $b_n = O(a_n)$ and $a_n = O(b_n)$.

[26]Following Chen and White (1999), we refer to θ_n as such due to how the sieve space Θ_n is constructed using the outputs of our deep sieve network architecture.

function and the training objective function takes on $\mathcal{L}_n(\theta) = \frac{1}{n}\sum_{t=1}^{n} l(Z_t, \theta)$, we define the following:

$$\theta_0 \equiv \arg\sup_{\theta\in\Theta} \mathbb{E}_{P_0}[l(Z_t, \theta)] \tag{2.2}$$

$$\hat{\theta}_n \equiv \arg\sup_{\theta\in\Theta_n} \mathcal{L}_n(\theta) \tag{2.3}$$

Here, $l(\cdot,\cdot)$ is a quasi-log-likelihood function[27] belonging to a class of general loss functions, permitting a variety of nonparametric/semiparametric regression problems. For example, in nonparametric mean regression ($f_0 \equiv \theta_0$) we have $l(Z_t, \theta) = -\frac{1}{2}(Y_t - \theta(X_t))^2$, where the solution to the population problem is the conditional expectation $\theta_0 = \mathbb{E}[Y_t \mid X_t = x]$ given $\mathbb{E}[\epsilon_t \mid X_t] = 0$. In the case of nonparametric quantile regression (using the check loss; see Koenker, 2005), $l(Z_t, \theta) = -|Y_t - \theta(X_t)|[\tau \mathbf{1}_{\{Y_t \geq \theta(X_t)\}} + (1-\tau)\mathbf{1}_{\{Y_t < \theta(X_t)\}}]$, where for the τ-th quantile and given the inverse CDF $F^{-1}(\cdot)$, we have the maximizer $\theta_0(x) = F^{-1}_{Y|X}(\tau \mid x)$ satisfying $P(Y_t \leq \theta_0(x) \mid X_t = x) = \tau$.

2.3 KASN Architecture

The network architecture of our proposed KASN features three main parameters: the number of layers (L_n), the network width (W_n; corresponding to the number of nodes W_n^l in a given layer $l \in [1, \ldots, L_n]$) and the size of the B-spline grid (G_n) for the learnable activation functions that make the network fully tractable. Unlike how MLPs place fixed activation functions on nodes, the learnable activation functions belong on *edges* (or, "in-between" layers) so that the KAN nodes simply perform summation. We define the number of total *potential* edges in our KASN architecture as:

$$r_n \equiv d\, W_n + (L_n - 2)\, W_n^2 + W_n = O\big(n^{\zeta_d} \log n\big), \tag{2.4}$$

[27] A real-valued mapping $l : \mathcal{Z} \times \Theta \to \mathbb{R}$ that scores how well a given θ fits an observation Z_t, without requiring correct specification of the full likelihood.

so that under $d = O(n^{\zeta_d})$, the potential edge count is dominated by the input layer. Throughout the KASN framework, r_n is a key parameter[28] essential to the tractability our architecture achieves. Central to controlling the metric entropy growth (see the bound established in Theorem 2; where it is demonstrated how dependence on d is logarithmic in nature) in this framework is obtaining a subset of *active* edges ($\hat{s}_n$) through our regularization procedure.

Due to how the expressiveness KANs are instead linked to the flexibility of the splines (i.e., is capable of being increased by simply expanding the grid size), this architecture facilitates the controlled growth of the sieve space. In these structures, the width determines the number of interactions between input variables and by extension, the diversity of interactions within a given layer. This makes KANs well-suited for estimation via the method of sieves in high-dimensions, in addition to how they do not suffer from the black box nature of MLPs. Likewise, stacking additional layers[29] in a KAN architecture permits richer representations of f_0. To this end, in the context of compositional target functions, Schmidt-Hieber (2020) points out that allowing the length of DNNs to grow concurrently with the sample size is intuitive. Similarly, the concluding remarks of Hornik et al. (1989) addresses the natural connection between multilayer feedforward networks and the method of sieves. The tractability of KAN architectures significantly contributes to this framework that harmonizes DNNs with a sieve estimation approach for nonparametric regression and semiparametric inference.

To ensure the domain is bounded[30] between $[0, 1]$, a CDF normalization is applied to inputs $X_t \equiv [X_{t,1}, \dots X_{t,d}]' \in \mathbb{R}^d$ (inducing compact support on the probability measure, a common assumption in sieve estimation frameworks) such that:

$$\tilde{X}_{t,j} = \hat{F}_n^j(X_{t,j}), \;\; j = 1, \dots, d \tag{2.5}$$

[28] In the case of the shallow sieve networks of Chen and White (1999), $r_n \equiv W_n$ simply represents the number of hidden units; as outlined in Section A.5 of the Appendix.

[29] Where each layer is a matrix of univariate functions Φ_l; see Appendix A.4 for an extended comparison with MLP architectures.

[30] e.g., as required by the KART.

where $\hat{F}_n^j(\cdot)$ is the (marginal) empirical distribution function (EDF). This introduces an approximation error[31] of $O(n^{-1/2}\log n)$: a term negligible with respect to the resulting KASN approximation error established in Theorem 1. Although commonly applied in empirical finance applications (e.g., copula models), the use of CDF normalization of inputs in deep learning architectures remains underexplored. Employing this normalization in the place of commonly used standardization techniques facilitates the use of a fixed spline grid: define $\mathcal{I}_\epsilon \equiv [-\epsilon, 1+\epsilon]$ for a small $\epsilon > 0$ padding term, so that $[0,1] \subset \mathcal{I}_\epsilon$ by construction. WLOG, let $\mu \equiv \tilde{\mu}$ denote the distribution of the normalized inputs $\tilde{X}_t \in [0,1]^d$, with all univariate function spaces are defined on $\mathcal{I}_\epsilon$ henceforth.

Let $\Theta \equiv A \times \mathcal{F}_d^{m+1}$ be the parameter space where $A \subset \mathbb{R}^a$ is a compact set representing the finite-dimensional space of our linear parameters of interest and Θ is equipped with the pseudometric $\|\cdot\|$. Given B-spline order $k \geq m+1$ and coefficient vector $\boldsymbol{c}_n \equiv \{\boldsymbol{c}_j\}_{j=1}^{r_n}$, the finite-dimensional sieve space is given by $\Theta_n \equiv A \times \mathcal{K}_n$, where the KASN class is defined as:

$$\mathcal{K}_n = \Bigg\{ f(\tilde{X}_t;\boldsymbol{c}_n) \;\Bigg|\; f(\boldsymbol{x};\boldsymbol{c}_n) = \Phi_{L_n} \circ \big(\Phi_{L_n-1} + h\big) \circ \cdots \circ \big(\Phi_1 + h\big) \circ \Phi_0(\boldsymbol{x}),\ \boldsymbol{c}_n \in \mathbb{R}^{r_n \times G_n}, \\ \sum_{j=1}^{r_n} \|\boldsymbol{c}_j\|_2 \leq \Delta_n,\ \|\boldsymbol{c}_n\|_{0,2} \leq s_n,\ \|f\|_\infty \leq \mathcal{B}_n,\ \max_{|\bar{\lambda}| \leq m+1} \|D^{\bar{\lambda}} f\|_\infty \leq \mathcal{D}_n \Bigg\}, \tag{2.6}$$

where $h : \mathbb{R}^{W_n^{l-1}} \to \mathbb{R}^{W_n^l}$ is the identity residual connection[32] $h(x) = x$, each $\Phi_l : \mathbb{R}^{W_n^{l-1}} \to \mathbb{R}^{W_n^l}$ (with $\Phi_0 : [0,1]^d \to \mathbb{R}^{W_n^1}$ and $\Phi_{L_n} : \mathbb{R}^{W_n^{L_n}} \to \mathbb{R}$) is a layer of univariate B-spline activation functions with coefficients $\boldsymbol{c}_j = (c_{j,1}, \ldots, c_{j,G_n})' \in \mathbb{R}^{G_n}$ for each potential edge $j \in \{1, \ldots, r_n\}$. Explicitly, each univariate edge takes on the form $\phi(x) = \sum_{g=1}^{G_n} c_g B_g(x)$.

The class $\mathcal{K}_n$ is delimited by four constraints. The coefficient radius Δ_n, defined in (2.10), imposes the $\ell_{1,2}$-constraint $\|\boldsymbol{c}_n\|_{1,2} \leq \Delta_n$ that our Group Lasso regularization targets. The deterministic sparsity bound s_n, defined in (2.7) below, restricts the number of active edges

[31] As established in Lemma 1, which addresses each case under our relevant assumption individually.

[32] A parameter-free mapping that supports gradient stability by bounding gradients away from zero, in a manner similar to ReLU activations in MLPs. Note that the KASN width W_n is fixed across every hidden layer $\ell \in [2, \ldots L_n]$. See Section A.4 in the Appendix for an expanded discussion, including a formal definition of the recursive B-spline functions.

through the group sparsity (pseudo-)norm $\|\boldsymbol{c}_n\|_{0,2} \equiv \#\{j \leq r_n : \boldsymbol{c}_j \neq \mathbf{0}\}$. The remaining two constraints are standard in the sieve literature (Chen, 2007): the supremum norm (sup-norm) bound $\mathcal{B}_n$ is constant when s_0 is fixed and grows logarithmically under K.3(b) (delivering the uniform boundedness used in the NLO sample-splitting procedure; see Section A.3), while the derivative bound $\mathcal{D}_n$ is polylogarithmic and is required for the derivative-based functional of Example 2.

Together these constraints render $\mathcal{K}_n$ compact. For each of the finitely many admissible active supports, the four constraints define a closed and bounded subset of $\mathbb{R}^{r_n \times G_n}$, which is compact by the Heine-Borel theorem, where $\mathcal{K}_n$ (and by extension Θ_n) is the continuous image of the finite union of these sets. Compactness in turn guarantees that the supremum defining $\hat{\theta}_n$ is attained exactly, so that the KASN estimator is an *exact* (rather than *approximate*) sieve extremum estimator.

We formally define the deterministic sparsity bound as:

$$s_n \equiv \lceil C_{s_n}\, \bar{s}_0 \rceil = O\big((\log n)^2\big), \qquad C_{s_n} \geq C_J, \tag{2.7}$$

where s_n renders $\mathcal{K}_n$ a deterministic class of functions, as required for the empirical process theory of Chen and Shen (1998), while remaining large enough to contain the approximation constructed in Theorem 1 (see the proof of this theorem for the definition of C_J and derivation of C_{s_n}). In practice, the Group Lasso penalty together with the active edge threshold (below) serves as the computational realization of this constraint: the penalized solution is sparse at the group level and edges falling below the threshold are pruned. As demonstrated in Theorem 2, r_n enters the metric entropy only through $\log(r_n/s_n) = O(\log n)$, so that the covariate dimension affects the complexity of the sieve space only logarithmically.

Next, we define the following adaptive threshold for what constitutes an active edge: Δ_n/r_n, which corresponds to a baseline network where coefficients are evenly distributed[33]

[33]See Appendix A.4 for an discussion on entropy regularization, a method frequently applied in reinforcement learning (Ahmed et al., 2019) in order to encourage exploration via selection of increasingly stochastic policies. We discuss how this method is used heuristically in KAN implementations (Liu et al., 2024) and

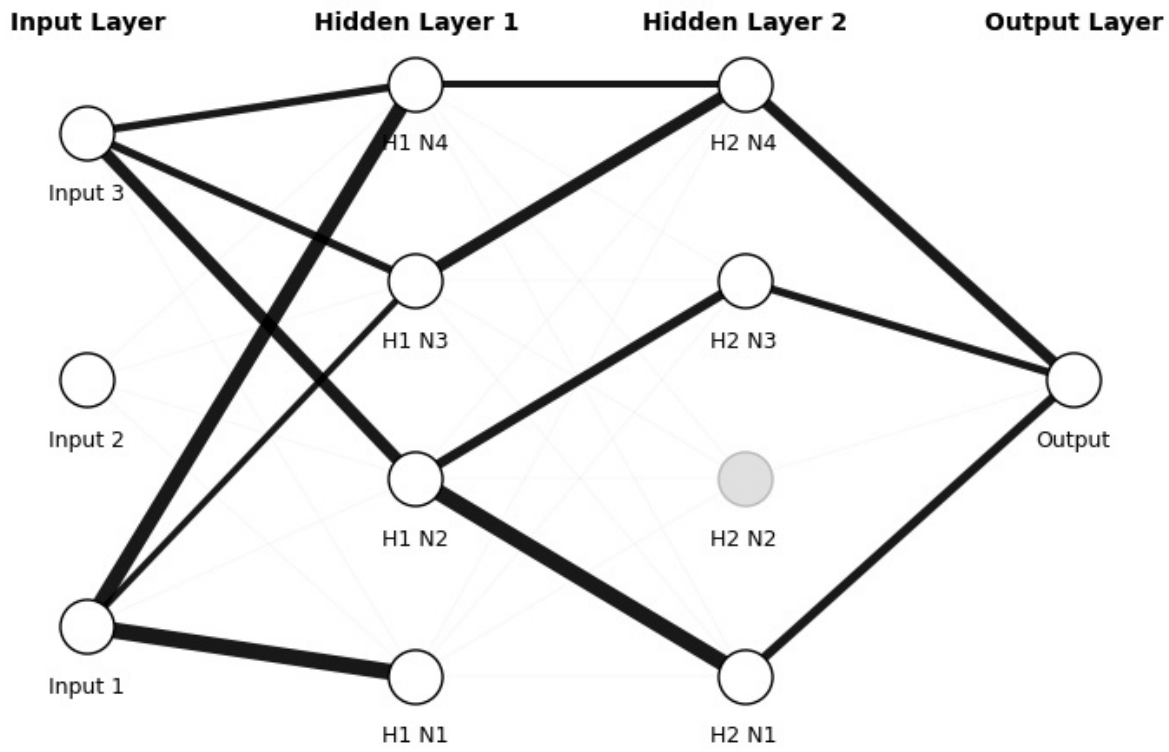


Figure 1: A KASN With Active Edges Highlighted

across all potential edges. This tractable and data-adaptive criterion addresses the potential issues associated with arbitrary fixed thresholds commonly applied in KAN architectures, particularly in the presence of many covariates. We note that it is possible for activation functions with non-zero coefficients that do not belong to the active paths (i.e., edges disconnected from the network output) to be pruned. See Figure 1 for an example of such a function, corresponding to the Node H1-N1; where the magnitude of each learned activation function corresponds to the thickness of each connecting line in the plot. In order to highlight the robustness of our regularization approach centered around the constrained set of coefficients $\sum_{j=1}^{r_n}\|\boldsymbol{c}_j\|_2 \equiv \|\boldsymbol{c}_n\|_{1,2}$, we only prune edges falling below the Δ_n/r_n threshold. The vast majority of active edges after training contribute directly to the network output across Monte Carlo simulations and empirical applications, justifying this conservative choice. Our theoretical results in Section 3 (in particular, the metric entropy bound established in Theorem 2) are built upon this adaptive threshold.

contrast it with our theoretically grounded regularization method.

The KASN estimator is defined as the constrained maximizer $\hat{\theta}_n = \arg\sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta)$,[34], where $\mathcal{K}_n$ imposes the $\ell_{1,2}$-constraint $\|\boldsymbol{c}_n\|_{1,2} \leq \Delta_n$ with radius $\Delta_n \asymp n^{\zeta_\delta}$ (see (2.10)). In practice, optimization is performed using the penalized objective function $\mathcal{L}_n(\theta) - \lambda\|\boldsymbol{c}_n\|_{1,2}$ over Θ_n, where $\hat{\boldsymbol{c}}(\lambda)$ denotes the maximizer for a given penalty weight $\lambda > 0$ (not to be confused with our multi-index definition earlier). Writing $R_n(\lambda) \equiv \|\hat{\boldsymbol{c}}(\lambda)\|_{1,2}$ for the realized coefficient norm at weight λ, the penalty weight associated with a given radius is

$$\lambda_n \equiv \inf\{\lambda > 0 : R_n(\lambda) \leq \Delta_n\}. \tag{2.8}$$

Selecting λ_n through (2.8) is equivalent to selecting Δ_n through the hyperparameter ζ_δ, since the map $\lambda \mapsto g_n(\lambda)$ is non-increasing on the convex penalized subproblem within each active support and is empirically monotone in our implementation. The asymptotic theory in Section 3 is developed for any deterministic sequence $\Delta_n \asymp n^{\zeta_\delta}$ with a fixed ζ_δ satisfying the condition stated below; the use of cross-validation in finite samples does not affect the asymptotic properties of $\hat{\theta}_n$. In practice, we apply holdout validation (for an overview see Tashman, 2000) and demonstrate the robustness of holdout validation (standard in time-series forecasting applications) in our Monte Carlo simulations. In Lemma 5, we establish the membership of $\hat{\theta}(\lambda_n)$ in $\mathcal{K}_n$ and its optimality, given that the rate of Theorem 3 is stated for a maximizer of $\mathcal{L}_n$ over $\mathcal{K}_n$.

2.4 Key Parameters and Assumptions

For cases where A is a nonempty set (e.g., in Example 1), we have: $\theta_0 \equiv (\alpha_0, f_0) \in \Theta$ and $\theta_n \equiv (\alpha_n, f(\cdot; \boldsymbol{c}_n)) = (\alpha_n, \pi_n f_0) \in \Theta_n$, where π_n is a projection operator representing the best possible approximation of f_0 in the sieve space. Specifically, π_n is the orthogonal projection under the L^2-norm which ensures that the approximation is the unique element in

[34]This supremum is attained exactly: the compactness of $\mathcal{K}_n$ and continuity of $\mathcal{L}_n(\theta)$ under our assumptions stated in the following section guarantee attainment by the extreme value theorem, so that $\mathcal{L}_n(\hat{\theta}_n) \geq \sup_{\theta \in \Theta_n} \mathcal{L}_n(\theta) - O(\tilde{\epsilon}_n^2)$ holds with $\tilde{\epsilon}_n = 0$ for any measurable maximizer.

the sieve space closest to the true θ_0: $\pi_n\theta_0 \equiv \arg\min_{\theta\in\Theta_n}\|\theta-\theta_0\|$. Existence follows from the compactness of $\mathcal{K}_n$ and the continuity of the L^2-norm. Under the pseudometric[35] $\|\cdot\|$ on Θ, we define Θ_n as dense in Θ: for any $\theta \in \Theta$, there exists a $\pi_n\theta \in \Theta_n$ such that $\|\theta - \pi_n\theta\| \to 0$ as $n \to \infty$. This ensures that the approximation error (Theorem 1) induced by the sieve space Θ_n becomes asymptotically negligible relative to the estimation error (Theorem 3).

We define the KASN architecture parameters[36] (L_n, W_n, G_n) as follows:

$$L_n \asymp \log\ n, \quad W_n \asymp s_0 \asymp \log n, \quad G_n \asymp n^{\gamma} \tag{2.9}$$

$$\Delta_n \asymp n^{\zeta_\delta}, \quad \zeta_\delta > \frac{\gamma}{2}, \quad \frac{1}{4(m+1)} < \gamma < \frac{1}{2} \tag{2.10}$$

where γ guarantees sufficient approximation power to deliver our $o_P(n^{-1/4})$ convergence rate. The lower bound on γ reflects that spline approximation is inherently limited by smoothness: a target in W_2^{m+1} is approximated at rate $G_n^{-(m+1)}$ regardless of the spline order, so that $k \geq m+1$ ensures the splines possess sufficient degree (without impacting/improving the rate). The lower bound $\zeta_\delta > \gamma/2$ ensures the coefficient radius Δ_n is large enough for the sieve space to contain the approximation constructed in Theorem 1, as we verify explicitly in its proof. Since Δ_n enters the entropy bound of Theorem 2 only logarithmically, this lower bound does not incur additional costs with respect to the estimation error established in Theorem 3. In order to reflect the deep compositional structure of the KASN architecture (Assumptions K.3-4), we define the *true total active edge count* $\bar{s}_0$ as follows:

$$\bar{s}_0 = s_0 \times L_n = O((\log n)^2) \tag{2.11}$$

where the derivation of this condition arises naturally in the proof of Theorem 1. Likewise, it is shown throughout the proofs of the main theorems that $W_n \geq s_0$ is a sufficient condition

[35]Recall that a pseudometric satisfies the axioms of a metric with the exception that $\|\theta_1 - \theta_2\|$ does not necessarily imply that $\theta_1 = \theta_2$. In the method of sieves literature, $\|\cdot\|$ is typically used interchangeably with $d(\cdot,\cdot)$.

[36]More specifically, under B-spline order k: $p_n \equiv \sum_{l=0}^{L_n-1} W_l W_{l+1}(G_n + k - 1) \approx L_n W_n^2 G_n = r_n G_n$.

for guaranteeing that the network possesses the required capacity, in addition to how the bounds imposed on γ are obtained.

KASN Assumptions:

K.1(a): *The strictly stationary process* $\{Z_t\}_{t=1}^n = (Y_t, X_t')'$ *with compact support is i.i.d. or M-dependent (for* $M \in \mathbb{N}$*), or*

K.1(b): *β-mixing stationary with exponential decay* $\beta(j) \leq \beta_0 e^{-cj}$ *for some* $\beta_0 > 0$, $c > 0$, *or*

K.1(c): *φ-mixing (uniform mixing) stationary with* $\phi(j) \leq \phi_0 j^{-\xi}$ *for some* $\phi_0 > 0$, $\xi > 2$.

K.2(a): *The parameter space* Θ *is equipped with the pseudometric* $\|\cdot\|$, Θ_n *(with network parameters* p_n*, restricting* $G_n \geq 2$*) is dense in* Θ *and* $\mathcal{F}_d^{m+1} \equiv W_2^{m+1}(\mu)$ *is a Sobolev space. Furthermore, the pseudometric* $\|\cdot\|$ *is equivalent to* $K^{1/2}$ *on* Θ*, where* $K(\theta_0, \theta) \equiv E_0[l(Z_t, \theta_0) - l(Z_t, \theta)]$ *is the Kullback-Leibler-type divergence: there exist constants* $0 < c_1 \leq c_2 < \infty$ *such that*

$$c_1 K^{1/2}(\theta_0, \theta) \leq \|\theta - \theta_0\| \leq c_2 K^{1/2}(\theta_0, \theta) \quad \forall\, \theta \in \Theta$$

K.2(b): (Identification) *For all small* $\epsilon > 0$ *and for some constant* $C_1 > 0$*:*

$$\sup_{\{\theta \in \Theta_n,\, \|\theta - \theta_0\| \leq \epsilon\}} \mathrm{Var}[l(Z_t, \theta) - l(Z_t, \theta_0)] \leq C_1 \epsilon^2$$

K.2(c): (QMLE Regularity) *There exists a function* $U(\cdot)$ *and some* $\nu \in (0, 2)$ *such that for any* $\delta > 0$*:*

$$\sup_{\{\theta \in \Theta_n, \|\theta - \theta_0\| \leq \delta\}} |l(\theta) - l(\theta_0)| \leq \delta^\nu U(Z_t)$$

with $E[|U(Z_t)|^\kappa] \leq C_2$*, for some* $C_2 > 0$, $\kappa > 2$.

K.2(d): (Penalty rate) *The penalty weight satisfies* $\lambda_n \bar{s}_0 = O(\delta_n^2)$*, where* $\bar{s}_0 = s_0 \times L_n$ *as defined in* (2.11) *and* δ_n *is the rate of Theorem 3.*

Our initial set of assumptions are standard in the literature on sieve extremum estimators. Assumption K.1 permits data dependence structures beyond the baseline i.i.d. case used to

derive the entropy bounds of DNNs. Since M-dependence is generally an unrealistic assumption for economic and financial time-series[37], the β-mixing and uniform mixing assumptions (see Section A.2 of the Appendix for full definitions) are necessary for controlling serial dependence in the context of empirical processes. As Chen and Shen (1998) highlight, exponential decay rates in stationary β-mixing processes is satisfied by many nonlinear Markov processes (e.g., a broad class of nonlinear ARCH models; Doukhan, 1994).

We do not claim support recovery (e.g., as in Negahban et al., 2012), as the only nondegeneracy condition invoked involves the population curvature assumption[38] in K.2(a). The variance regularity condition in Assumption K.2(b) guarantees that θ_0 is identifiable (noting that the sieve space $\mathcal{K}_n$ is compact[39] by construction) and that $l(\cdot)$ is well-behaved in a neighborhood surrounding it. Correspondingly, the smoothness of the likelihood is dictated by the parameter ν in Assumption K.2(c) (where $\nu = 1$ for the quadratic loss function), thus ensuring that the empirical process remains well-behaved as the sieve becomes increasingly fine as $n \to \infty$.

KASN Assumptions (continued):

K.3(a): (Sparse Compositional Structure) *The target function* $f_0 \in \mathcal{F}_d^{m+1} \equiv W_2^{m+1}(\mu)$, $f_0 : [0,1]^d \to \mathbb{R}$ *takes on the following sparse compositional form, where* $s_0 \ll 2d+1$*:*

$$f_0(X_t) = \sum_{i=1}^{s_0} \Phi_i \left(\sum_{j \in \mathcal{J}_i} \phi_{i,j}(X_{t,j}) \right)$$

with $|\mathcal{J}_i| \leq J_{\max} < \infty$, $J_{\max} \ll d$, *and* $d = O(n^{\zeta_d})$ *for some* $\zeta_d > 0$.

K.3(b): (Sparsity Growth) *For best approximation* $\pi_n \theta_0$*:*

i. $s_0 = O(1)$, *or*

[37] To see why this is the case, consider a Moving Average process of dependence order M (denoted $MA(M)$) with a sequence of i.i.d. shocks ϵ_t defined by $X_t = \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \cdots + \theta_n \epsilon_{t-M}$. For a finite M, this implies that the autocorrelation of this process is zero for all lags greater than M, thus ruling out the long-term persistence common in macroeconomic shocks, to give an example.

[38] Rather than a restricted eigenvalue condition, for example.

[39] Where $\hat{\theta}_n$ is the *exact* sieve extremum estimate; see Chen (2007) for an expanded discussion in the context of ill-posed problems.

ii. $s_0 = O(\log n)$

K.4(a): *Each univariate edge function* $\phi_{i,j}, \Phi_i$ *in the representation of* f_0 *belongs to the Sobolev space* $W_2^{m+1}(\mathcal{I}_\epsilon)$ *with uniformly bounded norm* $B < \infty$.

K.4(b): (Lipschitz Continuity) *The univariate edge functions* $\phi_{i,j}, \Phi_i$ *and their derivatives up to order* $m+1$ *are Lipschitz continuous, with uniform constant* $L < \infty$.

K.4(c): (Compositional Regularity) *Each outer function* Φ_i *in the sparse compositional representation of* f_0 *admits a depth-*$(L_n - 1)$ *univariate compositional representation:*

$$\Phi_i = \psi_{i,L_n} \circ \psi_{i,L_n-1} \circ \cdots \circ \psi_{i,2}$$

where each $\psi_{i,l} : \mathbb{R} \to \mathbb{R}$ *belongs to* $W_2^{m+1}(\mathcal{I}_\epsilon)$ *with a uniformly bounded norm and satisfies Assumption K.4(b) with Lipschitz constant* $L_\psi \leq 1$.

K.4(d): (Range Compatibility) *For each* $i = 1, \ldots, s_0$*:*

i. the inner aggregate satisfies $\sum_{j \in \mathcal{J}_i} \phi_{i,j}(x_j) \in [0,1]$ *for all* $x \in [0,1]^d$*, and*

ii. each partial composition $\psi_{i,l} \circ \cdots \circ \psi_{i,2}$ *maps* $[0,1]$ *into* $[0,1]$ *for every* $l = 2, \ldots, L_n$.

The second set of assumptions focuses on the structural properties of the target function and the regularity of its components. Assumption K.3(a) formalizes the sparse compositional structure motivated by the KART (see Definition 1) that allows the KAN sieve extremum to bypass the CoD. An interpretable set of primitive conditions under which K.3(a) naturally arises is the latent factor model:

$$Y_t = g_0(F_t) + \epsilon_t, \quad F_t = h(X_{t,1}, \ldots X_{t,S}), \quad S \ll d \tag{2.12}$$

where $F_t \in \mathbb{R}^S$ is a low-dimensional latent factor depending on only S covariates and ϵ_t satisfies $\mathbb{E}[\epsilon_t | X_t] = 0$. The conditional expectation $f_0(X_t) = \mathbb{E}[Y_t | X_t] = g_0(F_t)$ then depends on X_t through the subset of cardinality S, so that by the KART, f_0 admits a sparse compositional representation with an active edge count of $O(S)$ rather than $O(d)$. From this

perspective, the sparsity level $s_0 \asymp \log n$ is therefore directly interpretable as controlling the number of (active) covariates entering F_t, with the set of relevant factors slowly growing as n increases. The KASN embeds the approximate factor model of Stock and Watson (2002), whose framework accommodates $S \to \infty$ as $n \to \infty$. Crucially, K.3(a) is no more restrictive than the assumptions commonly found in the econometrics literature: beyond sparse linear regression[40] and sparse additive models (discussed earlier in the context of sparse GAMs), multiple-index models impose $f_0(X_t) = G(X_t'\beta_1, \dots X_t'\beta_S)$ with $S \ll d$. This corresponds to a KASN where the inner activation functions reduce to linear projections $\phi_j(X_{t,j}) = \beta_j X_{t,j}$ and the outer aggregation (with respect to the unknown function G, as in Ichimura and Lee, 1991) remains nonparametric. The KASN generalizes this by making both components fully nonparametric while permitting deeper compositional representations.

Assumptions K.4(a) and K.4(b) are regularity conditions on the univariate edge functions, analogous to assumptions commonly found in deep MLP frameworks. The Lipschitz continuity of the derivatives further ensures that the sieve extremum objective is stable[41]. We emphasize that the smoothness index $m + 1$ is a condition on the univariate edge functions rather than on the d-dimensional target: by the chain rule for compositions of smooth functions (Faà di Bruno's formula; see the proof of Lemma 2), the composition of components with bounded derivatives up to order $m + 1$ inherits at least W_2^{m+1} regularity. Therefore $f_0 \in \mathcal{F}_d^{m+1}$ is a floor guaranteed by the components rather than a ceiling imposed on the target, which the sparse compositional form permits to be far smoother than any individual component. In practice we fix cubic B-splines ($k = 4$) throughout, the standard basis in applied spline estimation; since our theoretical results require $k \geq m + 1$ (or $k \geq m + 2$ for the derivative functional of Corollary 1), this accommodates edge smoothness up to $m \leq 2$, with each additional order absorbed by a single additional spline order.

Assumption K.4(c) is a formalization of the structural assumptions implicit in KAN archi-

[40] Where sparsity is imposed on the vector of coefficients and the model is approximated by a shallow KASN with linear activation functions.

[41] Aiding in the mitigation of the aforementioned exploding gradients problem, as well as functional oscillations that unconstrained KANs are potentially prone to.

tectures generalizing the KART to deep architectures (Liu et al., 2024), extended to Lipschitz continuity[42]. This assumption reflects the structural intuition motivated by the literature on multiple-index model, where the inner index functions are linear (and therefore, trivially regular in general) while the outer aggregation G captures the complex nonlinear dependence. Assumption K.4(c) generalizes this by allowing nonparametric inner functions while imposing the required regularity on the outer compositional structure Φ_i to ensure compatibility with the entropy bound established in Theorem 2. Settings where this assumption is most credible involve individual covariate transformations that are smooth but interactions that are complex, such as the factor model structure motivating Assumption K.3(a). Restricting the intermediate arguments in the compositional architecture, Assumption K.4(d) requires all of the inner aggregates and partial compositions to be range-preserving maps on $[0, 1]$. Assumption K.5 imposes primitive smoothness and sparsity conditions on the Riesz representer v^*, extending the existing assumptions concerning the target function.

KASN Assumptions (continued):

K.5: (Representer Regularity) *Let Γ be a bounded linear functional on the closed linear span $\bar{V}$ of $\{\theta - \theta_0 : \theta \in \Theta\}$ under $\|\cdot\|$, with Riesz representer $v^* \in \bar{V}$ satisfying $\Gamma(v) = \langle v, v^* \rangle$ for all $v \in \bar{V}$. Assume:*

i. $\|v^*\|_\infty \leq B_v < \infty$*;*

ii. v^* admits a sparse compositional representation satisfying K.3(a) and K.4, with constants (s_0^v, J^v, B^v, L^v) of the same orders as those of f_0.

The main advantage of our deep architecture (and overparameterization in general) is in the use of additional layers to achieve more efficient representations of the outer functions Φ_i. As a result, our estimator benefits from the improved performance of deeper representations (as discussed in Liu et al. 2024 in comparison to shallow KANs) and the metric entropy growth is controlled via sparsity (through the $s_n \asymp \bar{s}_0$ active edges across the L_n layers).

[42]Where this constraint can be loosened at the cost of adjusted constants, as discussed in the proof of Theorem 1.

This is a key focus in our Monte Carlo simulations under DGPs that involve complex, compositional functions. Assumption K.4(d) ensures that every argument passed to an outer or compositional component lies in the domain $[0,1] \subset \mathcal{I}_\epsilon$ on which the regularity conditions in Assumptions K.4 are imposed. We emphasize that this is a normalization rather than a substantive restriction on the target function. Since affine maps are exactly representable by B-splines of order $k \geq 2$, any bounded sparse compositional representation[43] can be brought into this form (by absorbing affine rescalings). As a result, this alters the norm bound B and the Lipschitz constants only by fixed factors, consistent with the loosening of $L_\psi \leq 1$ discussed above. The padding $\epsilon > 0$ in $\mathcal{I}_\epsilon$ absorbs the asymptotically negligible approximation and normalization errors, ensuring all spline evaluations are contained in this domain for sufficiently large n. As discussed earlier, the interval $[0,1]$ is chosen WLOG and is consistent with earlier aNN frameworks that centered around fixed squashing activation functions (e.g., sigmoid).

With respect to selecting a suitable width such that $W_n \geq s_0$, the use of holdout validation or block-based CV (e.g., Racine, 2000) is both practical and consistent with existing deep learning approaches. In theory, overparameterization in any of the parameters p_n does not harm the model performance due to the adaptability of our sieve space and regularization approach. We verify this using our sparsity ratio (defined in terms of the Δ_n/r_n threshold for active edges) in the simulation analysis across many DGPs, defining $W_n = C_W \log n$ for some constant $C_W > 0$ (similarly for the KASN length, define $C_L > 0$) and evaluating a wide range of potential values. In the case of $s_0 = O(1)$, we simply let $W_n = C_W$ and focus our attention on the case above in proofs involving the metric entropy of this KAN architecture. The requirement that the width of the KASN grows at a rate equivalent to that of s_0 arises from our sparse compositional structure on $f_0 \in \mathcal{F}_d^{m+1}$ (Assumption K.3(a)). This ensures that the network possesses enough potential univariate functions to approximate the active edge functions in θ_0, even as $n \to \infty$. For simplicity and consistency with standard

[43] See Schmidt-Hieber (2020) for an analogous normalization in the context of MLP weights.

approaches, we set $L_n = \lfloor \log n \rfloor$ and use the stochastic optimizer Adam (Kingma and Ba, 2014) in practice (see Appendix A.4).

Under these assumptions, the KASN extremum estimator delivers the convergence rate: $\|\hat{\theta}_n - \theta_0\| = o_P(n^{-1/4})$. We note that while this estimator does not suffer from the CoD as demonstrated in the proofs, d is naturally still relevant to our network parameters through s_0 and the subset $\mathcal{J}_i$. This is despite how estimation methods relying on splines are notorious for failing to produce precise estimates in high-dimensional settings due to the CoD (Chen, 2007). Although there are existing spline estimation approaches that attempt to exploit functional sparsity in high-dimensional environments[44] similar to our proposed KAN sieve estimator, these methods are likely to perform poorly on the truly compositional structures (i.e., where imposing additivity is insufficient) found in economic and financial applications.

3 Theoretical Results

To simplify the notation, we assume $\theta_0 \equiv f_0 \in \Theta$ and denote the sieve space as $\mathcal{K}_n \equiv \Theta_n$ (i.e., $A \subset \mathbb{R}^a$, the finite-dimensional parameter space, is an empty set) for Theorems 1-3. In other words, the estimator is defined as $\hat{\theta}_n = \arg\sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta)$ with $\hat{\theta}_n \in \mathcal{K}_n$ by construction. Proofs of each of the following theorems can be found in Section A.1 of the Appendix. Verification of the conditions from Chen and Shen (1998) required for establishing the asymptotic normality of plug-in estimators (Theorem 4) follows after the main proofs in Section A.2. Unless stated otherwise, let the network parameters correspond to the conditions outlined in (2.9)-(2.11). In practice, we make use of cubic B-spline activations ($k = 4$) across all simulations and empirical applications, however the following theoretical results hold for all $k \geq m + 1$.

[44] e.g., Meier et al. (2009).

3.1 Rates of Convergence

Theorem 1. *Under Assumptions K.2(a) and K.3-4, with* $\bar{s}_0 = s_0 \times L_n$*, given B-splines of order* $k \geq m+1$ *with uniform knots on* $\mathcal{I}_\epsilon$*, and given* $\zeta_\delta > \gamma/2$ *as in* (2.10)*, there exists a sieve approximation* $\pi_n\theta_0 \in \mathcal{K}_n$ *such that:*

$$\|\pi_n\theta_0 - \theta_0\|_{L_2(\mu)} \leq \|\pi_n\theta_0 - \theta_0\|_\infty \leq C_{\bar{s}_0}\, G_n^{-(m+1)},$$

where $C_{\bar{s}_0} = O(\bar{s}_0) = O((\log n)^2)$ *is independent of* d*. Consequently,* $\gamma > \frac{1}{4(m+1)}$ *implies* $\|\pi_n\theta_0 - \theta_0\| = o(n^{-1/4})$*. Furthermore, if the inputs are normalized by the marginal CDFs as in* (2.5)*, then under K.1 the normalization error satisfies*

$$\left\|\pi_n\theta_0 \circ \hat{F}_n - \pi_n\theta_0 \circ F\right\| = O_P\big(n^{-1/2}(\log n)^2\big) = o_P(n^{-1/4}),$$

so that the total approximation error remains $o_P(n^{-1/4})$*.*

The proof of Theorem 1 relies on the sparse compositional structure imposed on the target function and the smooth univariate edge functions (Assumptions K.3 and K.4, respectively; in addition to Lemma 1, which establishes the supremum norm error of the CDF normalization given the EDF $\hat{F}_n$). Using results from spline approximation theory (de Boor, 1978; DeVore and Lorentz, 1993), we show that each edge function is approximated with error $O(G_n^{-(m+1)})$, the rate dictated by the smoothness of the target function. The resulting approximation error of the KASN is independent of d, a key result for mitigating the CoD (with d entering only through the CDF normalization at a asymptotically negligible rate). The proof further verifies that the sieve approximation $\pi_n\theta_0$ respects the $\ell_{1,2}$-constraint defining $\mathcal{K}_n$, which is the source of the lower bound $\zeta_\delta > \gamma/2$ in (2.10).

Corollary 1. *Under the conditions of Theorem 1, restricted to B-splines of order* $k \geq m+2$*,*

the sieve projection $\pi_n\theta_0 \in \mathcal{K}_n$ additionally satisfies:

$$\|\pi_n\theta_0 - \theta_0\|_{W_2^{m+1}(\mu)} \leq C'_{\bar{s}_0}\, G_n^{-1},$$

where $C'_{\bar{s}_0} = O\big(\bar{s}_0\,(\log n)^{c_m}\big)$ for a fixed integer c_m depending only on m, and the bound is taken with respect to the normalized inputs. Consequently, $\gamma > \frac{1}{4}$ implies $\|\pi_n\theta_0 - \theta_0\|_{W_2^{m+1}(\mu)} = o(n^{-1/4})$.

Remark: Corollary 1 supplies the approximation component for derivative-based functionals, such as the average derivative of Example 2: the first-derivative rate $G_n^{-(m+1)}$ controls the approximation component, while the corresponding estimation error control is delivered separately by interpolation (Lemma 6) rather than by the Sobolev bound itself. Sobolev norm control of the approximation error is therefore a component of (rather than necessary or sufficient for) the normality of derivative functionals. In both Theorem 1 and Corollary 1, the deterministic approximation terms dominate the stochastic $O_P(n^{-1/2}(\log n)^2)$ normalization error introduced through the normalization in (2.5).

Theorem 2. *Under Assumptions K.2-K.4, the sieve space $\mathcal{K}_n$ defined in (2.6)-(2.7) satisfies, for all $0 < \epsilon \leq \Delta_n$:*

$$\log \mathcal{N}\big(\epsilon, \mathcal{K}_n, \|\cdot\|_\infty\big) \leq C_1\, s_n \log\left(\frac{r_n}{s_n}\right) + C_2\, s_n G_n \left[\log\left(\frac{\Delta_n}{\epsilon}\right) + (\log n)^2\right],$$

where r_n is given by (2.4). Since $\|\cdot\|_{L_2(\mu)} \leq \|\cdot\|_\infty$ for any μ, the same bound holds for $\log \mathcal{N}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)})$.

The bound comprises a combinatorial term arising from the selection of the active support and a coefficient-complexity term arising from the $\ell_{1,2}$-constrained B-spline coefficients on the active edges. The combinatorial term is $O(s_n \log r_n) = O((\log n)^3)$ under (2.4), so the covariate dimension enters the entropy only through $\log d$. The $(\log n)^2$ inflation of the coefficient term reflects the continuity constant of the coefficient-to-function map (established

in Lemma 4), which compounds across the L_n layers; it is polylogarithmic and is absorbed without consequence in the entropy integral of Theorem 3. The bound remains conservative, since it allocates the full coefficient budget Δ_n to every active edge in order to simplify the analysis. The residual connections h are fixed, non-parameterized maps and contribute nothing to the covering number itself.

Corollary 2. *Under Assumptions K.2-K.4, the $L_2(\mu)$-bracketing entropy of $\mathcal{K}_n$ satisfies, for all $0 < \epsilon \leq \Delta_n$:*

$$\begin{aligned}\log \mathcal{N}_{[]}\big(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)}\big) &\leq \log \mathcal{N}\big(\epsilon/2, \mathcal{K}_n, \|\cdot\|_{\infty}\big) \\ &\leq C_1\, s_n \log\left(\frac{r_n}{s_n}\right) + C_2\, s_n G_n \left[\log\left(\frac{2\Delta_n}{\epsilon}\right) + (\log n)^2\right].\end{aligned}$$

Theorem 3. *Under Assumptions K.1-K.4, given the sieve space $\mathcal{K}_n$ and letting $\frac{1}{4(m+1)} < \gamma < \frac{1}{2}$:*

$$\|\hat{\theta}_n - \pi_n \theta_0\| = o_P(n^{-1/4})$$

The proof of Theorem 3 involves splitting the entropy integral at $\epsilon_0 = n^{-1}$ and showing that each term is $o_P(n^{-1/4})$. Obtaining convergence rates in the Sobolev norm would lead to a condition that $m > d$ as we show in the Appendix, which would re-introduce the CoD. Instead, the estimation error achieves a convergence rate of $o_P(n^{-1/4})$ which holds uniformly over $d = O(n^{\zeta_d})$.

Corollary 3. *Under Assumptions K.1-K.4, given the sieve space $\mathcal{K}_n$ and letting $\gamma < \frac{1}{2}$, the KASN extremum estimator achieves $\|\hat{\theta}_n - \theta_0\| = o_P(n^{-1/4})$.*

To see this, note that by the triangle inequality:

$$\|\hat{\theta}_n - \theta_0\| \leq \|\hat{\theta}_n - \pi_n\theta_0\| + \|\pi_n\theta_0 - \theta_0\|$$

$$\|\hat{\theta}_n - \theta_0\| = o_P(n^{-1/4}) + o(n^{-1/4}) = o_P(n^{-1/4})$$

Remark: By letting $p_n \to \infty$ and $\Delta_n \to \infty$ as $n \to \infty$, we obtain a sequence of increasingly flexible network architectures; making this analogous to the universal approximator result from White (1990) for MLPs, now extended to KANs under functional sparsity. For sparse compositional target functions, KASNs represent a class of *connectionist sieve networks* with multiple hidden layers that guarantees a dimension-free convergence rate, the first of its kind.

While Gouriéroux and Monfort (2025) correctly note that the KART is non-unique at the component level (with respect to $\phi_{i,j}$ and Φ_i), this does not affect the identification of the target function itself. The KASN framework formalizes identification of θ_0 as the exact maximizer of the population criterion rather than identification of the individual component functions. The compact sieve space $\mathcal{K}_n$ and the identification condition in Assumption K.2(b) guarantee the existence of a measurable maximizer $\hat{\theta}_n$ of the empirical criterion, which is sufficient for the convergence results in Theorems 1-3 and the asymptotic normality result (independent of the underlying architecture) in Theorem 4. Beyond measurability, the identification issues they raise surrounding the CoD are resolved directly: Corollary 3 establishes that the convergence rate $o_P(n^{-1/4})$ holds uniformly over $d = O(n^{\zeta a})$, rendering dimensional scaling irrelevant to the estimation problem.

In Theorem 4, we establish the asymptotic normality of plug-in estimators using KASNs. Following Chen and Shen (1998, Theorem 2), we consider a linear functional $\Gamma : \Theta \to \mathbb{R}$ with $v^* \in \bar{V}$ (by the Riesz representation theorem) satisfying $\Gamma'_{\theta_0} = \langle \theta - \theta_0, v^* \rangle$ for all $\theta \in \Theta$. The existence of v^*, the Gateaux derivative of Γ, ensures that the functional is bounded (where $\bar{V}$ denotes the completion of the space spanned by $\Theta - \{\theta_0\}$ and the inner product $\langle \cdot, \cdot \rangle$ is induced by $\|\cdot\|$) and identifies the direction in the sieve space that characterizes the asymptotic variability of the estimator. We expand on these definitions and list the relevant conditions from Chen and Shen (1998) in the Appendix, before verifying each condition for the KASN.

Theorem 4. *Suppose Assumptions K.1-K.5 hold,* $\mathbb{E}[U_t \mid \tilde{X}_t] = 0$, $\mathbb{E}[U_t^2 \mid \tilde{X}_t] \leq \bar{\sigma}^2 < \infty$,

and $\sup_t \mathbb{E}|U_t|^\kappa < \infty$ *for some* $\kappa > 2$. *Let* $\epsilon_n = \delta_n$ *from Theorem 3, then:*

$$\sqrt{n}\left(\Gamma(\hat{\theta}_n) - \Gamma(\theta_0)\right) \xrightarrow{d} \mathcal{N}\left(0, \sigma_*^2\right), \qquad \sigma_*^2 = \mathrm{Var}\left(l'[v^*, Z_1]\right) + 2\sum_{j=1}^{\infty} \mathrm{Cov}\left(l'[v^*, Z_1],\, l'[v^*, Z_{1+j}]\right),$$

where the long-run variance is finite under K.1 and the stated moment conditions.

In ultra-high-dimensional settings ($d \gg n$), estimates of this linear component (e.g., the parameter of interest in the partially linear regression model introduced in Example 1) are likely to suffer from regularization bias, leading us to consider an alternative approach centered around the DML framework (Chernozhukov et al., 2018) designed to address this. We illustrate the main theoretical results established in this section through the following examples, where the implementation of Example 3 is left to future work as an extension of the KASN framework.

For the partially linear regression model in Example 1 and our DML simulations, define the partition $X_{t,\mathcal{D}} \in \mathbb{R}^{d_1}$ and $X_{t,\mathcal{X}} \in \mathbb{R}^{d_2}$, where $\mathcal{D} = \{1, \ldots, d_1\}$ and $\mathcal{X} = \{d_1 + 1, \ldots, d\}$ and $d_1 + d_2 = d \in \mathbb{N}$. For simplicity, let $D_t \equiv X_{t,\mathcal{D}}$ (representing the linear component) and $W_t \equiv X_{t,\mathcal{X}}$ (noting that μ now corresponds to $W_t \in \mathbb{R}^{d_2}$) in these semiparametric applications.

Example 1 (Partially Linear Regression Model)**.** Suppose $\{Y_t, D_t, W_t'\}$ satisfies the model[45]

$$Y_t = D_t\alpha_0 + g_0(W_t) + U_t, \qquad \mathbb{E}[U_t|D_t, W_t] = 0,$$

where D_t is the (treatment) variable of interest and g_0 is an unknown nonparametric function of controls W_t. Assume $g_0 \in W_2^{m+1}$ admits a sparse compositional form (Assumption K.3(a)), where $\theta \equiv (\alpha_0, g_0) \in \Theta$ and μ is the distribution of W_t. Let $\hat{g}_n \in \mathcal{K}_n$ be the KASN estimator for g_0, and let $\hat{\alpha}_n$ be the resulting plug-in estimator of α_0 obtained by partialling out $\hat{g}_n$

[45] See the partition of X_t defined earlier in the context of semiparametric structural models.

(Robinson, 1988). Then, under Assumptions K.1-4 for $\hat{\theta}_n \equiv (\hat{\alpha}, \hat{g})$, we have:

$$(i) \quad \|\hat{g}_n - g_0\|_{L_2(\mu)} = o_P(n^{-1/4}), \text{ and}$$
$$(ii) \quad \sqrt{n}\big(\hat{\alpha}_n - \alpha_0\big) \rightsquigarrow N(0, \Sigma_\alpha),$$

where $\Sigma_\alpha \equiv \sigma_*^2$ is the asymptotic variance characterized in Theorem 4.

Example 2 (Weighted Average Derivative)**.** Let θ_0 be the regression function on the normalized inputs given $\{Y_t, \tilde{X}_t\}$ and let the estimand be the average derivative with respect to the original covariate X_j. By the chain rule, this equals a density-weighted derivative on the normalized scale:

$$\Gamma_j(\theta_0) \equiv \mathbb{E}\Big[\frac{\partial\, \theta_0(F(X_t))}{\partial X_{t,j}}\Big] = \mathbb{E}_\mu\big[w_j(\tilde{X}_{t,j})\, \partial_j \theta_0(\tilde{X}_t)\big], \qquad w_j \equiv f_j \circ F_j^{-1}, \tag{3.1}$$

where f_j is the original marginal density. The weight w_j is generated by the normalization in (2.5) rather than chosen, and is what renders the functional $\sqrt{n}$-estimable (see Powell et al., 1989). Let $\hat{\Gamma}_j = \frac{1}{n}\sum_{t=1}^n w_j(\tilde{X}_{t,j})\, \partial_j \hat{\theta}_n(\tilde{X}_t)$ be the plug-in estimator. Then under Assumptions K.1-K.4 and the additional regularity conditions on the functional (E.1-4, stated under Proposition 2 in Appendix A.2), we have:

$$(i) \quad \|\partial_j(\hat{\theta}_n - \theta_0)\|_{L_2(\mu)} = o_P(n^{-1/4}), \text{ and}$$
$$(ii) \quad \sqrt{n}\big(\hat{\Gamma}_j - \Gamma_j\big) \rightsquigarrow N(0, \sigma_\psi^2),$$

where σ_ψ^2 is the long-run variance of the influence sequence ψ_t.

Example 3 (Dynamic Discrete Choice Model Estimation)**.** Consider a dynamic discrete choice model[46] where an agent chooses an action $d_t \in \{1, \ldots, J\}$ each period to maximize

[46]See Norets (2012) for an example of the application of sieve aNNs in this setting. An extension of Norets (2012) centered around inference procedures facilitated by the KASN extremum estimator, including but not limited to counterfactual analysis, is left to future work. In this example, the KASN estimator permits time-series dependence under Assumption K.1, unlike the aforementioned application.

expected discounted utility. The per-period utility is $u(s_t, d_t; q)$, where s_t is the state vector and q is a vector of parameters. The value function satisfies:

$$V(s_t; q) = \max_{d_t} \big\{ u(s_t, d_t; q) + \beta \, \mathbb{E}[V(s_{t+1}; q) \mid s_t, d_t] \big\},$$

with discount factor $\beta \in (0, 1)$. Let $F(s_t, d_t; q) \equiv \mathbb{E}[V(s_{t+1}; q) \mid s_t, d_t]$ be the expected value function. Assume further that F_0 admits a sparse compositional representation as in Assumption K.3(a) (e.g., the utility function $u(s_t, d_t; q)$ is additively separable across states). Let $\hat{F}_n \in \mathcal{K}_n$ be the KASN estimator obtained using $l(Z_t, F) = -(V(s_{t+1}; q) - F(s_t, d_t; q))^2/2$ (where $V(s_{t+1}; q)$ is obtained as in Norets, 2012). Then under Assumption K.1-4, we have:

$$\begin{aligned} &(i) \quad \|\hat{F}_n - F_0\|_{L_2(\mu)} = o_P(n^{-1/4}), \text{ and} \\ &(ii) \quad \sqrt{n} \left(\frac{1}{n} \sum_{t=1}^{n} \hat{F}_n(s_t, d_t; q) - \mathbb{E}\big[F_0(s_t, d_t; q)\big] \right) \rightsquigarrow N(0, \Sigma_F), \end{aligned}$$

where μ is the joint distribution of (s, d) and $\Sigma_F \equiv \sigma_*^2$ is the asymptotic variance.

3.2 Double/Debiased Machine Learning

In this section, we show how our KASN extremum estimator can be applied within the DML framework of Chernozhukov et al. (2018) to obtain $\sqrt{n}$-consistent and asymptotic normal estimates of linear parameters of interest. A similar DML application appears in Farrell et al. (2025), although the convergence rate of the ReLU DNN they propose explicitly depends on the heterogeneity dimension (d_c). In addition to the presence of the CoD, their near-minimax $O(n^{-m/(m+d_c)} \log^8 n)$ convergence rate depends on the smoothness of the target function; highlighting the tradeoff arising from the highly useful economic interpretability of their proposed estimator. To the best of our knowledge, the KASN is the first sieve network architecture since Chen and White (1999) capable of delivering a dimension-free convergence rate of $o_P(n^{-1/4})$.

Chernozhukov et al. (2018) address how the naive application of machine learning algorithms in semiparametric models with low-dimensional causal parameters can lead to biased estimates. This arises from the regularization methods commonly employed in these models for nuisance function estimation, which perform well by reducing variance (overfitting) in a manner consistent with the standard variance-bias tradeoff. As a result of this overfitting and regularization bias, estimates of the causal parameter obtained by using the fitted nuisance function as a plug-in estimator will suffer from a heavy degree of bias. Furthermore, they demonstrate how the Donsker conditions typically imposed in semiparametric models to control complexity break down in ultra-high-dimensional settings. The DML framework resolves these issues through two key ingredients: Neyman orthogonality and sample splitting.

We consider the partially linear regression model[47] (Robinson, 1988) in our DML simulations and assume $Z_i = \{Y_i, X_i\} = \{Y_i, D_i, W_i\}$ is i.i.d. (satisfying K.1(a) here) for this application. This model is defined in terms of the following equations:

$$Y_i = D_i\alpha_0 + g_0(W_i) + U_i, \qquad E[U_i|W_i, D_i] = 0 \tag{3.2}$$

$$D_i = m_0(W_i) + V_i, \qquad E[V_i|W_i] = 0 \tag{3.3}$$

where $V_i = D_i - m_0(W_i)$ and the sample splitting procedure (in the case of two folds) entails estimating $m_0 \in \mathcal{G}$ and $g_0 \in \mathcal{G}$ (with $\mathcal{G}$ representing the nonparametric function space in Θ) using half of the dataset, before forming residuals on the other half. In this model, D_i represents the treatment variable (or more generally, the primary regressor of interest) with a linear causal effect α_0 estimated after controlling for the confounding effects of W_i. Neyman orthogonality is the condition that ensures moment conditions defining α_0 are insensitive to perturbations[48] in the nuisance functions $\eta_0 = (g_0, m_0) \in \mathrm{T}$ (where T is a convex subset of a normed vector space with m_0 satisfying the same conditions as g_0 and $\mathrm{T} \equiv \mathcal{G} \times \mathcal{G}$).

[47] Which is the leading example in Chernozhukov et al. (2018); see our partition of X_i defined earlier.

[48] With respect to errors in estimating both the outcome model and the treatment/propensity model.

Definition 2 (Neyman Orthogonality)**.** The score $\psi(\cdot;\cdot)$ satisfies Neyman orthogonality if:

$$\partial_\eta E[\psi(Z;\alpha_0,\eta_0)][\eta-\eta_0]=0 \quad \text{for all } \eta\in \mathrm{T} \tag{DML.1}$$

The main advantage of our procedure estimating nuisance functions using KASNs over widely used machine learning methods (e.g., Lasso, random forests) is how it is capable of capturing complex compositional structures[49] in the DGP. In their empirical example, Chernozhukov et al. (2018, Section 6.1) discuss how "Deep Learning methods" were experimented with but that the results were omitted due to computational and stability issues encountered. We observe no such issues across all of our Monte Carlo simulations and compare the results against those of a Lasso model using a penalty λ_L selected via cross-validation. By Theorem 4.1 of Chernozhukov et al. (2018), the following result holds when σ^2 is replaced with $\hat{\sigma}^2$.

Proposition 1. *Under Assumptions K.1(a)-K.4, the DML estimator $\hat{\alpha}$ where $(\hat{g}_n,\hat{m}_n)\in\mathcal{K}_n$ satisfies:*

$$\sigma^{-1}\sqrt{n}(\hat{\alpha}-\alpha_0)\rightsquigarrow N(0,1), \quad \sigma^2=\frac{E[V_i^2U_i^2]}{(E[V_i^2])^2}, \tag{DML.2}$$

where by Corollary 3, $\hat{g}_n$ and $\hat{m}_n$ each achieve $o_P(n^{-1/4})$ in $L_2(\mu)$, so that

$$\|\hat{g}_n-g_0\|\cdot\|\hat{m}_n-m_0\|=o_P(n^{-1/2}),$$

thus satisfying the rate condition of Theorem 4.1 of Chernozhukov et al. (2018). We implement the K-fold cross-fitting procedure of Chernozhukov et al. (2018), estimating $(\hat{g}_n,\hat{m}_n)$ on each complementary fold and forming residuals on the held-out fold.

The i.i.d. assumption underlying (DML.2) is appropriate for the baseline design but excludes the weakly dependent processes the KASN framework is centered around. When $\{Z_i\}$

[49] Nonlinear or otherwise.

is instead β-mixing under Assumption K.1(b), the complementary folds of standard cross-fitting are no longer independent of the held-out fold, and the i.i.d. variance in (DML.2) is no longer the relevant one. We therefore replace standard cross-fitting with the neighbors-left-out (NLO) scheme of Semenova et al. (2023). Partitioning the contiguous indices $\{1, \ldots, n\}$ into $K \geq 4$ blocks, we estimate $(\hat{g}_n, \hat{m}_n)$ on the quasi-complement of each block (obtained by removing the target block together with its two immediate neighbors; analogous to the buffering principle of the hv-block procedure of Racine, 2000) and form residuals on the target block. The omitted neighbors separate the training and evaluation folds by at least $\lfloor n/K \rfloor$ periods, rendering them approximately independent at a rate governed by the mixing coefficients.

Proposition 3 in Appendix A.3 establishes that the resulting NLO-DML estimator satisfies

$$\sqrt{n}(\hat{\alpha} - \alpha_0) \rightsquigarrow N(0, \sigma^2_{\mathrm{LR}}),$$

where σ^2_{LR} is the long-run variance of the score process $V_t U_t$, and gives the precise conditions under which the result holds. The product-rate condition is satisfied exactly as in Proposition 1, since Corollary 3 delivers $\|\hat{g}_n - g_0\| \cdot \|\hat{m}_n - m_0\| = o_P(n^{-1/2})$ on each quasi-complement (whose effective size is of order n for fixed K). The verification of the high-level conditions of Theorem 5.1 of Gonçalves et al. (2026) for the KASN nuisance is provided in Appendix A.3. A Newey-West HAC estimator (Newey and West, 1987) $\hat{\sigma}^2_{\mathrm{LR}}$ formed from the cross-fitted residuals is used to account for serial dependence in the score.

4 Monte Carlo Simulations

In this section, we analyze the performance of the KASN in Monte Carlo simulations. We first consider a basic nonparametric regression model where the target function takes on a sparse additive form and a sparse compositional (non-additive) form under two different regimes: a moderately high-dimensional setting ($d = 100$) and a genuinely high-dimensional

one ($d = 500$). We then evaluate our proposed estimator in a DML application (using the PLR model outlined in Eq. (3.2)-(3.3) and Example 1) against the performance of other ML methods. Results from the moderately high-dimensional setting are reported in the Appendix ($R = 400$ replications) while the high-dimensional nonparametric regression results and the DML performance ($R = 1000$ replications) are reported in the following section, with careful attention paid to the crucial sparsity ratio. This is the ratio of inactive edges relative to the total number of potential edges (r_n), using the Δ_n/r_n threshold for what constitutes an active edge (i.e., learned univariate functions with a magnitude greater than those in a hypothetical fully dense network) defined earlier.

We benchmark the KASN against a collection of relevant nonparametric and penalized estimators, each isolating a distinct source of the KASN performance across DGPs. The first benchmark model is the shallow sieve aNN of Chen and White (1999) denoted as "SLFN" since it corresponds to a single hidden layer feedforward network, which provides a natural comparison for our deeper B-spline architecture. The ReLU MLP of Farrell et al. (2021) discussed in Section 1.1, denoted "DNN", is the modern deep learning benchmark and controls for depth alone. For these two benchmark aNNs, we apply standard weight decay[50] in a manner consistent with common practice, which is generally insufficient for sparse target functions, as we demonstrate. The GAM we evaluate in these applications makes use of penalized splines (Servén and Brummitt, 2018), where the degree and number of basis functions match those of the learned activation functions in the KASN. Given that this estimator is misspecified under the DGP involving the compositional target[51], the performance gap between it and the KASN highlights the benefit of the compositional architecture. Lastly, we implement the Lasso and PPR (Friedman and Stuetzle, 1981), where the smoothness of the ridge functions applied in the latter also matches the edges in the KASN. For all regularized estimators evaluated, the penalty term is selected from the same grid using the validation

[50]Weight decay is equivalent to L_2 regularization, adding a penalty proportional to the sum of squared network weights to the loss function in order to prevent overfitting. See Goodfellow et al. (2016).

[51]In the case of the additive target, one would theoretically anticipate identical performance between the penalized splines GAM and the KASN.

set[52] under each DGP.

4.1 Finite-Sample Performance

In our baseline nonparametric regression, we evaluate ability of the KASN to recover a sparse additive signal with a target function dependent on $S = 30$ out of $d = 100$ total covariates and $n = 5000$ observations with a $0.5 : 0.25 : 0.25$ ratio for the train, test, and validation sets[53] respectively. The KASN parameters are set consistent with the conditions outlined in Section 2.4 and the target function takes on the form

$$f_0(X) = \sum_{j=1}^{S} \sin(X_j) + \epsilon, \quad \epsilon \sim N(0, 0.25). \tag{4.1}$$

In all four settings we consider, the covariates follow a stationary Gaussian AR(1) process[54] with $\rho_X = 0.7$ ensuring they are β-mixing, and each sample is split chronologically to avoid (look-ahead) data leakage. Alongside this moderately high-dimensional design, we run a truly high-dimensional counterpart where: $n = 2{,}000$ ($n_{\text{train}} = 1000$), $d = 500$, $S = 50$. In every case the KASN hyperparameters (γ, depth L_n, width W_n, grid G_n, penalty λ_n) are selected on the validation set, with the values chosen in each design reported in the corresponding table note. With the exception of PPR, the penalty term (including λ_n) for each model is selected via the minimized validation loss from a grid:

$$\lambda \in \{10^{-5}, 10^{-3}, 10^{-2}, 5 \times 10^{-2}, 10^{-1}, 1.0, 5.0\}.$$

In order to ensure a fair comparison with the other methods, the number of basis functions in the PPR model is selected using the validation set: $r \in \{2, 4, 6, 8, 10\}$.

[52] In order to ensure a fair comparison, the number of PPR basis functions is selected in an identical manner.

[53] The same ratio is used for the DGP featuring the compositional target function.

[54] Where $X_j = \rho_X X_{j-1} + v_j,\ v_j \sim \mathcal{N}(0, 1 - \rho_X^2)$, for all $j = 1, \ldots, d$.

Table 1: Moderate-Dimensional Nonparametric Regression: Additive Target

Method	**Train**		**Val.**		**Test**		$\|\hat{f}_n - f_0\|$	**Parameters**	**Fit (s)**
	RMSE	R^2	**RMSE**	R^2	**RMSE**	R^2			
KASN	0.4916	0.9848	0.6146	0.9763	0.6133	**0.9764**	**0.3525**	1,464/22,710	72.61
GAM	0.3495	0.9923	0.6822	0.9708	0.6799	0.9710	0.4612	1,208 (EDF)	11.69
DNN	2.2068	0.6905	2.7120	0.5365	2.7333	0.5292	2.6875	28,189	1.54
SLFN	2.3105	0.6642	2.4161	0.6335	2.4230	0.6314	2.3712	173	1.84
PPR	2.1171	0.7182	2.5436	0.5929	2.5442	0.5927	2.4947	$r = 2$	47.40
LASSO	2.2262	0.6885	2.3326	0.6585	2.3268	0.6602	2.2726	84/100	0.01
Oracle	0.4999	0.9843	0.5011	0.9842	0.4995	0.9843	0.0000	—	0.00

Note: Monte Carlo means over $R = 400$ replications; $n = 5{,}000$ split 2,500/1,250/1,250 chronologically, $d = 100$, $s_0 = 30$. "Parameters" is active/total edges after Group-Lasso pruning for the KASN (sparsity ratio 0.933), EDF for the GAM, and non-zero/total coefficients for the LASSO. KASN selection: $\gamma = 0.35$, $L_n = 4$, $W_n \approx 83$, $G_n = 15$, $\lambda_n \approx 0.05$, $\Delta_n \approx 27$, learning rate 10^{-3}.

As seen in Table 1, the KASN achieves a Test R^2 of 0.976 and a Test RMSE of 0.613, where both metrics are comparable to the performance on the training data ($R^2 = 0.985$, RMSE $= 0.492$) and sit just below the oracle floor of 0.984. The narrow gap between these train and test figures indicates minimal overfitting of the KASN, despite our novel and relatively aggressive regularization method. As one would expect, the correctly-specified GAM is competitive here (Test $R^2 = 0.971$), yet the KASN still attains the lowest held-out L^2 error against f_0 (0.353 versus the GAM's 0.461), showing that there is no cost with respect to accuracy for its additional flexibility when the true DGP happens to be additive. The remaining benchmarks trail substantially: the shallow SLFN (reliant on a fixed smooth activation function in this implementation) underfits (Test $R^2 = 0.631$) as it lacks the localized flexibility of the B-spline grid[55], and neither the deep ReLU MLP (0.529), PPR (0.593), nor the LASSO (0.660) closes the gap. This spread is indicative of the superior approximation power possessed by the KASN, demonstrating how the learned B-spline edges directly adapt to the smoothness and periodicity of the target function.

[55]We note that Chen and White (1999) permit more flexible activation functions in the form of B-splines and wavelets, however the implementation of this is beyond the scope of this comparative exercise.

Table 2: High-Dimensional Nonparametric Regression: Additive Target

Method	Train		Val.		Test		$\|\hat{f}_n - f_0\|$	Parameters	Fit (s)
	RMSE	R^2	**RMSE**	R^2	**RMSE**	R^2			
KASN	1.1717	0.9394	1.5593	0.8975	1.5954	**0.8927**	**1.5074**	10,681/54,399	414.46
GAM	0.0040	1.0000	4.0585	0.3712	4.0649	0.3692	4.0327	999 (EDF)	1243.60
DNN	2.8373	0.6532	4.2964	0.2897	4.3571	0.2695	4.3271	60,589	1.88
SLFN	3.9724	0.3818	4.6175	0.1834	4.6589	0.1687	4.6307	119	2.48
PPR	1.6612	0.8837	10.3518	−10.9671	11.2380	−13.1039	11.2183	$r = 2$	276.74
LASSO	5.0335	0.0344	5.1027	0.0072	5.1050	0.0063	5.0793	4/500	0.01
Oracle	0.4998	0.9905	0.5005	0.9904	0.5005	0.9904	0.0000	—	0.00

Note: Monte Carlo means over $R = 1{,}000$ replications; $n = 2{,}000$ split 1,000/500/500, $d = 500$, $S = 50$; columns as in Table 1. KASN sparsity ratio 0.799; selection $\gamma = 0.40$, $L_n = 4$, $W_n \approx 81$, $G_n = 15$, $\lambda_n \approx 0.06$, $\Delta_n \approx 50$, learning rate 10^{-3}.

The most critical finding in this setting is the effective dimensionality reduction achieved through the Group Lasso penalization. Out of the $r_n = 22{,}710$ potential edges, the KASN retains only $\hat{s}_n = 1{,}464$ active univariate functions: a sparsity ratio of 93.3%. This means that the network successfully identified the narrow subset of active edges required to represent the $S = 30$ relevant[56] covariates. This prevents the overfitting noise common in insufficiently regularized deep networks and is crucial for maintaining a controlled metric entropy as discussed earlier. The higher computational runtime relative to the shallow benchmarks is consistent with the main drawback of our architecture and its learned activation functions, but it is repaid in statistical efficiency. These results verify that even in cases where the true model is additive rather than deeply compositional, the KASN extremum estimator is robust and highly accurate, with an underlying architecture that is naturally robust to the CoD.

This robustness is demonstrated further as the covariate dimension grows. In the high-dimensional additive DGP ($d = 500$, $S = 50$, $n = 2{,}000$; Table 2), the GAM saturates despite correct specification: its effective degrees of freedom reach the training-sample size and it interpolates the training data (Train $R^2 = 1.000$) yet generalizes poorly (Test $R^2 = 0.369$), indicating severe overfitting. The KASN, by contrast, sustains a Test R^2 of 0.893 and again

[56]For brevity, we refer to covariates upon which f_0 depends on as 'relevant'.

the lowest L^2 error (1.507). The additive advantage that the GAM benefited from at $d = 100$ diminishes once the design is high-dimensional relative to n, while the sparsity-driven control of the KASN prevents the performance from degrading under a shorter amount of time spent fitting the model (414 versus 1244 seconds, excluding hyperparameter selection).

Table 3: Moderate-Dimensional Nonparametric Regression: Compositional Target

Method	**Train**		**Val.**		**Test**		$\|\hat{f}_n - f_0\|$	**Parameters**	**Fit (s)**
	RMSE	R^2	**RMSE**	R^2	**RMSE**	R^2			
KASN	0.4837	0.9852	0.5968	0.9775	0.5981	**0.9774**	**0.3263**	807/24,762	78.92
GAM	1.1457	0.9172	2.2453	0.6822	2.2428	0.6829	2.1860	1,208 (EDF)	11.70
DNN	2.4747	0.6097	2.9854	0.4363	3.0047	0.4290	2.9628	28,189	1.47
SLFN	2.6556	0.5546	2.7742	0.5143	2.7742	0.5143	2.7287	173	1.82
PPR	2.3304	0.6574	2.8409	0.4901	2.8292	0.4943	2.7845	$r = 2$	51.73
LASSO	2.5588	0.5871	2.6812	0.5468	2.6708	0.5503	2.6233	86/100	0.01
Oracle	0.4997	0.9843	0.5003	0.9842	0.5003	0.9842	0.0000	—	0.00

Note: Monte Carlo means over $R = 400$ replications; $n = 5{,}000$ split 2,500/1,250/1,250, $d = 100$, $s_0 = 30$; columns as in Table 1. KASN sparsity ratio 0.965; selection $\gamma = 0.35$, $L_n = 4$, $W_n \approx 88$, $G_n = 15$, $\lambda_n \approx 0.09$, $\Delta_n \approx 18$, learning rate 10^{-3}.

The compositional target,

$$f_0(X) = \frac{1}{1 + \exp\Big(- \sum_{j=1}^{S} \sin X_j \Big)} + 0.1 \sum_{j=1}^{S} |X_j| + \epsilon, \quad \epsilon \sim N(0, 0.25), \tag{4.2}$$

which places an outer link on an inner nonlinear index, separates the methods further because it lies outside the additive class the GAM (and, up to linear indices, PPR) can represent. At $d = 100$ (Table 3) the misspecified GAM attains only Test $R^2 = 0.683$, while the KASN recovers the compositional structure at 0.977 (close to the oracle floor of 0.984) at a 96.5% sparsity ratio. The same ordering holds in high dimension (Table 4): 0.733 for the KASN[57] against 0.297 for the GAM, near-zero for the DNN, SLFN, and LASSO, and divergent for PPR. Across all four designs the KASN attains the smallest $\|\hat{f}_n - f_0\|$ and the compositional gap is exactly where the compositional architecture of our sieve extremum estimator is designed to pay off. It is this structure that motivates the DML application of the following

[57]The validation loss suggests this performance could be improved by defining hyperparameter grids that are more optimal for selection.

subsection, whose nuisance functions are compositional by construction.

Table 4: High-Dimensional Nonparametric Regression: Compositional Target

Method	Train		Val.		Test		$\|\hat{f}_n - f_0\|$	Parameters	Fit (s)
	RMSE	R^2	RMSE	R^2	RMSE	R^2			
KASN	1.4667	0.9122	2.5689	0.7438	2.6210	**0.7333**	**2.5722**	11,113/50,842	400.71
GAM	0.0048	1.0000	4.2914	0.2983	4.2942	0.2974	4.2650	999 (EDF)	1240.56
DNN	4.9068	0.0852	5.1046	0.0083	5.1374	−0.0045	5.1133	60,589	1.53
SLFN	4.9533	0.0708	5.0937	0.0126	5.1230	0.0012	5.0987	119	1.70
PPR	1.1998	0.9266	24.3923	−25.9306	26.7846	−31.4722	26.7790	$r = 3$	148.08
LASSO	5.1410	0.0008	5.1463	−0.0079	5.1463	−0.0079	5.1220	0/500	0.01
Oracle	0.4996	0.9905	0.4994	0.9905	0.4994	0.9905	0.0000	—	0.00

Note: Monte Carlo means over $R = 1{,}000$ replications; $n = 2{,}000$ split 1,000/500/500, $d = 500$, $S = 50$; columns as in Table 1. KASN sparsity ratio 0.774; selection $\gamma = 0.40$, $L_n = 4$, $W_n \approx 76$, $G_n = 15$, $\lambda_n \approx 0.06$, $\Delta_n \approx 61$, learning rate 10^{-3}.

4.2 DML Simulations

In our DML simulations, we compare the performance of KASNs against Lasso in estimating $\alpha_0 = 1.5$. Across each DGP, we generate $d = 100$ covariates in total with $S = 30$ relevant covariates and set use five folds in the sample spliting procedure. The test set serves an important purpose in making sure our fitted models generalize well to unseen data and we report these loss metrics. We fix the KASN parameter determining the grid size as $\gamma = 0.499$ for all simulations. In each DGP, the true partially linear regression model is generated using the following nonlinear compositional functions and computed orthogonal residuals:

$$m_0(\boldsymbol{X}) = \sin\left(\sum_{j=1}^{S} \frac{1}{1+e^{-X_j}}\right) + 0.1\sum_{j=1}^{S} X_j^2 + \epsilon_m, \quad \epsilon_m \sim N(0, 0.25) \tag{4.3}$$

$$g_0(\boldsymbol{X}) = \frac{1}{1+e^{-\sum_{j=1}^{S}\sin(X_j)}} + 0.1\sum_{j=1}^{S}|X_j| + \epsilon_g, \qquad \epsilon_g \sim N(0, 0.25) \tag{4.4}$$

where $\hat{V} = D - \hat{m}(\boldsymbol{X})$, and $\hat{U} = Y - \hat{g}(\boldsymbol{X})$ are used to obtain $\hat{\alpha} = \frac{\sum \hat{V}_i(Y_i - \hat{g})}{\sum \hat{V}_i D_i}$. For our main simulation DGP, we have a KASN with $L_n = 6$ and $W_n = 201$ (KASN #1). We also consider an additional KASN with $W_n = 101$ (KASN #2).

Table 5: DML Simulation Results: ($\alpha_0 = 1.5$)

Metric	**KASN #1**	**KASN #2**	**SLFN**	**Lasso**
Causal Parameter ($\hat{\alpha}$)				
$\hat{\alpha}$	1.5024	1.4804	1.6660	1.7318
95% CI	[1.4594, 1.5454]	[1.4368, 1.5240]	[1.6324, 1.6996]	[1.7077, 1.7559]
RMSE	0.0657	0.0461	0.1672	0.2323
Nuisance Function Performance				
Train R^2 ($\hat{m}_0$ / $\hat{g}_0$)	0.80 / 0.82	0.84 / 0.85	0.09 / 0.07	0.18 / 0.16
Test R^2 ($\hat{m}_0$ / $\hat{g}_0$)	0.39 / 0.45	0.36 / 0.43	-0.04 / -0.06	-0.13 / -0.17
Sparsity Ratio (%) ($\hat{m}_0$ / $\hat{g}_0$)	92.5 / 92.9	88.8 / 89.0	NA / NA	8.35 / 4.99
Runtime (s)	2302.98	1070.20	0.17	0.04

Across the DML simulations, the KASN demonstrates a strong ability to recover the true causal parameter $\alpha_0 = 1.5$ despite the complex compositional structure of the nuisance functions in the DGP. The wider network (KASN #1 in Table 2) achieves a near-zero bias, yielding $\hat{\alpha} = 1.5024$ while the narrower architecture (KASN #2) yields $\hat{\alpha} = 1.4804$. In contrast, the SLFN and Lasso estimators exhibit substantial upward bias, with estimates of $\hat{\alpha} = 1.666$ and $\hat{\alpha} = 1.7318$, respectively. Despite the relatively strong performance of the SLFN in the earlier nonparametric regression (including its ability to generalize to unseen data, captured by the Test R^2), it fails to learn the compositional structures of m_0 and g_0. This highlights how shallow networks generally struggle to decompose complex hierarchical nonlinearities, similar to the the Lasso model which imposes a linear structure: resulting in a failure to eliminate regularization bias even under Neyman orthogonality and sample-splitting. Conversely, both iterations of the KASN display significant predictive power on the test set; confirming our theory that the learned B-spline activations effectively decomposes compositions into univariate sieve components, whereas the internal structure is unidentifiable to other architectures.

We observe that overparameterization in W_n does not degrade performance. Despite the larger width of KASN #1, it maintains high sparsity ratios, indicating the success of our regularization method combining the Group Lasso with the constraint coefficient Δ_n. As emphasized in our theoretical results, this structural sparsity is crucial for controlling

the KASN metric entropy in high-dimensions while preserving approximation power; thus preventing the explosion in complexity that standard MLPs are prone to. Furthermore, the stability of our estimator across different widths reinforces the robustness of the sieve grid resolution G_n. Even when the width is halved, the KASN extremum estimator maintains strong performance and ability to generalize to unseen data. These results provide empirical evidence that the KASN is a superior ML method for estimating nuisance functions in high-dimensional settings characterized by compositional covariate effects.

5 Empirical Application

We evaluate our proposed KASN extremum estimator in an empirical application featuring a large set of covariates relative to the overall sample size. Rapach and Zhou (2021) demonstrate how principal component analysis (PCA) can be applied to 118 macroeconomic and financial variables in order to identify interpretable risk factors. The resulting components (e.g., housing, yields, credit spreads) are linear combinations of subsets of the aforementioned variables obtained from the Federal Reserve Economic Data Monthly Database (FRED-MD, 1963-2024), which are then applied to estimate risk premia for a large set of test assets. See the online appendix of Rapach and Zhou (2021) for an outline of the standard transformations applied to the data. We split the $n = 726$ observations using a $0.8 : 0.1 : 0.1$ train/validation/test ratio.

In our initial application, we perform a simple forecasting exercise evaluating the ability of the KASN estimator to forecast changes in interest rates. We consider the 1-year constant maturity Treasury yield (GS1) from FRED-MD, constructing a 12-month lag structure over the full sample. For this exercise alone, we do not make use of early stopping (see Section A.4 in the Appendix), with KASN hyperparameters selected consistent with what is outlined in Section 2.4. Table 6 reports the results, the KASN achieves a low train RMSE, consistent with the overparameterization outlined earlier.

Table 6: Comparative Performance: LASSO, SLFN, and KASN (12-Month Lag, GS1)

Metric	**LASSO**	**SLFN**	**KASN**
Train RMSE	0.4525	0.3559	0.0265
Test R^2	0.1154	-0.2200	0.5266
Sparsity (%)	88.89	N/A	98.32
Training Time (s)	1.40	9.76	117.03
DM Statistic (vs. KASN)	2.0695	2.5543	N/A
p-value (5% sig.)	0.0385	0.0106	N/A

We find that in terms of the out-of-sample generalization in the form of the Test R^2, the KASN significantly outperforms our two comparison models. The robustness of this result is supported by additional specifications (increasing the size of the test set, evaluating 18-month lags, using FEDFUNDS as the target variable instead of GS1) and these results can be found in the Appendix. Formal comparison is provided by Diebold-Mariano (DM) tests (Diebold and Mariano, 1995) of equal predictive accuracy against the KASN. The DM statistic for both the LASSO (2.070) and the SLFN (2.554) comparison models are rejected at the 5% level, confirming that the out-of-sample advantage of the KASN is statistically significant and not attributable to sampling variation.

The findings of Rapach and Zhou (2021) suggests the existence of sparse linear relationships between the raw macro-finance variables, which we use (in addition to the aforementioned key sparse components) to evaluate our model against Lasso in a simple nonparametric regression. Given how this is the type of setting that the Lasso model traditionally excels in, this serves as a robustness test for the KASN involving noisy time-series financial data. Selecting $\gamma = 0.499$ and $\zeta_\delta = 0.9$, the KASN parameters are given by: $L_n = 6$, $W_n = 243$, $G_n = 24$ and $\Delta_n = 307.91$.

As shown in Table 7, the KASN achieves a Test R^2 of 0.51, which is nearly identical to the that of the Lasso model. This is of significant as it provides empirical evidence of the robustness of our sieve extremum estimator in high-dimensional settings characterized by (approximate) linearity. In regimes involving linear or additive DGPs, overparameterized

Table 7: Sparse Macro-Finance Factors

Metric	**LASSO**	**SLFN**	**KASN**
Train RMSE	4.49	3.85	3.94
Train R^2	0.38	0.54	0.52
Test RMSE	4.70	4.53	4.75
Test R^2	0.52	0.56	0.51
Runtime (s)	1.85	58.11	534.46
Sparsity (%)	89.3	N/A	97.7

MLP architectures generalize poorly (an issue addressed by Schmidt-Hieber, 2020) due to the unconstrained parameter space. Conversely, the KASN architecture effectively contracts or collapses to the linear approximation. Combined with our DML simulation results, this displays robustness to both extremes (with respect to the target function complexity) in its ability to generalize. Given this demonstrated robustness of our proposed estimator, the tradeoff resulting in increased computation time is justified. The Lasso model produces non-zero coefficient estimates for 12 covariates, while the KASN demonstrates a sparsity ratio of 97.7%. By pruning a significant proportion of redundant edges, the KASN minimizes the risk of capturing spurious nonlinearities; which is crucial for preserving consistency in finite-samples, especially when n is small relative to the total number of covariates as it is in this application.

6 Conclusion

This paper introduces the KASN estimator, a novel sieve extremum estimator that integrates the adaptive functional flexibility of KANs with the method of sieves. Our framework addresses a critical gap in the deep learning literature by providing an architecture that is innately robust to the CoD under structural sparsity and delivers a convergence rate of $o_P(n^{-1/4})$. This is achieved through the use of learnable B-spline activation functions and a novel regularization method involving the Group Lasso.

The established theoretical results demonstrate that the KASN maintains a controlled

entropy under a sparse compositional assumption imposed on the target function. This enables consistent estimation in high-dimensional settings, where the number of covariates is large relative to the sample size. Furthermore, KASN estimator converges quickly enough to deliver asymptotic normal plug-in estimators. Unlike standard MLPs, which often suffer from overparameterization and poor generalization in sparse environments, the KASN effectively prunes redundant univariate functions to identify and recover the underlying signal of the target function. Our Monte Carlo simulations and empirical application further substantiates these theoretical claims, with the KASN consistently producing a high ratio of inactive edges relative to the total potential edges (across a wide range of architectural sizes, dictated by network lengths and widths). The KASN performs especially well in DML simulations involving highly complex compositional nuisance functions relative to other ML methods evaluated.

In summary, the KASN estimator introduced in this paper represents a tractable and statistically efficient alternative for nonparametric estimation. By bridging the gap between GAMs and deep compositional networks, it provides a robust and flexible approach to approximation that rigorously controls the complexity of the function space in high-dimensional settings.

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A Appendix

Table 8: Summary of Notation

Symbol	Description
General / DGP	
n	Sample size
d	Dimension of covariates
$Z_t = (Y_t, X_t')'$	Strictly stationary observed process
P_0	True joint probability measure of $\{Z_t\}$
μ	Marginal probability measure of covariates
$\beta(j)$	β-mixing coefficient at lag j (K.1(b))
$\phi(j)$	ϕ-mixing (uniform mixing) coefficient at lag j (K.1(c))
Parameter Spaces and Norms	
Θ	Infinite-dimensional parameter space
θ_0	True parameter of interest (e.g., f_0 or (α_0, g_0))
$\hat{\theta}_n$	KASN sieve extremum estimator
$\pi_n\theta_0$	Best $L_2(\mu)$-projection of θ_0 onto Θ_n
$\mathcal{K}_n$	KASN sieve class (nonparametric component)
$W_2^{m+1}(\mu)$	Sobolev space with derivatives up to order $m+1$ in $L_2(\mu)$
m	Smoothness order of target function
$\|\cdot\|$	$L_2(\mu)$ norm (default unless specified)
$\|\cdot\|_\infty$	Supremum norm on $\mathcal{I}_\epsilon$
$\|\cdot\|_{W_2^{m+1}(\mu)}$	Sobolev norm (sum of L_2 norms of derivatives)
$\mathcal{I}_\epsilon$	Padded domain $[-\epsilon, 1+\epsilon]$ for B-spline evaluations
Model and Component Functions	
$l(Z_t, \theta)$	Quasi-log-likelihood (criterion) function
$\mathcal{L}_n(\theta)$	Empirical criterion: $\frac{1}{n}\sum_{t=1}^n l(Z_t, \theta)$
$\phi_{i,j}$	Univariate inner edge function, branch i, covariate j
Φ_i	Outer aggregation function for branch i
$\psi_{i,l}$	Compositional layer function: $\Phi_i = \psi_{i,L_n} \circ \cdots \circ \psi_{i,2}$
B, L	Uniform Sobolev norm bound and Lipschitz constant (K.4)
L_ψ	Lipschitz constant of $\psi_{i,l}$; $L_\psi \leq 1$ under K.4(c)
Q_n	Quasi-interpolant (spline projection operator)
$h(x)$	Residual identity connection: $h(x) = x$
DML / Semiparametric Notation	
D_t	Treatment variable (linear component)
W_t	Controls (nonlinear/nonparametric component)
$g_0(W_t)$	Outcome nuisance function in PLR
$m_0(W_t)$	Treatment nuisance function in PLR ($E[D_t \mid W_t]$)
$\psi(Z_t; \alpha, \eta)$	Neyman-orthogonal score
J_0	Jacobian of score with respect to α: $E[\psi^a(Z_t; \eta_0)]$
$\Gamma(\cdot)$	Smooth linear functional (e.g., average derivative)
v^*	Riesz representer of Γ

Table 9: KASN Parameter Definitions and Growth Rates

Parameter	**Condition**
Number of Layers	$L_n \asymp \log n$
Width	$W_n \asymp s_0 \asymp \log n$
Grid Size	$G_n \asymp n^{\gamma}, \quad \frac{1}{4(m+1)} < \gamma < \frac{1}{2}$
Coefficient Radius	$\Delta_n \asymp n^{\zeta_\delta}, \quad \zeta_\delta > \frac{\gamma}{2}$
Sparsity Cap (deterministic)	$s_n = \lceil C_{s_n}\, \bar{s}_0 \rceil = O((\log n)^2)$
Realized Active Edges	$\hat{s}_n = \|\hat{\boldsymbol{c}}_n\|_{0,2} \le s_n$
Total Potential Edges	$r_n = (d+1)\, W_n + (L_n - 1)\, W_n^2$
Total Parameters	$p_n \approx r_n G_n$

A.1 Proofs of Main Results (Convergence)

Lemma 1. *Let $\mathcal{J} \equiv \bigcup_{i=1}^{s_0} \mathcal{J}_i$ denote the set of active coordinates, with $|\mathcal{J}| \le s_0 J_{\max} = O(\log n)$ under K.3. Suppose each marginal distribution function F^j is continuous. Then under K.1:*

$$\max_{j \in \mathcal{J}} \|\hat{F}_n^j - F^j\|_\infty = O_P\big(n^{-1/2} \log n\big).$$

Proof of Lemma 1: Fix $j \in \mathcal{J}$ and note that $\hat{F}_n^j - F^j$ is the empirical process indexed by the class of (half-line) indicators $\{\mathbf{1}_{\{\cdot \le x\}} : x \in \mathbb{R}\}$, a VC class of index one (van der Vaart and Wellner, 1996). Under K.1(a), the Dvoretzky-Kiefer-Wolfowitz inequality (in the i.i.d. case[58]; van der Vaart, 1998, Theorem 19.3) yields the tail bound $P(\|\hat{F}_n^j - F^j\|_\infty > t) \le C_M \exp(-c_M n t^2)$. Under K.1(b), we apply the Bernstein-type inequality for geometrically β-mixing bounded sequences from Theorem 1 of Merlevède et al. (2011). Specifically, for the centered indicators $\mathbf{1}_{\{X_{t,j} \le x\}} - F^j(x)$ on a grid of n quantile points of F^j (consisting of $x_k = (F^j)^{-1}(k/n)$, $k = 1, \ldots, n$) and extending to the supremum by the monotonicity of $\hat{F}_n^j$ and F^j, we have:

$$P\big(\|\hat{F}_n^j - F^j\|_\infty > t\big) \le C\, n \exp\Big(-\frac{c\, n t^2}{\log n}\Big), \qquad t \ge c' n^{-1/2}, \tag{L1.1}$$

[58] Or its M-dependent analogue, obtained by splitting the sample into $M+1$ i.i.d. subsequences.

with constants depending only on (β_0, c) from K.1(b). In either case, applying the union bound over the active coordinates and (L1.1):

$$\begin{aligned} P\Big(\max_{j\in\mathcal{J}}\|\hat{F}_n^j - F^j\|_\infty > t\Big) &\le \sum_{j\in\mathcal{J}} P\big(\|\hat{F}_n^j - F^j\|_\infty > t\big) \\ &\le |\mathcal{J}| \cdot C\, n \exp\Big(-\frac{c\, n t^2}{\log n}\Big) \\ &\le C'\, n \log n\, \exp\Big(-\frac{c\, n t^2}{\log n}\Big), \end{aligned} \tag{L1.2}$$

absorbing $|\mathcal{J}| = O(\log n)$ into C' under K.3. Setting $t = A\, n^{-1/2} \log n$, the exponent in (L1.2) equals $-cA^2 \log n$, which dominates $\log(n \log n)$ for A sufficiently large, so that (L1.2) vanishes and $\max_{j\in\mathcal{J}}\|\hat{F}_n^j - F^j\|_\infty = O_P(n^{-1/2} \log n)$. Under K.1(c), the moment bound of Yokoyama (1980) for stationary mixing sequences with $\xi > 2$ gives $\mathbb{E}\|\hat{F}_n^j - F^j\|_\infty^p \le C_p\, n^{-p/2}$ for some $p > 2$, and by Markov's inequality with the same $t = A\, n^{-1/2} \log n$:

$$\begin{aligned} P\Big(\max_{j\in\mathcal{J}}\|\hat{F}_n^j - F^j\|_\infty > t\Big) &\le \sum_{j\in\mathcal{J}} \frac{\mathbb{E}\|\hat{F}_n^j - F^j\|_\infty^p}{t^p} \\ &\le |\mathcal{J}|\, C_p\, n^{-p/2} \left(A\, n^{-1/2} \log n\right)^{-p} \\ &= C_p'\, A^{-p}\, (\log n)^{1-p} = o(1), \end{aligned} \tag{L1.3}$$

since $p > 2 > 1$. □

Proof of Theorem 1: Throughout, all univariate norms are taken on $\mathcal{I}_\epsilon$ and $\|\cdot\|_\infty$ denotes the sup-norm on the relevant domain. We first construct the sieve approximation $\pi_n\theta_0$ on the normalized inputs $\tilde{X}_t = F(X_t) \in [0,1]^d$ and establish the deterministic bound, with the EDF perturbation handled in the concluding step.

Under K.3(a), θ_0 admits a sparse compositional structure with inner edges given by $\phi_{i,j}$ for $j \in \mathcal{J}_i$ (noting that $\forall j \notin \mathcal{J}_i$, $\phi_{i,j} \equiv 0$), which are sufficiently smooth to be approximated by B-splines of order $k \ge m+1$. Fix a univariate component $\phi \in \{\phi_{i,j}, \psi_{i,l}\}$. By K.4(a), $\phi \in W_2^{m+1}(\mathcal{I}_\epsilon)$ with norm B at most, and the (one-dimensional) Sobolev embedding gives

$\phi \in C^m(\mathcal{I}_\epsilon)$ with $\|\phi\|_\infty \leq C_e B$. By K.4(b), $\phi^{(m)}$ is Lipschitz with constant L, so that $\phi^{(m+1)}$ exists almost everywhere with $\|\phi^{(m+1)}\|_\infty \leq L$; hence $\phi \in W_\infty^{m+1}(\mathcal{I}_\epsilon)$.

Let $S_k(G_n)$ denote the space of splines of order $k \geq m+1$ on the uniform knot sequence over $\mathcal{I}_\epsilon$ with mesh $\delta_T = O(G_n^{-1})$. Furthermore, let Q_n denote the quasi-interpolant of de Boor (1978, Chapter XII), which reproduces polynomials of degree less than k. Applying the error bound for the quasi-interpolant (de Boor, 1978, Theorem XII.1), the Jackson-type inequality $\omega_k(\phi, \delta_T)_\infty \leq 2^{k-m-1}\, \delta_T^{m+1} \|\phi^{(m+1)}\|_\infty$ (valid for $k \geq m+1$), and the derivative bound from K.4(b):

$$\begin{aligned} \|\phi - Q_n\phi\|_\infty &\leq C_k'' \, \omega_k(\phi, \delta_T)_\infty \\ &\leq C_k'' \, 2^{k-m-1} \, \delta_T^{m+1} \, \|\phi^{(m+1)}\|_\infty \\ &\leq C_k \, L \, G_n^{-(m+1)}, \end{aligned} \tag{T1.1}$$

where $C_k > 0$ depends only on k, absorbing the factor 2^{k-m-1} and the mesh constant.

We next verify that the approximation $\pi_n\theta_0$ belongs to the sieve space $\mathcal{K}_n$. By the stability of the B-spline basis (de Boor, 1978, Corollary XI.1), the coefficient vector $\boldsymbol{c}(\phi)$ of $Q_n\phi = \sum_g c_g B_g$ satisfies $\|\boldsymbol{c}(\phi)\|_\infty \leq C_k' \|\phi\|_\infty \leq C_k' C_e B$ (where $C_e > 0$ is the Sobolev embedding constant), so that:

$$\begin{aligned} \|\boldsymbol{c}(\phi)\|_2 &\leq (G_n + k - 1)^{1/2} \, \|\boldsymbol{c}(\phi)\|_\infty \\ &\leq (G_n + k - 1)^{1/2} \, C_k' C_e B \\ &\leq C_B \, G_n^{1/2}. \end{aligned} \tag{T1.2}$$

The sieve projection constructed below assigns non-zero coefficients only to the active edges: at most $J_{\max}$ inner edges and one output edge per branch, and one compositional edge per branch per hidden layer, for a total of at most $s_0(J_{\max} + L_n) \leq C_J \bar{s}_0$ active edges (with all remaining potential edges set to zero). The compositional edges carry the coefficients

of $Q_n(\psi_{i,l} - \mathrm{id})$, whose sup-norm is bounded by $C_e B + (1 + 2\epsilon)$, so (T1.2) applies with an adjusted constant C'_B. Summing over the active edges and applying (T1.2):

$$\begin{aligned}\|\boldsymbol{c}_n\|_{1,2} = \sum_{j=1}^{r_n} \|\boldsymbol{c}_j\|_2 &\le s_0(J_{\max} + L_n) \cdot C'_B \, G_n^{1/2} \\ &\le C_J \bar{s}_0 \cdot C'_B \, G_n^{1/2} \\ &= O\big((\log n)^2 \, n^{\gamma/2}\big) \le \Delta_n \end{aligned} \tag{T1.3}$$

for all n sufficiently large, since $\Delta_n \asymp n^{\zeta_\delta}$ with $\zeta_\delta > \gamma/2$ by (2.10).

The sieve class (2.6) composes hidden layers of the form $\Phi_l + h$ with $h(x) = x$. To realize the true compositions of K.4(c) within this class, the spline on each active compositional edge in layer $l \in \{2, \ldots, L_n\}$ targets the residual $\psi_{i,l} - \mathrm{id}$ (rather than $\psi_{i,l}$ itself), so that the realized layer map is

$$\tilde{\psi}_{i,l}(x) \equiv Q_n(\psi_{i,l} - \mathrm{id})(x) + x, \qquad x \in \mathcal{I}_\epsilon.$$

Since $\mathrm{id} \in W_2^{m+1}(\mathcal{I}_\epsilon)$ with $(\psi_{i,l} - \mathrm{id})^{(m+1)} = \psi_{i,l}^{(m+1)}$ for $m \ge 1$ (and $(\psi_{i,l} - \mathrm{id})' = \psi'_{i,l} - 1$ with an unchanged Lipschitz constant for $m = 0$), (T1.1) applies to $\psi_{i,l} - \mathrm{id}$ directly, yielding the per-layer error:

$$\begin{aligned}\big\|\tilde{\psi}_{i,l} - \psi_{i,l}\big\|_\infty &= \big\|\big(Q_n(\psi_{i,l} - \mathrm{id}) + \mathrm{id}\big) - \psi_{i,l}\big\|_\infty \\ &= \|Q_n(\psi_{i,l} - \mathrm{id}) - (\psi_{i,l} - \mathrm{id})\|_\infty \\ &\le C_k \, L \, G_n^{-(m+1)}. \end{aligned} \tag{T1.4}$$

Crucially, the error of each realized layer is measured against the true $\psi_{i,l}$, so that error propagation through subsequent layers is governed by the Lipschitz constants of the true components, for which $L_\psi \le 1$ under K.4(c).

Define $u_i = \sum_{j \in \mathcal{J}_i} \phi_{i,j}(\tilde{X}_{t,j})$ and $v_i = \sum_{j \in \mathcal{J}_i} Q_n \phi_{i,j}(\tilde{X}_{t,j})$. By the triangle inequality and

(T1.1):

$$\begin{aligned}\|u_i - v_i\|_\infty &\leq \sum_{j\in\mathcal{J}_i} \|\phi_{i,j} - Q_n\phi_{i,j}\|_\infty \\ &\leq |\mathcal{J}_i|\, C_k\, L\, G_n^{-(m+1)} \\ &\leq J_{\max}\, C_k\, L\, G_n^{-(m+1)}. \end{aligned} \tag{T1.5}$$

By K.4(d)(i), $u_i \in [0,1]$, and by (T1.5), $v_i \in \mathcal{I}_\epsilon$ for all n large enough that $J_{\max} C_k L G_n^{-(m+1)} \leq \epsilon$. Applying K.4(b) with Lipschitz constant L:

$$\begin{aligned}\|\Phi_i(u_i) - \Phi_i(v_i)\|_\infty &\leq L\, \|u_i - v_i\|_\infty \\ &\leq J_{\max}\, C_k\, L^2\, G_n^{-(m+1)}. \end{aligned} \tag{T1.6}$$

Under K.4(c), $\Phi_i = \psi_{i,L_n} \circ \cdots \circ \psi_{i,2}$, and we let $\hat{\Phi}_i = \tilde{\psi}_{i,L_n} \circ \cdots \circ \tilde{\psi}_{i,2}$ denote the realized composition of the layer maps constructed above. Applying the following decomposition:

$$\begin{aligned}\Phi_i - \hat{\Phi}_i = \sum_{l=2}^{L_n} &\Big[\psi_{i,L_n} \circ \cdots \circ \psi_{i,l+1} \circ \psi_{i,l} \circ \tilde{\psi}_{i,l-1} \circ \cdots \circ \tilde{\psi}_{i,2}\Big] \\ &- \Big[\psi_{i,L_n} \circ \cdots \circ \psi_{i,l+1} \circ \tilde{\psi}_{i,l} \circ \tilde{\psi}_{i,l-1} \circ \cdots \circ \tilde{\psi}_{i,2}\Big]\end{aligned}$$

Each term differs only in the layer-l map, evaluated at the common realized prefix $\tilde{\psi}_{i,l-1} \circ \cdots \circ \tilde{\psi}_{i,2}$, and propagates through the true tail composition $\psi_{i,L_n} \circ \cdots \circ \psi_{i,l+1}$, whose Lipschitz constant is at most $L_\psi^{L_n - l} \leq 1$ by K.4(c); K.4(d)(ii) together with the padding ϵ ensures every intermediate argument lies in $\mathcal{I}_\epsilon$ for all n sufficiently large, since the accumulated perturbation of intermediate values is $O(L_n G_n^{-(m+1)}) = o(1)$. Therefore:

$$\|\Phi_i - \hat{\Phi}_i\|_\infty \leq \sum_{l=2}^{L_n} L_\psi^{L_n - l} \left\|\tilde{\psi}_{i,l} - \psi_{i,l}\right\|_\infty. \tag{T1.7}$$

Substituting (T1.4) into (T1.7):

$$\begin{aligned}\|\Phi_i - \hat{\Phi}_i\|_\infty &\le C_k\, L\, G_n^{-(m+1)} \sum_{l=2}^{L_n} L_\psi^{L_n - l} \\ &= C_k\, L\, G_n^{-(m+1)} \cdot \frac{L_\psi^{L_n - 1} - 1}{L_\psi - 1}, \end{aligned} \tag{T1.8}$$

where since $L_\psi \le 1$ by K.4(c): if $L_\psi < 1$, the geometric sum satisfies

$$\sum_{l=2}^{L_n} L_\psi^{L_n - l} = (1 - L_\psi^{L_n - 1})/(1 - L_\psi) \le 1/(1 - L_\psi) < \infty,$$

and if $L_\psi = 1$, the sum equals $L_n - 1 = O(\log n)$. In both cases[59] the sum is $O(\log n)$, absorbed into the constant $C_3 = O(\log n) > 0$:

$$\|\Phi_i - \hat{\Phi}_i\|_\infty \le C_3\, L\, G_n^{-(m+1)}. \tag{T1.9}$$

The output node sums the s_0 branch outputs, and the discrepancy between $\hat{\Phi}_i(v_i)$ and $\Phi_i(u_i)$ is exactly the sum of the two bounded terms, by the triangle inequality and the Lipschitz continuity of Φ_i. Summing (T1.6) and (T1.9) over $i = 1, \ldots, s_0$:

$$\begin{aligned}\|\pi_n\theta_0 - \theta_0\|_\infty &\le \sum_{i=1}^{s_0} \left(\|\Phi_i(u_i) - \Phi_i(v_i)\|_\infty + \|\Phi_i - \hat{\Phi}_i\|_\infty \right) \\ &\le s_0 \left(J_{\max} C_k L^2 + C_3 L \right) G_n^{-(m+1)} \\ &= C_{\bar{s}_0}\, G_n^{-(m+1)}, \end{aligned} \tag{T1.10}$$

where $C_{\bar{s}_0} \equiv s_0(J_{\max} C_k L^2 + C_3 L) = O(s_0 \log n) = O(\bar{s}_0)$ since $C_3 = O(\log n)$ from earlier and $\bar{s}_0 = s_0 L_n = O((\log n)^2)$. Since μ is a probability measure supported on $[0,1]^d$, the L_∞ bound implies the $L_2(\mu)$ bound directly: $\|\pi_n\theta_0 - \theta_0\|_{L_2(\mu)} \le \|\pi_n\theta_0 - \theta_0\|_\infty$, which establishes

[59] As mentioned in the main text, the constraint $L_\psi \le 1$ may be loosened to the following (tail-product) condition: $\sup_l \prod_{j=l}^{L_n} L_{\psi_{i,j}} \le C_\Pi < \infty$ uniformly in n, under which every bound in this proof holds with constants multiplied by C_Π: the propagation factors in (T1.7) are bounded by C_Π rather than by one, and the geometric sum by $C_\Pi(L_n - 1) = O(\log n)$.

the deterministic bound. Under $G_n \asymp n^{\gamma}$ and $\bar{s}_0 = O((\log n)^2)$, we have:

$$C_{\bar{s}_0}\, G_n^{-(m+1)} = O\big((\log n)^2\, n^{-(m+1)\gamma}\big),$$

and $-(m+1)\gamma < -\frac{1}{4} \iff \gamma > \frac{1}{4(m+1)}$ is sufficient for the resulting $o(n^{-1/4})$ rate.

It remains to account for the CDF normalization. The realized approximation depends on the inputs only through the active coordinates $\mathcal{J} = \bigcup_i \mathcal{J}_i$. For each active inner edge, the quasi-interpolant satisfies the simultaneous approximation property $\|(Q_n\phi_{i,j})'\|_\infty \leq \|\phi'_{i,j}\|_\infty + o(1) \leq C(B, L)$ (de Boor, 1978, Chapter XII; see also Schumaker, 2007, Theorem 12.8), so each is Lipschitz with a constant independent of n. Replacing F^j with $\hat{F}_n^j$ therefore perturbs each inner-edge output by at most $C(B, L)\,\|\hat{F}_n^j - F^j\|_\infty$, the inner aggregate by at most $J_{\max}$ times the maximal such perturbation, and the branch output by the same order after propagation through the realized composition, whose Lipschitz constant is bounded as in (T1.7)-(T1.9). Summing over the s_0 branches and applying Lemma 1:

$$\begin{aligned}
\big\|\pi_n\theta_0 \circ \hat{F}_n - \pi_n\theta_0 \circ F\big\| &\leq C\, s_0 J_{\max} \max_{j\in\mathcal{J}} \|\hat{F}_n^j - F^j\|_\infty \\
&= O(\log n) \cdot O_P\big(n^{-1/2}\log n\big) \\
&= O_P\big(n^{-1/2}(\log n)^2\big), \qquad \text{(T1.11)}
\end{aligned}$$

which is $o_P(n^{-1/4})$ and of smaller order than the deterministic bound (T1.10) for every $\gamma < \frac{1}{2}$.

□

For the following Lemma proofs, it is useful to define $D \equiv \max\{C_e B,\, L,\, 1\}$, where C_e is the Sobolev embedding constant introduced in the proof of Theorem 1. The recursive application of the chain rule and the Faà di Bruno formula are standard tools in the analysis of deep compositional architectures; see Kohler and Langer (2021) for derivative propagation through smooth compositions and Schmidt-Hieber (2020) for the use of compositional structure (sparse in MLP weights) to mitigate the curse of dimensionality

Lemma 2. *Under Assumptions K.4, fix i and let $H_l \equiv \psi_{i,L_n} \circ \cdots \circ \psi_{i,l}$ denote the tail composition for $2 \le l \le L_n$. Then for all l:*

$$\|H_l'\|_\infty \le 1, \qquad \|H_l^{(r)}\|_\infty \le C_r(D)\, L_n^{r-1} \quad \text{for } 2 \le r \le m+2,$$

where $C_r(D) > 0$ depends only on r and D. In particular, all derivative bounds on the tail compositions (and by extension, on $\Phi_i = H_2$) grow at most at a polylogarithmic rate in n.

Proof of Lemma 2: Under K.4(c), and with all arguments lying in $[0,1] \subset \mathcal{I}_\epsilon$ by K.4(d)(ii), the first-derivative bound follows from the chain rule:

$$\|H_l'\|_\infty \le \prod_{j=l}^{L_n} \|\psi_{i,j}'\|_\infty \le \prod_{j=l}^{L_n} L_\psi \le 1. \tag{L2.1}$$

For $r \ge 2$, we induct on depth via $H_l = H_{l+1} \circ \psi_{i,l}$. By the Faà di Bruno formula:

$$(H_{l+1} \circ \psi_{i,l})^{(r)} = \sum_{\pi \in \mathcal{P}_r} H_{l+1}^{(|\pi|)}(\psi_{i,l}) \prod_{b \in \pi} \psi_{i,l}^{(|b|)},$$

where $\mathcal{P}_r$ denotes the set partitions of $\{1, \ldots, r\}$. The single partition with $|\pi| = r$ contributes at most $\|H_{l+1}^{(r)}\|_\infty \cdot L_\psi^r \le \|H_{l+1}^{(r)}\|_\infty$; every other partition has $|\pi| \le r-1$ and contributes at most $\max_{s<r} \|H_{l+1}^{(s)}\|_\infty D^r$, since each of its at most r factors $\psi_{i,l}^{(|b|)}$ is bounded by D. Since $|\mathcal{P}_r|$ is the Bell number $\mathcal{B}_r$ (a constant depending only on r, with $\mathcal{B}_r \le 15$ for $r \le 4$), we obtain the recursion:

$$\|H_l^{(r)}\|_\infty \le \|H_{l+1}^{(r)}\|_\infty + \mathcal{B}_r\, D^r \max_{s<r} \|H_{l+1}^{(s)}\|_\infty, \tag{L2.2}$$

which unwinds over at most L_n layers from the terminal value $\|H_{L_n}^{(r)}\|_\infty = \|\psi_{i,L_n}^{(r)}\|_\infty \le D$. For $r = 2$ the maximum in (L2.2) is over $s = 1$, bounded by one under (L2.1), so unwinding

yields:

$$\begin{aligned}\|H_l''\|_\infty &\le \|H_{L_n}''\|_\infty + (L_n - l)\,\mathcal{B}_2\,D^2 \\ &\le D + L_n\,\mathcal{B}_2\,D^2 \\ &\le C_2(D)\,L_n. \end{aligned} \tag{L2.3}$$

Substituting (L2.3) into (L2.2) for $r = 3$, the increment per layer is at most $\mathcal{B}_3 D^3\, C_2(D) L_n$, and unwinding gives $\|H_l'''\|_\infty \le C_3(D)\, L_n^2$. Proceeding inductively, suppose $\|H_l^{(s)}\|_\infty \le C_s(D) L_n^{s-1}$ for all $s < r$; then the maximum in (L2.2) is $O(L_n^{r-2})$, each of the L_n layers contributes an increment of order $\mathcal{B}_r D^r L_n^{r-2}$, and unwinding yields $\|H_l^{(r)}\|_\infty \le C_r(D) L_n^{r-1}$. The recursion therefore closes for every $r \ge 2$, with $C_r(D)$ depending only on r and D through the $\mathcal{B}_r$; the derivative orders $r \le m + 2$ required by Corollary 1 are the relevant special case. □

Lemma 3. *Let $F : \mathcal{I}_\epsilon \to \mathbb{R}$ satisfy $\max_{1\le s\le m+1}\|F^{(s)}\|_\infty \le M_F$ with $F^{(m+1)}$ Lipschitz with constant Λ_F, and let $g_1, g_2 : \mathcal{X} \to \mathcal{I}_\epsilon$ (with $\mathcal{X} = [0,1]^d$ or $\mathcal{I}_\epsilon$) have all partial derivatives up to order $m+1$ bounded by M_g, with*

$$\Delta_g \equiv \max_{|\bar\lambda|\le m+1} \left\|D^{\bar\lambda}(g_1 - g_2)\right\|_\infty.$$

Then:

$$\max_{|\bar\lambda|\le m+1} \left\|D^{\bar\lambda}\big(F\circ g_1 - F\circ g_2\big)\right\|_\infty \le C(m)\,(M_F + \Lambda_F)\,(1 + M_g)^{m+1}\,\Delta_g,$$

where $C(m)$ depends only on m.

Proof of Lemma 3: Fix a multi-index $\bar\lambda$ with $|\bar\lambda| = r \le m+1$. By the multivariate Faà di Bruno formula, $D^{\bar\lambda}(F\circ g)$ is a finite sum (with at most $C'(r)$ terms) of expressions of the form $F^{(s)}(g)\prod_{a=1}^s D^{\bar\lambda_a} g$ with $1 \le s \le r$ and $\sum_a \bar\lambda_a = \bar\lambda$. For each such term, adding and

subtracting the mixed expression:

$$F^{(s)}(g_1)\prod_a D^{\bar\lambda_a} g_1 - F^{(s)}(g_2)\prod_a D^{\bar\lambda_a} g_2 = \big[F^{(s)}(g_1) - F^{(s)}(g_2)\big]\prod_a D^{\bar\lambda_a} g_1 + F^{(s)}(g_2)\Big[\prod_a D^{\bar\lambda_a} g_1 - \prod_a D^{\bar\lambda_a} g_2\Big]. \tag{L3.1}$$

For the first bracket, $F^{(s)}$ is Lipschitz for every $s \le m+1$ with constant at most $M_F + \Lambda_F$, since $F^{(s+1)}$ is bounded by M_F for $s \le m$ and $F^{(m+1)}$ is Lipschitz with constant Λ_F; hence the first term of (L3.1) is bounded by $(M_F + \Lambda_F)\,\Delta_g\, M_g^s \le (M_F + \Lambda_F)(1 + M_g)^{m+1}\Delta_g$. For the second bracket, telescoping the product difference factor by factor:

$$\prod_{a=1}^{s} D^{\bar\lambda_a} g_1 - \prod_{a=1}^{s} D^{\bar\lambda_a} g_2 = \sum_{a=1}^{s} \Big(\prod_{b<a} D^{\bar\lambda_b} g_1\Big)\,\big(D^{\bar\lambda_a} g_1 - D^{\bar\lambda_a} g_2\big)\,\Big(\prod_{b>a} D^{\bar\lambda_b} g_2\Big),$$

so that each of the at most $m+1$ summands is bounded by $M_g^{s-1}\Delta_g \le (1+M_g)^{m+1}\Delta_g$, and the second term of (L3.1) is bounded by $M_F\,(m+1)\,(1+M_g)^{m+1}\Delta_g$. Summing over the $C'(r)$ terms and taking the maximum over $\bar\lambda$ yields the claim with $C(m) = (m+2)\max_{r\le m+1} C'(r)$. □

Proof of Corollary 1: We work throughout in the sup norms of derivatives on $\mathcal{I}_\epsilon$ and convert to $W_2^{m+1}(\mu)$ in our concluding step. The sieve projection is that of Theorem 1, whose membership in $\mathcal{K}_n$ was verified in (T1.3); only the error bounds are strengthened here using Lemma 2 and Lemma 3.

Fix a univariate component $\phi \in \{\phi_{i,j},\ \psi_{i,l} - \mathrm{id}\}$. As established above, $\|\phi^{(r)}\|_\infty \le D$ for all $0 \le r \le m+2$ under K.4, so $\phi \in W_\infty^{m+2}(\mathcal{I}_\epsilon)$. For splines of order $k \ge m+2$ on the uniform knot sequence with mesh $\delta_T = O(G_n^{-1})$, applying the simultaneous approximation property of the quasi-interpolant (Schumaker, 2007, Theorem 12.8; see also de Boor, 1978,

Chapter XII) at each derivative order $0 \le l \le m+1$:

$$\begin{aligned} \left\| (\phi - Q_n\phi)^{(l)} \right\|_\infty &\le C_k \, D \, G_n^{-(m+2-l)} \\ &\le C_k \, D \, G_n^{-1}. \end{aligned} \tag{C1.1}$$

With u_i and v_i as in the proof of Theorem 1, additivity across coordinates implies that all mixed partial derivatives of $u_i - v_i$ vanish, and pure partials reduce to univariate derivatives. By (C1.1) and the triangle inequality:

$$\begin{aligned} \max_{|\bar\lambda| \le m+1} \left\| D^{\bar\lambda}(u_i - v_i) \right\|_\infty &\le \max_{1 \le l \le m+1} \sum_{j \in \mathcal{J}_i} \left\| (\phi_{i,j} - Q_n\phi_{i,j})^{(l)} \right\|_\infty \\ &\le J_{\max} \, C_k \, D \, G_n^{-1}, \end{aligned} \tag{C1.2}$$

and the partial derivatives of u_i and v_i up to order $m+1$ are bounded by $J_{\max} D + o(1) \le 2 J_{\max} D$.

Applying Lemma 3 with $F = \Phi_i$, $g_1 = u_i$, and $g_2 = v_i$: by Lemma 2, $M_F \le C_{m+1}(D) \, L_n^m$ and the Lipschitz constant of $F^{(m+1)}$ satisfies $\Lambda_F \le C_{m+2}(D) \, L_n^{m+1}$, both polylogarithmic in n; K.4(d) ensures the arguments lie in $\mathcal{I}_\epsilon$ for n large. With $M_g \le 2 J_{\max} D$ and Δ_g given by (C1.2):

$$\max_{|\bar\lambda| \le m+1} \left\| D^{\bar\lambda}\big(\Phi_i(u_i) - \Phi_i(v_i)\big) \right\|_\infty \le C \, (\log n)^{m+1} \, G_n^{-1}. \tag{C1.3}$$

Under K.4(c), $\Phi_i = \psi_{i,L_n} \circ \cdots \circ \psi_{i,2}$, and we let $\hat\Phi_i = \tilde\psi_{i,L_n} \circ \cdots \circ \tilde\psi_{i,2}$ denote the realized composition of the layer maps $\tilde\psi_{i,l} \equiv Q_n(\psi_{i,l} - \mathrm{id}) + \mathrm{id}$ constructed in the proof of Theorem 1. Applying the same decomposition as in (T1.7), each term of $\Phi_i(v_i) - \hat\Phi_i(v_i)$ is of the form $H_{l+1} \circ g_{1,l} - H_{l+1} \circ g_{2,l}$, where $g_{1,l} = \psi_{i,l} \circ \hat h_l$ and $g_{2,l} = \tilde\psi_{i,l} \circ \hat h_l$ share the realized prefix composition $\hat h_l = \tilde\psi_{i,l-1} \circ \cdots \circ \tilde\psi_{i,2}$ applied to v_i. We apply Lemma 3 twice per term. First, with $F = \psi_{i,l} - \tilde\psi_{i,l}$ absorbed as the difference, the derivatives of $g_{1,l} - g_{2,l} = (\psi_{i,l} - \tilde\psi_{i,l}) \circ \hat h_l$

up to order $m+1$ are bounded, by the chain rule and (C1.1) applied to $\psi_{i,l} - \mathrm{id}$:

$$\begin{aligned}\max_{|\bar{\lambda}|\le m+1} \left\| D^{\bar{\lambda}}(g_{1,l} - g_{2,l}) \right\|_\infty &\le C\,(1 + M_{\hat{h}})^{m+1} \max_{0\le l'\le m+1} \left\| (\psi_{i,l} - \tilde{\psi}_{i,l})^{(l')} \right\|_\infty \\ &\le C\,(1 + M_{\hat{h}})^{m+1}\, C_k\, D\, G_n^{-1}, \end{aligned} \tag{C1.4}$$

where $M_{\hat{h}}$ bounds the derivatives of the realized prefix composition $\hat{h}_l$ and satisfies $M_{\hat{h}} \le C_{m+1}(D)\, L_n^m + o(1)$, since $\hat{h}_l$ is the true partial composition perturbed by accumulated approximation errors that vanish by Theorem 1. Second, with $F = H_{l+1}$: by Lemma 2, $M_F + \Lambda_F = O((\log n)^{m+1})$, and the derivatives of $g_{1,l}$, $g_{2,l}$ up to order $m+1$ are polylogarithmically bounded by the same prefix estimates. Combining, each telescope term is bounded by $C(\log n)^{c'_m} G_n^{-1}$ for a fixed integer c'_m depending only on m, and summing over the $L_n - 1 = O(\log n)$ layers:

$$\max_{|\bar{\lambda}|\le m+1} \left\| D^{\bar{\lambda}}\big(\Phi_i(v_i) - \hat{\Phi}_i(v_i)\big) \right\|_\infty \le C\,(\log n)^{c'_m+1}\, G_n^{-1}. \tag{C1.5}$$

Summing (C1.3) and (C1.5) over $i = 1, \dots, s_0$ bounds every partial derivative of $\pi_n\theta_0 - \theta_0$ up to order $m+1$, uniformly on $[0,1]^d$, by $C\, s_0 (\log n)^{c'_m+1} G_n^{-1}$. Since $\pi_n\theta_0 - \theta_0$ depends on the inputs only through the active coordinates $\mathcal{J}$ with $|\mathcal{J}| = O(\log n)$, all partial derivatives with respect to inactive coordinates vanish identically, and the number of non-vanishing multi-indices with $|\bar{\lambda}| \le m+1$ is $\binom{|\mathcal{J}|+m+1}{m+1} = O((\log n)^{m+1})$. Since μ is a probability measure:

$$\begin{aligned}\|\pi_n\theta_0 - \theta_0\|_{W_2^{m+1}(\mu)} &\le \Big(\textstyle\sum_{|\bar{\lambda}|\le m+1} \|D^{\bar{\lambda}}(\pi_n\theta_0 - \theta_0)\|_\infty^2 \Big)^{1/2} \\ &\le O\big((\log n)^{(m+1)/2}\big) \cdot C\, s_0\, (\log n)^{c'_m+1}\, G_n^{-1} \\ &\le C'_{\bar{s}_0}\, G_n^{-1}, \end{aligned} \tag{C1.6}$$

with $C'_{\bar{s}_0} = O(\bar{s}_0 (\log n)^{c_m})$ for $c_m \equiv c'_m + 1 + \lceil (m+1)/2 \rceil$. Under $G_n \asymp n^\gamma$, the bound

is $o(n^{-1/4})$ whenever $\gamma > \frac{1}{4}$, since the polylogarithmic factor is dominated by any strict inequality in the exponent. $\square$

Lemma 4. *Fix an active support $S \subseteq \{1, \dots, r_n\}$ with $|S| \le s_n$, and let $f(\cdot; \boldsymbol{c})$, $f(\cdot; \boldsymbol{c}')$ be elements of $\mathcal{K}_n$ with support S. Then:*

$$\|f(\cdot; \boldsymbol{c}) - f(\cdot; \boldsymbol{c}')\|_\infty \le \Lambda_n \sum_{j \in S} \|\boldsymbol{c}_j - \boldsymbol{c}'_j\|_2, \qquad \log \Lambda_n = O\big((\log n)^2\big).$$

Proof of Lemma 4: By the partition-of-unity property of the B-spline basis ($B_g \ge 0$, $\sum_g B_g \equiv 1$ on $\mathcal{I}_\epsilon$), a single edge satisfies:

$$\Big\| \sum_g (c_g - c'_g) B_g \Big\|_\infty \le \|\boldsymbol{c}_j - \boldsymbol{c}'_j\|_\infty \le \|\boldsymbol{c}_j - \boldsymbol{c}'_j\|_2. \tag{L4.1}$$

For the propagation constant, each edge spline has derivative bounded by $\|\sum_g c_g B'_g\|_\infty \le k\, C_k\, G_n \|\boldsymbol{c}_j\|_2$, since at most k basis functions are non-zero at any point and $\|B'_g\|_\infty \le C_k G_n$ on the uniform mesh. Each layer map (splines summed at nodes, plus the identity residual) is therefore Lipschitz with constant at most $\Lambda_{\text{lay}} \equiv 1 + W_n\, k\, C_k\, G_n \Delta_n$. Telescoping the perturbation layer by layer, a coefficient change in layer l alters the layer output by at most $\sum_{j \in S \cap \text{layer } l} \|\boldsymbol{c}_j - \boldsymbol{c}'_j\|_2$ by (L4.1), and propagates to the output through at most L_n downstream layers. Therefore:

$$\begin{aligned} \Lambda_n \le \Lambda_{\text{lay}}^{L_n}, \qquad \log \Lambda_n &\le L_n \log \Lambda_{\text{lay}} \\ &= O(\log n) \cdot O\big(\log\big(n^{\zeta_\delta + \gamma} \log n\big)\big) \\ &= O\big((\log n)^2\big), \end{aligned} \tag{L4.2}$$

since $\Lambda_{\text{lay}} = O(n^{\zeta_\delta + \gamma} \log n)$ under (2.10). $\square$

Proof of Theorem 2: An element of $\mathcal{K}_n$ is specified by an active support S with $|S| \le s_n$ and a coefficient vector supported on S satisfying the $\ell_{1,2}$-constraint; we cover the two

components in turn. The number of admissible supports satisfies $\sum_{s \le s_n} \binom{r_n}{s} \le \left(\frac{e\, r_n}{s_n}\right)^{s_n}$, so the support selection (by Lemma 17.5.1 of Cover and Thomas, 1991) contributes at most:

$$\begin{aligned} \log \sum_{s \le s_n} \binom{r_n}{s} &\le s_n \log\left(\frac{e\, r_n}{s_n}\right) \\ &\le 2\, s_n \log\left(\frac{r_n}{s_n}\right) \end{aligned} \tag{T2.1}$$

to the log covering number, where the final inequality holds since $s_n = O((\log n)^2)$ while $r_n \gtrsim d \to \infty$ at a polynomial rate under (2.4), so that $s_n < r_n/e$ for all n large.

For a fixed support S, the coefficient vector lies in the product of the $|S| \le s_n$ Euclidean balls $\mathcal{B}_j = \{\boldsymbol{c}_j \in \mathbb{R}^{G_n} : \|\boldsymbol{c}_j\|_2 \le \Delta_n\}$. Set the coefficient resolution $\epsilon' \equiv \epsilon/(s_n \Lambda_n)$, with Λ_n from Lemma 4. By Corollary 4.2.13 of Vershynin (2018), each ball admits an ϵ'-net of log cardinality at most $G_n \log(3\Delta_n/\epsilon')$. Furthermore, log covering numbers add across the coordinates of a product set, so the product over the edges in S admits a net of log cardinality at most:

$$\begin{aligned} s_n\, G_n \log\left(\frac{3\Delta_n}{\epsilon'}\right) &= s_n\, G_n \log\left(\frac{3\Delta_n s_n \Lambda_n}{\epsilon}\right) \\ &= s_n\, G_n \left[\log\left(\frac{\Delta_n}{\epsilon}\right) + \log(3 s_n) + \log \Lambda_n\right] \\ &\le C_2\, s_n G_n \left[\log\left(\frac{\Delta_n}{\epsilon}\right) + (\log n)^2\right], \end{aligned} \tag{T2.2}$$

absorbing $\log(3s_n) = O(\log \log n)$ and $\log \Lambda_n = O((\log n)^2)$ from (L4.2) into the bracketed term. By Lemma 4, if two coefficient vectors on the common support S lie within ϵ' of one another edgewise in ℓ_2, the corresponding functions satisfy:

$$\|f(\cdot; \boldsymbol{c}) - f(\cdot; \boldsymbol{c}')\|_\infty \le \Lambda_n \sum_{j \in S} \|\boldsymbol{c}_j - \boldsymbol{c}'_j\|_2 \le \Lambda_n\, s_n\, \epsilon' = \epsilon,$$

so the image of the coefficient net is an ϵ-net for the functions supported on S. Taking the union of the function nets over the admissible supports and combining (T2.1) with (T2.2)

yields the stated bound with $C_1 = 2$. The fixed residual maps h are common to every element and contribute nothing. The $L_2(\mu)$ statement follows since a sup-norm ϵ-net is an $L_2(\mu)$ ϵ-net for any probability measure μ. □

Proof of Corollary 2: Let $\{f_i\}$ be an $(\epsilon/2)$-net for $\mathcal{K}_n$ in the sup norm, as provided by Theorem 2. The brackets $[f_i - \epsilon/2,\ f_i + \epsilon/2]$ cover $\mathcal{K}_n$, and each bracket has $L_2(\mu)$-size

$$\big\|(f_i + \epsilon/2) - (f_i - \epsilon/2)\big\|_{L_2(\mu)} = \|\epsilon\|_{L_2(\mu)} = \epsilon$$

for any probability measure μ (van der Vaart and Wellner, 1996, Section 2.7). Substituting $\epsilon/2$ into the bound of Theorem 2 completes the proof. □

Proof of Theorem 3: Since $\hat{\theta}_n \in \mathcal{K}_n$ by construction and $\mathcal{K}_n$ is compact under the constraints including Δ_n, Theorem 1 of Chen and Shen (1998) applies directly (i.e., $\hat{\theta}_n = \arg\sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta)$ is the exact maximizer under K.2(c), such that Equation 2.1 of Chen and Shen, 1998 holds with $\tilde{\epsilon}_n = 0$). Under Assumptions K.1–K.2, their conditions A.1, A.2 and A.4 are satisfied (Condition A is outlined in its entirety in Section A.2). Condition A.1 is satisfied by all cases of K.1: trivially under K.1(a) ($\beta(j) = 0$ for $j > M$), under K.1(b) with block size $a_{n1} = \lfloor n^{1/2} \rfloor$, $a_{n2} = \lfloor n/(2a_{n1}) \rfloor$ (as required by Lemma 1 of Chen and Shen, 1998), and under K.1(c) by Corollary 1 of Chen and Shen (1998), which extends Theorem 1 to uniform mixing processes under the same block construction. A.2 is implied by K.2(b), which in turn holds automatically under quadratic loss[60] and A.4 holds under K.2(c) with $\nu \in (0, 2)$, envelope $U(Z_t)$, and $E|U(Z_t)|^\kappa \leq C_2 < \infty$ for $\kappa > 2$. For A.3, it is sufficient to verify the following integral condition:

$$\int_0^{\delta_n} \sqrt{\frac{\log \mathcal{N}_{[]}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)})}{n}}\, d\epsilon \leq C_3 \delta_n^2, \tag{T3.1}$$

[60] Writing $g \equiv \theta - \theta_0$, the loss difference is exactly $g[U_t - \frac{1}{2}g]$, and $\|g\|_\infty \leq 2\mathcal{B}_n$ since $\|\theta_0\|_\infty \leq \mathcal{B}_n$ under K.3-K.4. This yields $\mathrm{Var}[\cdot] \leq 2(\bar{\sigma}^2 + \mathcal{B}_n^2)\|g\|^2$, where the constant is fixed under K.3(b)(i) and $O((\log n)^2)$ under K.3(b)(ii), inflating δ_n in (T3.9) by only polylogarithmic factors.

for $\delta_n = n^{-1/4-\eta}$ and some fixed $\eta > 0$. Applying Corollary 2, the combinatorial term satisfies $C_1 s_n \log(r_n/s_n) = O((\log n)^3)$ and is therefore dominated by the coefficient-complexity term (of order $(\log n)^4 n^\gamma$), so that both are absorbed into a single constant $C_2 > 0$. Applying the subadditivity of the square root to separate the ϵ-dependent component of the resulting bound:

$$\begin{aligned}\sqrt{\frac{\log \mathcal{N}_{[]}(\epsilon, \mathcal{K}_n, \|\cdot\|)}{n}} &\leq \sqrt{\frac{C_2 (\log n)^2 n^\gamma \big[\log(2\Delta_n/\epsilon) + (\log n)^2\big]}{n}} \\ &\leq \sqrt{C_2 (\log n)^2 n^{\gamma-1} \log(2\Delta_n/\epsilon)} + \sqrt{C_2 (\log n)^4 n^{\gamma-1}} \\ &= C(\log n)\, n^{(\gamma-1)/2} \big[\log(2\Delta_n/\epsilon)\big]^{1/2} + C(\log n)^2\, n^{(\gamma-1)/2}, \qquad \text{(T3.2)}\end{aligned}$$

where the second term in (T3.2) is constant in ϵ and reflects the continuity constant of the coefficient-to-function map established in Lemma 4. We decompose the integral by splitting at $\epsilon_0 = n^{-1}$ and show that both terms are $o_P(n^{-1/4})$:

$$\int_0^{\delta_n} \sqrt{\frac{\log \mathcal{N}_{[]}(\epsilon, \mathcal{K}_n, \|\cdot\|_{L_2(\mu)})}{n}} d\epsilon = \underbrace{\int_0^{\epsilon_0} \cdots d\epsilon}_{(I)} + \underbrace{\int_{\epsilon_0}^{\delta_n} \cdots d\epsilon}_{(II)}$$

Term (I): We first evaluate the contribution of the ϵ-dependent component of (T3.2) via the substitution $u = \log(2\Delta_n/\epsilon)$, so $\epsilon = 2\Delta_n e^{-u}$ and $d\epsilon = 2\Delta_n e^{-u}\, du$. When $\epsilon = \epsilon_0 = n^{-1}$, $u = \log(2\Delta_n n) \leq C_\zeta \log n \equiv a$; as $\epsilon \to 0^+$, $u \to \infty$. Therefore:

$$\text{(I)} = \sqrt{C_2 (\log n)^2 n^{\gamma-1}} \int_a^\infty u^{1/2} \cdot 2\Delta_n e^{-u}\, du \qquad \text{(T3.3)}$$

Integrating by parts with $v = u^{1/2}$, $dw = e^{-u} du$, we obtain:

$$\int_a^\infty u^{1/2} e^{-u} du \leq C a^{1/2} e^{-a} \qquad \text{(T3.4)}$$

for $a \geq 1$. Substituting $a = \log(2\Delta_n n)$ yields $e^{-a} = (2\Delta_n n)^{-1}$ and $\Delta_n = O(n^{\zeta_\delta})$:

$$\begin{aligned}(\mathrm{I}) &\leq C(\log n)\, n^{(\gamma-1)/2} \cdot 2n^{\zeta_\delta} \cdot (C_\zeta \log n)^{1/2} \cdot \frac{1}{2n^{\zeta_\delta} \cdot n} \\ &= O\big(n^{(\gamma-1)/2-1}(\log n)^{3/2}\big)\,, \end{aligned} \tag{T3.5}$$

where ζ_δ cancels exactly. The constant component of (T3.2) contributes:

$$\begin{aligned}\int_0^{\epsilon_0} C(\log n)^2 n^{(\gamma-1)/2}\, d\epsilon &= C(\log n)^2\, n^{(\gamma-1)/2} \cdot n^{-1} \\ &= O\big(n^{(\gamma-1)/2-1}(\log n)^2\big)\,, \end{aligned} \tag{T3.2a}$$

so that the two components of Term (I) are of the same polynomial order and (I) $= O(n^{(\gamma-1)/2-1}(\log n)^2)$. For $\gamma < \frac{1}{2}$: $\frac{\gamma-1}{2} - 1 < -\frac{3}{4} < -\frac{1}{2} - 2\eta$, therefore (I) $= o(\delta_n^2)$.

Term (II): for $\epsilon \in [\epsilon_0, \delta_n]$, bound $\log(2\Delta_n/\epsilon) \leq \log(2\Delta_n \cdot n) \leq C_\zeta \log n$ uniformly, so that the integrand in (T3.2) is bounded by $C(\log n)^{3/2} n^{(\gamma-1)/2} + C(\log n)^2 n^{(\gamma-1)/2} \leq C'(\log n)^2 n^{(\gamma-1)/2}$. Then:

$$\begin{aligned}(\mathrm{II}) \leq \int_{\epsilon_0}^{\delta_n} C'(\log n)^2\, n^{(\gamma-1)/2}\, d\epsilon &= C'(\log n)^2 n^{(\gamma-1)/2} \int_{\epsilon_0}^{\delta_n} d\epsilon \\ &\leq C'(\log n)^2 n^{(\gamma-1)/2} \cdot \delta_n \end{aligned} \tag{T3.6}$$

For (T3.1) to hold, we require (II) $\leq C_3 \delta_n^2$. This requires:

$$n^{(\gamma-1)/2+1/4+\eta}(\log n)^2 \leq C_3, \tag{T3.7}$$

where $\frac{\gamma-1}{2} + \frac{1}{4} + \eta = \frac{\gamma}{2} - \frac{1}{4} + \eta$. For (T3.7) to hold as $n \to \infty$, we require the left side to be bounded, i.e., $\frac{\gamma}{2} - \frac{1}{4} + \eta < 0$, which gives $\gamma < \frac{1}{2} - 2\eta$. Since $\eta > 0$ may be taken arbitrarily small, the binding condition is $\gamma < \frac{1}{2}$, under which $n^{\gamma/2-1/4+\eta}(\log n)^2 \to 0$ and (T3.7) is satisfied.

Since both Term (I) and Term (II) satisfy the fixed-point condition (T3.1) under $0 <$

$\gamma < \frac{1}{2}$, Theorem 1 of Chen and Shen (1998) with block size $a_{n1} = \lfloor n^{1/2} \rfloor$, $a_{n2} = \lfloor n/(2a_{n1}) \rfloor$ (under K.1(b), trivially under K.1(a), and by Corollary 1 of Chen and Shen (1998) under K.1(c)) yields:

$$\|\hat{\theta}_n - \pi_n \theta_0\| = O_P(\delta_n) = o_P(n^{-1/4}) \tag{T3.8}$$

since $\delta_n = n^{-1/4-\eta}$ for arbitrary $\eta > 0$. Solving (T3.6)-(T3.7) for the smallest admissible δ_n rather than verifying the condition at a given δ_n identifies the rate delivered by Condition A.3 explicitly:

$$\delta_n \asymp n^{(\gamma-1)/2} (\log n)^2, \tag{T3.9}$$

so that matching exponents in $\delta_n = n^{-1/4-\eta}$ admits any fixed $\eta < \frac{1-2\gamma}{4}$, which is positive precisely when $\gamma < \frac{1}{2}$. □

Lemma 5. *Let $\hat{\theta}(\lambda_n)$ maximize $\mathcal{L}_n(\theta) - \lambda_n \|\boldsymbol{c}_n\|_{1,2}$ over Θ_n, with λ_n defined by (2.8) and $\lambda_n > 0$. Define the slack $\bar{\Delta}_n \equiv \Delta_n - R_n(\lambda_n) \geq 0$. Then:*

(i) $\hat{\theta}(\lambda_n) \in \mathcal{K}_n$;

(ii) $\mathcal{L}_n(\hat{\theta}(\lambda_n)) \geq \sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta) - \lambda_n \bar{\Delta}_n$, so that $\hat{\theta}(\lambda_n)$ is the exact constrained maximizer whenever R_n is continuous at λ_n (i.e., where $\bar{\Delta}_n = 0$);

(iii) irrespective of the continuity of R_n, $\mathcal{L}_n(\hat{\theta}(\lambda_n)) \geq \mathcal{L}_n(\pi_n \theta_0) - \lambda_n \|\boldsymbol{c}(\pi_n \theta_0)\|_{1,2}$.

Under Assumption K.2(d), the tolerance in (iii) is $O(\delta_n^2)$, so $\hat{\theta}(\lambda_n)$ is an approximate sieve extremum estimator over $\mathcal{K}_n$ in the sense of Chen and Shen (1998) and the conclusion of Theorem 3 applies to it.

Proof of Lemma 5:

For (i), the definition of λ_n as an infimum together with the right continuity of the non-increasing map R_n gives $R_n(\lambda_n) \leq \Delta_n$, which is the $\ell_{1,2}$ constraint defining $\mathcal{K}_n$. The

remaining constraints defining $\mathcal{K}_n$ are imposed on Θ_n directly and are therefore inherited.

For (ii), fix any $\theta \in \mathcal{K}_n$ with coefficient vector $\boldsymbol{c}(\theta)$. Since $\mathcal{K}_n \subseteq \Theta_n$, the point θ is feasible in the penalized problem, so optimality of $\hat{\theta}(\lambda_n)$ yields

$$\mathcal{L}_n(\hat{\theta}(\lambda_n)) - \lambda_n R_n(\lambda_n) \;\geq\; \mathcal{L}_n(\theta) - \lambda_n \|\boldsymbol{c}(\theta)\|_{1,2}. \tag{A.1}$$

Rearranging and using $\|\boldsymbol{c}(\theta)\|_{1,2} \leq \Delta_n$, which holds because $\theta \in \mathcal{K}_n$, gives

$$\mathcal{L}_n(\hat{\theta}(\lambda_n)) \;\geq\; \mathcal{L}_n(\theta) - \lambda_n\big(\|\boldsymbol{c}(\theta)\|_{1,2} - R_n(\lambda_n)\big) \;\geq\; \mathcal{L}_n(\theta) - \lambda_n \bar{\Delta}_n. \tag{A.2}$$

Taking the supremum over $\theta \in \mathcal{K}_n$ delivers the stated inequality. When R_n is continuous at λ_n we have $R_n(\lambda) > \Delta_n$ for every $\lambda < \lambda_n$ by the definition of the infimum, so continuity forces $R_n(\lambda_n) = \Delta_n$ and hence $\bar{\Delta}_n = 0$. The constraint then binds, the penalty difference against any competitor in $\mathcal{K}_n$ is non-positive, and $\hat{\theta}(\lambda_n)$ attains the constrained supremum exactly.

For (iii), apply the same optimality inequality at the single point $\theta = \pi_n \theta_0$, which lies in $\mathcal{K}_n \subseteq \Theta_n$ by Assumption K.2(d), and discard the non-negative term $\lambda_n R_n(\lambda_n)$ on the left. This yields the stated bound without reference to $R_n(\lambda_n)$ and therefore without any continuity requirement. Under Assumption K.3(b), the projection $\pi_n \theta_0$ has at most $s_n = O(\bar{s}_0)$ active edges and Assumption K.2(c) bounds the coefficient norm of each active edge, so $\|\boldsymbol{c}(\pi_n \theta_0)\|_{1,2} = O(\bar{s}_0)$. The tolerance is then $O(\lambda_n \bar{s}_0)$, which is $O(\delta_n^2)$ under Assumption K.2(b). □

Remark. Our implementation sets $\lambda_n \asymp \sqrt{\log(L_n W_n^2)/n}$, which under $W_n = C_W \lfloor \log n \rfloor$ and $L_n \asymp \log n$ is of order $n^{-1/2}(\log\log n)^{1/2}$. With $\bar{s}_0 = O((\log n)^2)$, Assumption K.2(d) requires $n^{-1/2}(\log n)^2(\log\log n)^{1/2} = O(\delta_n^2)$, which holds for any rate of the form $\delta_n \asymp n^{-1/4}(\log n)^a$ with $a > 1$. The condition is therefore a restriction on the logarithmic factors carried by δ_n rather than on its polynomial order, it does not bind on the $o_P(n^{-1/4})$ conclusion of Theorem 3.

A.2 Proof of Main Results (Asymptotic Normality)

Definition 3 (β-mixing). *A sequence $\{Z_t\}$ is β-mixing if its mixing coefficient $\beta(j) \to 0$ as $j \to \infty$, where given σ-fields $\mathcal{F}_{-\infty}^{t}, \mathcal{F}_{t+j}^{\infty}$ generated by $(Z_{-\infty}, \ldots, Z_t)$ and $(Z_{t+j}, \ldots Z_{\infty})$ (respectively):*

$$\beta(j) \equiv \sup_t \mathbb{E} \sup_{B \in \mathcal{F}_{t+j}^{\infty}} |P(B \mid \mathcal{F}_{-\infty}^{t}) - P(B)|$$

Definition 4 (ϕ-mixing). *A sequence $\{Z_t\}$ is uniform mixing (ϕ-mixing) if its mixing coefficient $\phi(j) \to 0$ as $j \to \infty$, where given σ-fields from **Definition 3** above:*

$$\phi(j) \equiv \sup_t \sup_{A \in \mathcal{F}_{-\infty}^{t}, P(A)>0} \sup_{B \in \mathcal{F}_{t+j}^{\infty}} |P(B \mid A) - P(B)|.$$

A brief note on notation. Since the supremum over $\mathcal{K}_n$ is attained exactly under compactness and K.2(c), the slackness condition from Chen and Shen (1998, Equation 2.1) used to define *approximate* sieve extremum estimators is equal to zero (written in terms of $\tilde{\epsilon}_n$ in this framework to avoid confusion and discussed earlier[61]; the KASN estimator is an *exact* sieve extremum estimator with $\tilde{\epsilon}_n = 0$). The reused term ϵ_n appearing in Conditions B.1-B.3 (stated in full below) of Chen and Shen (1998) now corresponds to the sieve approximation error, bounded in Theorem 1.

Condition A (Chen and Shen, 1998):

A.1. $\{Y_i\}_{i=1}^{n}$ is a stationary β-mixing sequence with $\beta(j) \leq \beta_0 j^{-\xi}$ for some $\beta_0 > 0$, $\xi = \kappa - 2 > 2$.

A.2. For all small $\epsilon > 0$

$$\sup_{\{\theta \in \Theta_n : d(\theta_0, \theta) \leq \epsilon\}} \text{var}(l(\theta, Y_t) - l(\theta_0, Y_t)) \leq C_1 \epsilon^2.$$

[61] See the proof of Theorem 3 in the previous section.

A.3. Let $\mathcal{F}_n = \{l(\theta, Y_t) - l(\theta_0, Y_t) : d(\theta_0, \theta) \leq \delta, \theta \in \Theta_n\}$. There exists $\delta_n \in (0, 1)$ such that

$$\delta_n = \sup \left\{ \delta > 0 : \delta^{-2} \int_{b\delta^2}^{a\delta} H^{1/2}(w, \mathcal{F}_n)\, dw \leq C_2 n^{1/2} \right\}.$$

A.4. For any $\delta > 0$, there exist a constant $\nu \in (0, 2)$ and a measurable function $U_n(\cdot)$ such that

$$\sup_{\{\theta \in \Theta_n : d(\theta_0, \theta) \leq \delta\}} |l(\theta, Y_t) - l(\theta_0, Y_t)| \leq \delta^{\nu} U_n(Y_t),$$

with $\sup_n E[U_n(Y_t)]^{\kappa} \leq C_2$ for $\kappa > 2$.

Condition B (Chen and Shen, 1998):

B.1.

$$\sup_{\{\theta \in \Theta_n : \|\theta - \theta_0\| \leq \delta_n\}} \mu_n \left(r[\theta - \theta_0, Y] - r[P_n(\theta^*(\theta, \epsilon_n)) - \theta_0, Y] \right) = O_p(\epsilon_n^2).$$

B.2.

$$\sup_{\{\theta \in \Theta_n : 0 < \|\theta - \theta_0\| \leq \delta_n\}} \left(K(\theta_0, P_n \theta^*(\theta, \epsilon_n)) - K(\theta_0, \theta) - \frac{1}{2} \left[\|\theta^*(\theta, \epsilon_n) - \theta_0\|^2 - \|\theta - \theta_0\|^2 \right] \right) = O(\epsilon_n^2).$$

B.3.

$$\sup_{\{\theta \in \Theta_n : 0 < \|\theta - \theta_0\| \leq \delta_n\}} \|\theta^*(\theta, \epsilon_n) - P_n(\theta^*(\theta, \epsilon_n))\| = O(\delta_n^{-1} \epsilon_n^2);$$

in addition,

$$\sup_{\{\theta \in \Theta_n : \|\theta - \theta_0\| \leq \delta_n\}} \mu_n \left(l'_{\theta_0}[\theta^*(\theta, \epsilon_n) - P_n(\theta^*(\theta, \epsilon_n)), Y] \right) = O_p(\epsilon_n^2).$$

B.4.

$$\sup_{\{\theta \in \Theta_n : \|\theta - \theta_0\| \leq \delta_n\}} \mu_n \left(l'_{\theta_0}[\theta - \theta_0, Y] \right) = O_p(\epsilon_n).$$

B.5.

$$n^{1/2}\mu_n\left(l'_{\theta_0}[U^*, Y]\right) \xrightarrow{P_{\theta_0}} \mathcal{N}(0, \sigma_v^2)$$

with

$$\sigma_v^2 = \lim_{n\to\infty} n^{-1}\text{var}_0\left(\sum_{t=1}^{n} l'_{\theta_0}[U^*, Y_t]\right) > 0.$$

Proof of Theorem 4: We begin verifying conditions B.1-B.5 by defining the class $\mathcal{F}_n \equiv \{\theta \in \mathcal{K}_n : \|\theta - \theta_0\| \leq \epsilon_n\}$. Note that under Assumption K.5, we have:

$$\|v^* - \pi_n v^*\|_\infty = O\big((\log n)^2\, n^{-(m+1)\gamma}\big) = O\big(n^{-1/4-\eta_v}\big) \tag{T4.0}$$

For quadratic loss the remainder difference is given by:

$$r[\theta - \theta_0, Z] - r[\pi_n\theta_0 - \theta_0, Z] = -\tfrac{1}{2}\left(\theta - \pi_n\theta_0\right)\big(\theta + \pi_n\theta_0 - 2\theta_0\big)(\tilde{X}). \tag{T4.1}$$

Consider the class $\mathcal{R}_n \equiv \{r[\theta - \theta_0, Z] - r[\pi_n\theta_0 - \theta_0, Z] : \theta \in \mathcal{F}_n\}$. Its envelope is bounded in sup norm by $C\,\mathcal{B}_n^2$, and applying Hölder's inequality together with the coupling $\eta \leq \eta_0$ (so that $\|\theta - \pi_n\theta_0\| \leq \epsilon_n + o(n^{-1/4}) \leq 2\epsilon_n$), each element satisfies:

$$\begin{aligned}\big\|r[\theta - \theta_0, Z] - r[\pi_n\theta_0 - \theta_0, Z]\big\|_{L_2(\mu)} &\leq \tfrac{1}{2}\,\|\theta + \pi_n\theta_0 - 2\theta_0\|_\infty\,\|\theta - \pi_n\theta_0\|_{L_2(\mu)} \\ &\leq C\,\mathcal{B}_n \cdot 2\epsilon_n \\ &\leq C'\,\mathcal{B}_n\,\epsilon_n. \end{aligned} \tag{T4.2}$$

The map $\theta \mapsto r[\theta - \theta_0, Z] - r[\pi_n\theta_0 - \theta_0, Z]$ is Lipschitz in the sup norm with constant $C\,\mathcal{B}_n$, so brackets (in the sup-norm) for $\mathcal{K}_n$ transform into brackets for $\mathcal{R}_n$, and by Corollary 2:

$$\log \mathcal{N}_{[]}\big(\epsilon, \mathcal{R}_n, L_2(\mu)\big) \leq C_1 s_n \log\left(\frac{r_n}{s_n}\right) + C_2 s_n G_n\Big[\log\left(\frac{4\mathcal{B}_n\Delta_n}{\epsilon}\right) + (\log n)^2\Big]. \tag{T4.3}$$

Applying Lemma 1 of Chen and Shen (1998) to $\mathcal{R}_n$ at radius $C'\mathcal{B}_n\epsilon_n$, the entropy integral

computation is identical to that of the proof of Theorem 3, with Δ_n replaced by $4\mathcal{B}_n\Delta_n$ and the radius rescaled by $\mathcal{B}_n$, both of which alter only polylogarithmic factors. Since $\epsilon_n = \delta_n$ solves that integral condition by construction:

$$\sup_{\theta\in\mathcal{F}_n}\Big|\mu_n\big(r[\theta-\theta_0,Z]-r[\pi_n\theta_0-\theta_0,Z]\big)\Big| = O_P\big(\epsilon_n^2\big). \tag{T4.4}$$

and condition B.1 is satisfied.

To see how condition B.2 holds, note that $\mathbb{E}[r[\theta-\theta_0,Z]] = -\frac{1}{2}\|\theta-\theta_0\|^2$, so for $\theta\in\mathcal{F}_n$, factoring the difference of squares and applying the triangle inequality:

$$\begin{aligned}\big|\mathbb{E}\,r[\theta-\theta_0,Z]-\mathbb{E}\,r[\pi_n\theta_0-\theta_0,Z]\big| &= \tfrac{1}{2}\Big|\|\theta-\theta_0\|^2-\|\pi_n\theta_0-\theta_0\|^2\Big| \\ &= \tfrac{1}{2}\big(\|\theta-\theta_0\|+\|\pi_n\theta_0-\theta_0\|\big)\Big|\|\theta-\theta_0\|-\|\pi_n\theta_0-\theta_0\|\Big| \\ &\le \tfrac{1}{2}\big(\|\theta-\theta_0\|+\|\pi_n\theta_0-\theta_0\|\big)^2 \\ &\le C\,\epsilon_n^2, \end{aligned}\tag{T4.5}$$

using $\|\pi_n\theta_0-\theta_0\| = O(n^{-1/4-\eta_0}) \le C\epsilon_n$ under the coupling $\eta\le\eta_0$.

By the Cauchy-Schwarz inequality and (T4.0):

$$\begin{aligned}\sup_{\theta\in\mathcal{F}_n}\big|\langle\theta-\theta_0,\,v^*-\pi_nv^*\rangle\big| &\le \epsilon_n\,\|v^*-\pi_nv^*\| \\ &= \epsilon_n\,O\big(n^{-1/4-\eta_v}\big) \\ &= O\big(\epsilon_n^2\big), \end{aligned}\tag{T4.6}$$

where the final equality uses $\epsilon_n = n^{-1/4-\eta}$ with $\eta\le\eta_v$ and condition B.3 is verified.

In the case of B.4, the process $\mu_n\big(l'[v^*-\pi_nv^*,Z]\big) = \mu_n\big(U\,(v^*-\pi_nv^*)(\tilde{X})\big)$ has mean zero since $\mathbb{E}[U_t\mid\tilde{X}_t]=0$. Write $g_n\equiv v^*-\pi_nv^*$, a fixed (non-random) function for each n with $\|g_n\|_\infty = O(n^{-1/4-\eta_v})$ by (T4.0). Applying Davydov's (covariance) inequality (see Doukhan,

1994, Section 1.2.2) with $\kappa > 2$ as in the theorem premise:

$$\begin{aligned}\operatorname{Var}\Big(n^{1/2}\mu_n\big(Ug_n(\tilde{X})\big)\Big) &\le C\,\|g_n\|_\infty^2 \sup_t \big(\mathbb{E}|U_t|^\kappa\big)^{2/\kappa} \sum_{j=0}^{\infty} \beta(j)^{1-2/\kappa} \\ &= O\big(n^{-1/2-2\eta_v}\big), \end{aligned} \tag{T4.7}$$

where the mixing series converges under the exponential decay of K.1(b) (and under K.1(c) provided $\xi(1-2/\kappa) > 1$, as permitted by the premise). By Chebyshev's inequality and (T4.7):

$$\begin{aligned}\mu_n\big(l'[g_n, Z]\big) &= O_P\big(n^{-1/2} \cdot n^{-1/4-\eta_v}\big) \\ &= O_P\big(n^{-3/4-\eta_v}\big) \\ &= O_P\big(\epsilon_n^2\big), \end{aligned} \tag{T4.8}$$

since $\epsilon_n^2 = n^{-1/2-2\eta}$ and $3/4 + \eta_v \ge 1/2 + 2\eta$ under $\eta \le \eta_v$.

Lastly, the sequence $\{l'[v^*, Z_t]\}_t = \{U_t v^*(\tilde{X}_t)\}_t$ is strictly stationary with mean zero, inherits the mixing properties of $\{Z_t\}$ under K.1 (as a measurable function of Z_t), and satisfies $\mathbb{E}|U_t v^*(\tilde{X}_t)|^\kappa \le B_v^\kappa \sup_t \mathbb{E}|U_t|^\kappa < \infty$ for $\kappa > 2$ by K.5 (and $\kappa > 4$ under K.1(c); given $B_v < \infty$). The central limit theorem for strongly mixing sequences (Doukhan, 1994, Theorem 1.7; alternatively Corollary 1 of Chen and Shen, 1998, under K.1(c)) then yields $n^{1/2}\mu_n(l'[v^*, Z]) \xrightarrow{d} \mathcal{N}(0, \sigma_*^2)$ with σ_*^2 as displayed, the covariance series converging absolutely by the same Davydov bound as in (T4.7). Combining B.1-B.5 through Theorem 2 of Chen and Shen (1998) completes the proof. □

Lemma 6. *Fix a coordinate j and suppose the conditional density $p_{j|-j}$ of $\tilde{X}_j$ given $\tilde{X}_{-j}$ satisfies $0 < c_p \le p_{j|-j} \le C_p < \infty$ uniformly. Let $g : [0,1]^d \to \mathbb{R}$ satisfy $\|\partial_j^l g\|_\infty \le \mathcal{D}$ for $0 \le l \le m+1$. Then, with a constant independent of d:*

$$\|\partial_j g\|_{L_2(\mu)} \le C(m, c_p, C_p)\, \|g\|_{L_2(\mu)}^{\frac{m}{m+1}} \big(\mathcal{D} + \|g\|_{L_2(\mu)}\big)^{\frac{1}{m+1}}.$$

Proof of Lemma 6: For fixed x_{-j}, write $h_{x_{-j}}(\cdot) \equiv g(\cdot, x_{-j})$ on $[0,1]$. The one-dimensional Gagliardo-Nirenberg inequality (Adams and Fournier, 2003, Theorem 5.2) gives $\|h'_{x_{-j}}\|_{L_2[0,1]} \leq K \|h_{x_{-j}}\|^{m/(m+1)}_{L_2[0,1]} \|h_{x_{-j}}\|^{1/(m+1)}_{W^{m+1,2}[0,1]}$, with $\|h_{x_{-j}}\|_{W^{m+1,2}[0,1]} \leq C(m)(\mathcal{D} + \|h_{x_{-j}}\|_{L_2[0,1]})$ by the derivative bounds. Squaring, integrating over x_{-j} against μ_{-j}, and applying Hölder's inequality with exponents $\frac{m+1}{m}$ and $m+1$:

$$\int \|h'_{x_{-j}}\|^2_{L_2[0,1]}\, d\mu_{-j} \leq C \left(\int \|h_{x_{-j}}\|^2_{L_2[0,1]}\, d\mu_{-j} \right)^{\frac{m}{m+1}} \left(\mathcal{D}^2 + \int \|h_{x_{-j}}\|^2_{L_2[0,1]}\, d\mu_{-j} \right)^{\frac{1}{m+1}}. \quad \text{(L5.1)}$$

The density bounds transfer between slice integrals and μ, since $\int \|h_{x_{-j}}\|^2_{L_2[0,1]} d\mu_{-j} \leq c_p^{-1} \|g\|^2_{L_2(\mu)}$ and $\|\partial_j g\|^2_{L_2(\mu)} \leq C_p \int \|h'_{x_{-j}}\|^2_{L_2[0,1]} d\mu_{-j}$. Chaining these through (L5.1) and taking square roots yields the claim. □

Proposition 2. *Consider the estimand Γ_j and plug-in estimator $\hat{\Gamma}_j$ of (3.1). Suppose Assumptions K.1-K.4 hold, together with:*

(E.1) f_j is continuously differentiable, strictly positive on the interior of its support, and vanishes at the boundary, with $w'_j = (f'_j/f_j) \circ F_j^{-1}$ satisfying K.4;

(E.2) the conditional density satisfies $0 < c_p \leq p_{j|-j} \leq C_p < \infty$, and the copula density p is continuously differentiable in its j-th argument;

(E.3) Assumption K.5 holds for the representer

$$v_j^*(\tilde{x}) = -\big(w'_j(\tilde{x}_j) + w_j(\tilde{x}_j)\, \partial_j \log p(\tilde{x})\big), \quad \text{(A.3)}$$

obtained by integration by parts, whose boundary terms vanish under E.1 since $w_j(0) = w_j(1) = 0$;

(E.4) $m \geq 1$, $k \geq m+2$, and $\frac{1}{4(m+1)} < \gamma < \frac{m}{2m+1}$.

Then $\sqrt{n}(\hat{\Gamma}_j - \Gamma_j) \xrightarrow{d} \mathcal{N}(0, \sigma^2_\psi)$, where σ^2_ψ is the long-run variance of the influence sequence $\psi_t = w_j(\tilde{X}_{t,j})\partial_j \theta_0(\tilde{X}_t) - \Gamma_j + U_t v_j^*(\tilde{X}_t)$.

Proof of Proposition 2: Decompose $\sqrt{n}(\hat{\Gamma}_j - \Gamma_j) = \text{(I)} + \text{(II)} + \text{(III)}$, where $\text{(I)} = \sqrt{n}\,\mu_n(w_j\partial_j\theta_0)$, $\text{(II)} = \sqrt{n}\,\mathbb{E}[w_j\partial_j(\hat{\theta}_n - \theta_0)]$, and $\text{(III)} = \sqrt{n}\,\mu_n(w_j\partial_j(\hat{\theta}_n - \theta_0))$. Term (I) joins the score through the influence sequence and is handled by the mixing CLT as in Condition B.5, using E.1 for the boundedness and smoothness of $w_j\partial_j\theta_0$. Term (II) equals $\sqrt{n}\langle\hat{\theta}_n - \theta_0, v_j^*\rangle$ by E.3; since Γ_j is linear with no remainder, Theorem 2 of Chen and Shen (1998) under the conditions verified for Theorem 4 (with K.5 instantiated at v_j^*) delivers $\sqrt{n}\langle\hat{\theta}_n - \theta_0, v_j^*\rangle = \sqrt{n}\mu_n(Uv_j^*(\tilde{X})) + o_P(1)$.

Term (III) is the stochastic equicontinuity term, where the derivative cap $\mathcal{D}_n$ and Lemma 5 are relevant. Applying Lemma 5 to $g = \hat{\theta}_n - \theta_0$ (whose derivatives are bounded by $2\mathcal{D}_n$ by (2.6)) together with Theorem 3 gives $\|\partial_j(\hat{\theta}_n - \theta_0)\| = O_P(\delta_n^{m/(m+1)}\mathcal{D}_n^{1/(m+1)})$. The derivative class $\{\partial_j\theta : \theta \in \mathcal{K}_n\}$ obeys the entropy bound of Theorem 2 with $\log\Lambda_n$ inflated by an additive $O((\log n)^2)$ (the sup norm of B-spline derivatives contributes a factor C_kG_n per layer), and multiplication by the bounded w_j preserves brackets, so Lemma 1 of Chen and Shen (1998) applied at radius $O_P(\rho_n)$ with the fixed-point rate $\delta_n \asymp n^{(\gamma-1)/2}(\log n)^2$ of Theorem 3 yields:

$$\sqrt{n}\left|\mu_n\big(w_j\partial_j(\hat{\theta}_n - \theta_0)\big)\right| = O_P\Big(\delta_n^{\frac{m}{m+1}}\mathcal{D}_n^{\frac{1}{m+1}}\sqrt{s_nG_n}\,(\log n)^c\Big) = O_P\Big(n^{\frac{(\gamma-1)}{2}\frac{m}{m+1}+\frac{\gamma}{2}}\,(\log n)^{c'}\Big), \tag{P1.1}$$

whose exponent is negative if and only if $\gamma < \frac{m}{2m+1}$, which is E.4, so $\text{(III)} = o_P(1)$. On the approximation side, (C1.1) at derivative order $l = 1$ gives $\|w_j\partial_j(\pi_n\theta_0 - \theta_0)\| = O(\text{polylog}) \cdot O(G_n^{-(m+1)}) = o(n^{-1/4})$ under $\gamma > \frac{1}{4(m+1)}$, consuming only first-derivative control rather than the full W_2^{m+1} rate of Corollary 1. Collecting terms, $\sqrt{n}(\hat{\Gamma}_j - \Gamma_j) = \sqrt{n}\mu_n(\psi) + o_P(1)$, and the mixing CLT applied to $\{\psi_t\}$ completes the argument. □

A.3 NLO Sample Splitting

We extend Proposition 1 to time-series settings through the NLO cross-fitting scheme of Semenova et al. (2023). We establish asymptotic normality by verifying the high-level conditions of Gonçalves et al. (2026, Theorem 5.1). The verification reduces to three assumptions, each satisfied in the KASN framework under K.1(b), the standard sieve envelope condition K.2(c), and Corollary 3.

Let $\{Z_t\}_{t=1}^n$ be strictly stationary under K.1(b), let $\alpha \in \mathbb{R}$ denote the scalar target parameter, and let $\eta \in \mathcal{T}$ denote an infinite-dimensional nuisance. The Neyman-orthogonal score satisfies $\mathbb{E}[\psi(Z_t; \alpha_0, \eta_0)] = 0$, and is linear in the target:

$$\psi(Z_t; \alpha, \eta) = \psi^a(Z_t; \eta)\,\alpha + \psi^b(Z_t; \eta), \qquad J_0 := \mathbb{E}[\psi^a(Z_t; \eta_0)]. \tag{A.4}$$

Partition $\{1, \ldots, n\}$ into $K \geq 4$ adjacent blocks[62] of size $T_b = \lfloor n/K \rfloor$, with $\mathcal{M}_k = \{(k-1)T_b + 1, \ldots, kT_b\}$ and any remainder appended to $\mathcal{M}_K$. Writing $N(k) = \{k-1, k, k+1\} \cap \{1, \ldots, K\}$, the quasi-complement is

$$\mathcal{M}_k^{qc} = \bigcup_{j \notin N(k)} \mathcal{M}_j,$$

which is non-empty for $K \geq 4$ and separated from $\mathcal{M}_k$ by at least T_b periods on each side. We adopt the construction of Gonçalves et al. (2026) such that the two endpoints are not treated as adjacent. For each k, let $\hat{\eta}_k$ be fit on $B_k^{qc} = \{Z_t : t \in \mathcal{M}_k^{qc}\}$. The NLO-DML estimator and Jacobian are

$$\hat{\alpha} \equiv -\hat{J}^{-1} \cdot \frac{1}{n} \sum_{k=1}^{K} \sum_{t \in \mathcal{M}_k} \psi^b(Z_t; \hat{\eta}_k), \qquad \hat{J} \equiv \frac{1}{n} \sum_{k=1}^{K} \sum_{t \in \mathcal{M}_k} \psi^a(Z_t; \hat{\eta}_k). \tag{A.5}$$

Gonçalves et al. (2026) state Theorem 5.1 for a doubly robust moment whose target

[62] Noting that the local index k in this construction is distinct from the spline order k defined in the main paper.

is the average response function $\theta_{0,h}$ and whose nuisances are the conditional mean $g_{0,h}$ and the density-ratio Riesz representer α_0. For the PLR model of Example 1, we have: $Y_t = D_t\alpha_0 + g_0(W_t) + U_t$ and $D_t = m_0(W_t) + V_t$ with $\mathbb{E}[U_t \mid D_t, W_t] = 0$ and $\mathbb{E}[V_t \mid W_t] = 0$. The orthogonal score is $\psi(Z_t; \alpha, \eta) = (Y_t - D_t\alpha - g(W_t))(D_t - m(W_t))$ with $\eta = (g, m)$. This gives $\psi^a(Z_t; \eta) = -D_t(D_t - m(W_t))$, $\psi^b(Z_t; \eta) = (Y_t - g(W_t))(D_t - m(W_t))$, $J_0 = -\mathbb{E}[V_t^2]$, and (oracle) score $\psi(Z_t; \alpha_0, \eta_0) = V_tU_t$.

Proposition 3. *Suppose K.1(b) and K.2–K.4 hold, with $K \geq 4$ fixed and $T_b = \lfloor n/K \rfloor$, and impose the conditions $0 < \mathbb{E}[V_t^2] < \infty$ and $\mathbb{E}[(V_tU_t)^4] < \infty$, together with a HAC bandwidth b_n satisfying $b_n \to \infty$ and $b_n/n \to 0$. Then the NLO-DML estimator (A.5) satisfies*

$$\sqrt{n}\,(\hat{\alpha} - \alpha_0) = -J_0^{-1}\,\frac{1}{\sqrt{n}}\sum_{t=1}^{n}\psi(Z_t; \alpha_0, \eta_0) + o_P(1) \rightsquigarrow N\big(0,\, \sigma_\psi^2/J_0^2\big)\,, \tag{A.6}$$

and $\sqrt{n}\,(\hat{\alpha} - \alpha_0)/\hat{\sigma} \rightsquigarrow N(0,1)$, where σ_ψ^2 is the long-run variance of $\psi(Z_t; \alpha_0, \eta_0)$ and $\hat{\sigma}^2$ is the Newey–West HAC estimator of σ_ψ^2/J_0^2 formed from the cross-fitted score residuals.

Proof. The estimator in (A.5) is the DR-NLO estimator of Gonçalves et al. (2026) tailored to the score (A.4). Their Theorem 5.1 delivers (A.6) once their Assumptions 1–3 hold, with the preliminary estimation of the nuisances leaving the (first-order) distribution unaffected through the orthogonality of the moment and the approximate independence between $\mathcal{M}_k$ and $\mathcal{M}_k^{qc}$ under NLO cross-fitting.

Their Assumption 1 requires $\{Z_t\}$ to be stationary and geometrically β-mixing, $\beta(j) \leq C\exp(-c_\beta j)$. This is simply K.1(b) with $C = \beta_0$ and $c_\beta = c$. Assumption 2 fixes some $q > 2$ and requires a bounded Riesz representer, a conditional q-th moment bound on the regression residual, and $\mathbb{E}[|\psi(Z_t; \alpha_0, \eta_0)|^{2+\epsilon}] < \infty$. Take $q = 4$ (noting that this is implied by our conditions on κ in the main text). The orthogonal nuisance is $V_t = D_t - m_0(W_t)$, and K.2(c) bounds m_0 (and its sieve approximant) uniformly given $0 < \mathbb{E}[V_t^2] < \infty$, so the representer direction is bounded. The conditional fourth-moment bound on $U_t = Y_t - D_t\alpha_0 - g_0(W_t)$ holds under the error-moment conditions of the main text, and $\mathbb{E}[|V_tU_t|^{2+\epsilon}] < \infty$ for $\epsilon = 2$

follows from $\mathbb{E}[(V_t U_t)^4] < \infty$ (since the oracle score equals $V_t U_t$). Lastly, their Assumption 3 requires $\hat{\eta}_k$ to lie in a shrinking neighborhood with probability tending to one, an L_q rate $o(1)$ on that neighborhood (their 3(i)), and the L_2 product rate $\sqrt{n}\, r_{g,n}\, r_{m,n} = o(1)$ (their 3(ii)). Define $\mathcal{G}_n = \{g \in \mathcal{K}_n : \|g - g_0\|_{L_2(\mu)} \leq \delta_n\}$ and $\mathcal{M}_n = \{m \in \mathcal{K}_n : \|m - m_0\|_{L_2(\mu)} \leq \delta_n\}$ with $\delta_n = o(n^{-1/4})$. Applying Theorems 1-3 on each $\mathcal{M}_k^{qc}$, whose effective sample size is $\tilde{n}_k \geq n(K-3)/K \asymp n$ since K is fixed, gives $\hat{g}_k \in \mathcal{G}_n$ and $\hat{m}_k \in \mathcal{M}_n$ (with probability tending to one). Under K.2(c) every element of $\mathcal{K}_n$ is uniformly bounded on the (normalized) domain, as are g_0 and m_0 under K.4(a), so $\sup_{g \in \mathcal{G}_n} \|g - g_0\|_\infty \leq 2B$. For any $q > 2$ and any $g \in \mathcal{G}_n$, we have

$$\|g - g_0\|_q \leq \|g - g_0\|_\infty^{1-2/q}\, \|g - g_0\|_2^{2/q} \leq (2B)^{1-2/q}\, \delta_n^{2/q} = o(1), \tag{A.7}$$

and the same bound holds over $\mathcal{M}_n$. Hence $r_{g,q,n}, r_{m,q,n} = o(1)$ for every fixed $q > 2$, in particular $q = 4$, so their 3(i) holds. For their 3(ii), Corollary 3 gives $\|\hat{g}_n - g_0\| = o_P(n^{-1/4})$ and $\|\hat{m}_n - m_0\| = o_P(n^{-1/4})$ in $L_2(\mu)$, so $\sqrt{n}\, \|\hat{g}_n - g_0\|\, \|\hat{m}_n - m_0\| = o_P(1)$.

With Assumptions 1–3 verified, Gonçalves et al. (2026, Theorem 5.1) yields the representation in (A.6). Consistency of the Newey–West estimator under mixing (Newey and West, 1987; Andrews, 1991), applied to the cross-fitted score residuals with bandwidth b_n, gives $\hat{\sigma}^2 \xrightarrow{P} \sigma_\psi^2 / J_0^2$, and a second application of Slutsky's theorem with $|J_0| \geq c_J > 0$ completes the studentized limit. □

Remark: Under K.1(c) the process is ϕ-mixing with $\phi(j) \leq \phi_0 j^{-\xi}$, so $\beta(j) \leq 2\phi(j) \leq 2\phi_0 j^{-\xi}$ by Volkonskii and Rozanov (1959). The geometric decay of Assumption 1 is replaced by this polynomial rate, which the argument of Gonçalves et al. (2026) accommodates whenever $\sum_j \beta(j)^{1-2/q} < \infty$, that is $\xi(1 - 2/q) > 1$. With $q = 4$ this requires $\xi > 2$, the condition already imposed in K.1(c). The central limit theorem then follows for ϕ-mixing sequences, and the verification of Assumptions 2–3 is unchanged.

A.4 KASN Optimization and Architecture Details

Define early stopping with patience ε_{tol} such that the computed estimator $\tilde{\theta}_n$ satisfies $\mathcal{L}_n(\tilde{\theta}_n) \geq \sup_{\theta \in \mathcal{K}_n} \mathcal{L}_n(\theta) - \varepsilon_{\text{tol}}$. By the triangle inequality and K.2(a):

$$\|\tilde{\theta}_n - \theta_0\| \leq \|\tilde{\theta}_n - \hat{\theta}_n\| + \|\hat{\theta}_n - \theta_0\| = O(\varepsilon_{\text{tol}}^{1/2}) + o_P(n^{-1/4}).$$

Setting $\varepsilon_{\text{tol}} = o(n^{-1/2})$ ensures the early stopping error[63] is $o(n^{-1/4})$ and does not affect the convergence rate of Theorem 3. This resulting condition is similar to the one in Chen and Shen (1998, Equation 2.1), however it does not concern the (deterministic) KASN class in (2.6) itself. In practice, the KASN estimator is optimized using the Adam optimizer (Kingma and Ba, 2014) with learning rate $\lambda_t \to 0$ satisfying $\sum_t \lambda_t = \infty$ and $\sum_t \lambda_t^2 < \infty$ (the standard stochastic approximation conditions of Robbins and Monro, 1951; see Zaheer et al. (2018) for the relevant discussion in the context of non-convex optimization problems). Under these conditions, Adam iterates converge to a stationary point of $\mathcal{L}_n$ over $\mathcal{K}_n$, and the early stopping criterion $\varepsilon_{\text{tol}} = o(n^{-1/2})$ ensures the resulting $\tilde{\theta}_n$ satisfies Theorem 3 without affecting the convergence rate. See the online supplementary material of Farrell et al. (2021) for a similar implementation in practice. The development of a novel stochastic gradient descent algorithm for penalized sieve neural networks (similar to the one proposed in Gouriéroux and Monfort, 2025) is left to future research.

Numerical optimization is performed using PyTorch with GPU acceleration via CUDA when available (CPU otherwise), which significantly reduces the severity of the main drawback of the KASN architecture. The local support properties of B-splines (see the definition of $\mathcal{I}_\epsilon$ in the main text) allow the per-edge spline evaluations to be expressed as a batched matrix operation. This vectorization ensures that increasing the grid resolution G_n does not require architectural changes to the computation graph, with runtime scaling linearly in G_n for fixed network width and depth.

[63] See the discussion in the Section 2.3 about a similar condition that appears in Shen (1997) involving ϵ_n.

The compositional structure of KAN architectures can be represented as follows, in contrast with the structure of a deep MLP:

$$\text{KAN}(\boldsymbol{x}) = \big(\boldsymbol{\Phi}_{L_n-1} \circ \boldsymbol{\Phi}_{L_n-2} \circ \cdots \circ \boldsymbol{\Phi}_1 \circ \boldsymbol{\Phi}_0\big)\boldsymbol{x}$$
$$\text{MLP}(\boldsymbol{x}) = \big(\boldsymbol{W}_{L-1} \circ \sigma \circ \boldsymbol{W}_{L-2} \circ \sigma \circ \ \cdots \circ \boldsymbol{W}_1 \circ \sigma \circ \boldsymbol{W}_0\big)\boldsymbol{x}$$

where σ denotes a fixed activation (e.g., ReLU) and $\boldsymbol{\Phi}_l$ represents a collection of learnable univariate spline functions at layer l in the form of a matrix. In other words, the typical weight matrices on the edges of MLP structures are replaced with univariate functions in the form of B-splines.

The difference between r_n and the number of *active* edges[64] can be seen in Figure 1, where the solid lines represent learned activation functions with non-zero B-spline coefficients under ℓ_1-regularization. In this example, the second input variable (x_2) does not contribute to the true target function in the DGP and there is an inactive node present in the second hidden layer, lacking any activation functions to sum as a result of the $\ell_{1,2}$-regularization. Using r_n, we define our total parameters in the KAN as $p_n \equiv O(r_n G_n)$ as in Liu et al. (2024) where $p = O(LW^2G)$.

The fact that the network learns activation functions instead of weight matrices is what allows us to fix W_n across hidden layers in the first place, attempting to do so with MLP structures[65] is incompatible with our sieve estimation approach. Under a hypothetical framework incorporating MLPs with sieve estimation, ensuring that the MLP simultaneously possesses a bounded entropy and a degree of approximation power sufficient for high-dimensional estimation (with a dimension-independent convergence rate of $o_P(n^{-1/4})$ or faster) is an extremely challenging task. This is due to how the approximation power of an MLP is tied to its width and depth, which are typically required to grow with the sample size in order to

[64]Denoted by s_n and determined by s_0 in addition to the B-spline coefficient constraint Δ_n: another key parameter in our sieve space definition.

[65]See Farrell et al. (2021) for a discussion on this class of "fixed width" networks, which are especially likely to struggle in high-dimensions.

achieve convergence rates comparable to the aforementioned minimax rates in nonparametric regression. As a result of this inherent dependency, it is difficult to control the metric entropy of such a sieve without inordinately restricting the approximation capabilities of the MLP.

In theory, the learnable activation functions $\phi(\cdot)$ can potentially include a basis function $b(x)$ in order to provide increased stability as part of our network initialization, as outlined in Liu et al. (2024). During training, the initial activation function would take on a weighted sum of the basis function and the B-spline function as follows:

$$\phi(x) = w_b b(x) + w_s \text{spline}(x)$$
$$b(x) = \text{silu}(x) = x/(1+e^{-x}), \quad \text{spline}(x) = \sum_{g=1}^{G_n} c_g B_g(x)$$

where $B_g(x) \equiv B_{g,k}(x)$ for a fixed spline order k and the weights (w_b, w_s) are updated during training in addition to the learned B-spline coefficients: with $w_b \to 0$ and $w_s \to 1$ early on in the training process, so that the resulting network derives its approximation capabilities purely from B-splines. Note that the basis function $b(x)$ is simply a combination of the ReLU and sigmoid activation functions, resulting in the "sigmoid linear unit", or "SiLU" activation.

We find that across Monte Carlo simulations, this optimization approach can lead to a persistent degree of overfitting, one that cannot be addressed adequately by our novel regularization method once the transition to $w_s = 1$ is complete. In other words, the residual influence of $b(x)$ can persist in the learned edge function in a way that the $\ell_{1,2}$ penalty on spline coefficients $\{\boldsymbol{c}_j\}$ does not directly penalize. To circumvent this issue, we instead introduce residual connections in the form of the identity mapping[66] $h : \mathbb{R}^{W_n^{l-1}} \to \mathbb{R}^{W_n^l}$, throughout the $l = 2, \ldots, L_n$ hidden layers. The residual connection $h(x) = x$ passes the input from the previous hidden layer, improving stability during training by mitigating the

[66] As denoted in the definition of the KASN class $\mathcal{K}_n$ in equation (2.6).

vanishing gradient problem[67] in deep networks similar to the ReLU/SiLU activation functions. For this reason, the implementation of residual connections is common in many deep MLP architectures (inspired by He et al., 2016) and the growing literature on KANs (e.g., Yu et al., 2024). Furthermore, the presence of this identity mapping eliminates the necessity for carefully selecting appropriate spline initializations, allowing for a wide range of initialization techniques to be applied in practice. Given the relatively aggressive regularization approach of our framework, ensuring gradient stability is crucial with respect to learning the relevant true $\phi(x) = \sum_{k=1}^{G_n} c_k B_k(x)$ edge functions.

The B-spline functions $B_{j,k}(x)$ are defined recursively[68]: for order k and a (non-decreasing) sequence of knot point t_j, the B-spline $B_{j,k}(x)$ is given by:

$$B_{j,k}(x) = \frac{x - t_j}{t_{j+k} - t_j} B_{j,k-1}(x) + \frac{t_{j+k+1} - x}{t_{j+k+1} - t_{j+1}} B_{j+1,k-1}(x)$$

with the base case for order $k = 1$ defined as:

$$B_{j,1}(x) = \begin{cases} 1 & \text{if } t_j \leq x < t_{j+1} \\ 0 & \text{otherwise} \end{cases}$$

In order to simplify the analysis in our key theorems/results, we fix the order of the B-splines as $k = 4$ (degree 3), with G_n intervals accordingly ($G_n + 1$ knot points in total: $t_0 < t_1 < \cdots < t_{G_n}$). We note that in practice, KANs typically use cubic splines for a wide range of general applications, so this formalization does not incur additional costs or tradeoffs beyond in certain settings outside of the scope of nonparametric/semiparametric regression (e.g., symbolic regression). Furthermore, our theoretical results extend to any $k \geq m + 1$, although the sensitivity analysis of higher order B-splines (across a wide range of network parameters) evaluating their stability in this setting is left to future work.

In the KASN framework, we do not apply entropy regularization, a method that pe-

[67] Through bounding the gradient magnitude away from zero.

[68] This definition is commonly referred to as the Cox-de Boor recursion formula: de Boor (1978).

nalizes the average magnitude of the activation functions commonly applied in KANs (see the approach outlined in Liu et al., 2024, Appendix C therein). While this may assist in reducing the number of redundant functions in general KAN architectures (in settings where $n \gg d$), this form of regularization effectively distributes the task of learning the structure of f_0 across the many potential edges. Liu et al. (2024) apply this in the opposite manner (with a negative penalty) in order to discourage dense networks. However, this heuristic approach fails to ensure against the explosion of the metric entropy in high-dimensions, unlike under the structural sparsity introduced in this framework. Instead, our constraint Δ_n in addition to penalizing entire vectors of B-spline coefficients across edges under our regularization method forces the network to use each active edge s_n as efficiently as possible *without* sacrificing approximation power; successfully controlling the metric entropy while balancing the approximation and estimation errors.

Under our proposed regularization method (that is novel for KAN architectures) and pruning based on the Δ_n/r_n threshold for active edges, KASNs can also be viewed[69] as a deep compositional generalization of the Multivariate Adaptive Regression Splines (MARS) estimator (Friedman, 1991). Unlike MARS, which requires manual specification of the maximum degrees of interaction[70], KASNs hierarchically learn complex interactions through a deep composition of univariate B-splines. The KART guarantees this representation exists, with Assumption K.3(a) imposing the sparse structure that controls combinatorial complexity at an $O(s_n) = O((\log n)^2)$ rate. Furthermore, the Δ_n/r_n threshold used to prune univariate edges is a sieve analogue of the backward deletion procedure (guided by Generalized CV) introduced by Friedman (1991), one that is formally derived from the entropy bound (Theorem 2) rather than heuristically.

[69] In addition to the illustrative comparisons throughout the main section of this paper.
[70] In order to control the complexity of basis functions.

A.5 Shallow Sieve Networks (Chen and White, 1999)

Let d_x denote the dimension of the covariates. Let $\mathcal{B}_{d_x}^m$ denote the weighted Sobolev space comprising all functions on $\mathbb{R}^{d_x}$ whose partial derivatives up to order m are both continuous and uniformly bounded (see the definitions in Section 2). The target function $f : \mathbb{R}^{d_x} \to \mathbb{R}$ is assumed to admit a Fourier representation such that it belongs to the functional class:

$$\mathcal{F}_{d_x}^{m+1} = \left\{ f : \mathbb{R}^d \to \mathbb{R} \;\middle|\; f(x) = \int \exp\big(ia^\top x\big)\, d\sigma_f(a),\ \|\sigma_f\|_{m+1} \equiv \int I(a)^{m+1}\, d|\sigma_f|_{\mathrm{tv}}(a) < \infty \right\} \tag{A.8}$$

This assumption imposes a general smoothness constraint (ensuring the Fourier transform possesses a bounded first moment) required by the class of SLFNs utilized in Chen and White (1999), denoted by $\mathcal{G}_n \equiv G_{d_x}^m(\psi, r_n, \Delta_n)$ and defined as:

$$\mathcal{G}_n = \Bigg\{ g : g(x) = \sum_{j=1}^{r_n} v_j l(a_j)^{-m} \psi(a_j' x + w_j), \qquad a_j \in \mathbb{R}^{d_x},\ (w_j, v_j) \in \mathbb{R},\ \sum_{j=1}^{r_n} |v_j| \le \Delta_n \Bigg\} \tag{A.9}$$

Here, $\psi \in \mathcal{B}_1^m$ represents the selected activation function (satisfying a Hölder condition) that is k-finite for some $k \ge m$. The parameter vector $\omega_j \equiv (a_j, w_j, v_j)$ contains the network weights, where r_n specifies the number of hidden units as discussed earlier in this paper. For any weight vector a_j, the term $l(a_j) \equiv \max\{(a_j' a_j)^{\frac{1}{2}}, 1\}$ normalizes the input layer weights, thereby controlling model complexity and enforcing the required smoothness of the approximation.